72. (b) $\omega_f = \omega_o + \alpha t$

Wheel stops $\omega_f = 0$

$$0 - \frac{2\pi \times 60}{60} = \alpha(60)$$

$$\alpha = \frac{\pi}{30}$$

Torque = $I\alpha$

$$= 2\frac{\pi}{30}$$

$$= \frac{\pi}{15}\text{ N-m}$$

73. (b) External Torque is zero

L = constant

$I_1\omega_1 = I_2\omega_2 \Rightarrow Mr^2\omega = (M + 4m)r^2\omega_1$

$$\omega_1 = \frac{M\omega}{M + 4m}$$

74. (b) When a child sits on a rotating disc, no external torque is introduced. Hence the angular momentum of the system is conserved. But the moment of inertia of the system will increase and as a result, the angular speed of the disc will decrease to maintain constant angular momentum.

[$\because$ Angular momentum = moment of inertia × angular velocity]

75. (c) According to law of conservation of angular momentum.

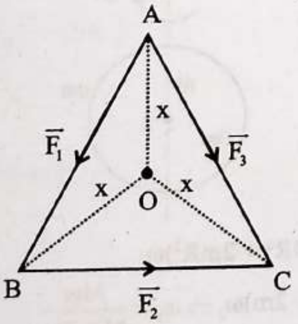
$I\omega = I'\omega'$

$Mr^2\omega = (Mr^2 + 2mr^2)\omega'$

$$\omega' = \frac{M\omega}{M + 2m}$$

76. (c) Torque will get balanced upon increasing the angle of inclination.

77. (a) From the centre distance of three sides are equal



$\therefore F_1X + F_2X - F_3X = 0$

$F_3 = F_1 + F_2$

$|\vec{F}_3| = |\vec{F}_1| + |\vec{F}_2|$

78. (d) A couple produces purely rotational motion.

79. (c) Force $\vec{F} = -3\hat{i} + \hat{j} + 5\hat{k}$ and distance of the point $\vec{r} = 7\hat{i} + 3\hat{j} + \hat{k}$

Torque $\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 3 & 1 \\ -3 & 1 & 5 \end{vmatrix} = 14\hat{i} - 38\hat{j} + 16\hat{k}$

80. (d) Torque

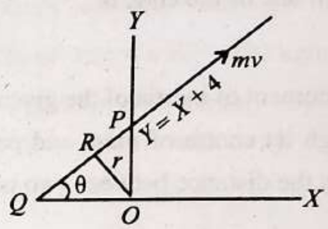
$\vec{\tau} = \vec{r} \times \vec{F}$

$\left(3\hat{i} + 2\hat{j} + 3\hat{k}\right) \times \left(2\hat{i} - 3\hat{j} + 4\hat{k}\right)$

$= 17\hat{i} - 6\hat{j} - 13\hat{k}.$

81. (a) Angular momentum, $\vec{L} = \vec{r} \times \vec{p}$

Y = X + 4 line has been shown in the figure.



When X = 0, Y = 4, so OP = 4.

The slope of the line can be obtained by comparing with the equation of line

y = mx + c

$m = \tan\theta = 1 \qquad \Rightarrow \theta = 45°$

$\angle OQP = \angle OPQ = 45°$

If we draw a line perpendicular to this line.

Length of the $\perp$ar

= OR = OP sin 45°

$$= \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

Angular momentum of particle along this line

$= r \times mv = 2\sqrt{2} \times 5 \times 3\sqrt{2} = 60$ units

82. (b) At the point of contact the frictional force acts tangentially and the weight of the sphere pass through if perpendicularly and there is no other force acting on the sphere. The moment of all forces acting on the body about point of contact is zero hence there is no torque about points of contact which means angular momentum about point of contact must be conserved.

83. (a) Work needed = change in kinetic energy

Here, Final KE = 0

Initial KE = $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{3}{4}mv^2 \quad \left(\because I = \frac{1}{2}mr^2 \text{ and } \omega = \frac{v}{r}\right)$

$$= \frac{3}{4} \times 100 \times \left(20 \times 10^{-2}\right)^2 = 3\text{J}$$

work needed = 3 J

84. (b) $\dfrac{K_t}{K_t + K_r} = \dfrac{\frac{1}{2}mv^2}{\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2}$

$$\Rightarrow \frac{\frac{1}{2}mv^2}{\frac{1}{2}mv^2 + \frac{1}{2}\times\frac{2}{5}mR^2\left(\frac{v^2}{R^2}\right)} = \frac{5}{7}$$

85. (a) External torque = zero

By conservation of angular momentum

$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$

$I_1 = I_2 = I$

$\omega = \frac{\omega_1 + \omega_2}{2}$

$KE_1 = \frac{1}{2}I\omega_1^2 + \frac{1}{2}I\omega_2^2$

$KE_2 = \frac{1}{2}(2I)\omega^2$

$KE_2 = I\left[\frac{\omega_1 + \omega_2}{2}\right]^2$

$\Delta KE = \frac{1}{4}I[\omega_1 - \omega_2]^2$

86. (b) Time taken by the body to reach the bottom when it rolls down on an inclined plane, $t = \sqrt{\frac{2l\left(1+\frac{K^2}{R^2}\right)}{8\sin\theta}}$

$$\frac{t_d}{t_s} = \sqrt{\frac{1+\frac{R^2}{2R^2}}{1+\frac{2R^2}{5R^2}}} = \sqrt{\frac{3}{2}\times\frac{5}{7}} = \sqrt{\frac{15}{14}}$$

$\Rightarrow t_d > t_s$

[t_d is time taken by disk and t_s is time taken by sphere]

87. (a) For rolling motion without slipping on inclined plane

$a_1 = \frac{g\sin\theta}{1+\frac{K^2}{R^2}}$ [θ is the angle of inclined plane with the ground]

And for slipping motion on inclined plane

$a_2 = g\sin\theta$

Required ratio $= \frac{a_1}{a_2} = \frac{1}{1+\frac{K^2}{R^2}} = \frac{1}{1+\frac{2}{5}} = \frac{5}{7}$

88. (a) From conservation of mechanical energy

$$= \frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right) = mgh \Rightarrow \frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right) = mg\left(\frac{3v^2}{4g}\right)$$

$\Rightarrow \frac{K^2}{R^2} = \frac{1}{2}$

∴ The object is a Disc

89. (b) We know that for the rolling body, the initial kinetic energy is equal to potential energy of the spring. K.E. and potential energy of spring $= \frac{1}{2}k(\Delta x)^2$

Kinetic energy of rolling body = Translational kinetic energy + rotational kinetic energy

Kinetic energy of rolling body $= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{3}{4}mv^2$

Now, $\frac{1}{2}k(\Delta x)^2 = \frac{3}{4}mv^2 \Rightarrow \Delta x = \sqrt{\frac{3\times3\times4^2}{2\times200}} = 0.6m$

90. (d) Time taken to reach the bottom of the inclined plane

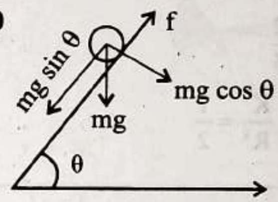
$$t = \sqrt{\frac{2l\left(1+\frac{k^2}{R^2}\right)}{g\sin\theta}}.$$

For, solid cylinder, $K^2 \ \frac{R^2}{2}$

For Holler cylinder, $K^2 = R^2$

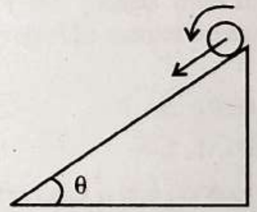
Hence, solid cylinder will reach first.

91. (a)



Frictional force required for pure rolling, i.e. rolling without slipping converts translational energy into rotational energy

92. (c) In case of pure rotation sphere have both rotational and translational K.E.



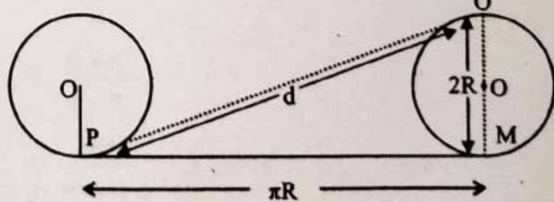
Conservation of energy

$$mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv^2 + \frac{1}{2}\frac{MR^2}{2}\frac{v^2}{R^2}$$

$Mgh = \frac{3}{4}Mv^2$

$v = \sqrt{\frac{4}{3}gh}$

93. (b)



Displacement $= \sqrt{(\pi R)^2 + (2R)^2} = \sqrt{\pi^2 + 4}$ ($\because R = 1$ m)

94. (d) The linear acceleration of COM will be $a = \frac{F}{m}$ wherever the force is applied. Hence, the acceleration will be same whatever the value of h may be.

95. (a) Velocity at top most point, $v_T = v_{cm} + \omega r$

Velocity at bottom, $v_B = v_{cm} - \omega r = 0 \Rightarrow v_{cm} = \omega r$

$v_T = v_{cm} + v_{cm} = 2v_{cm}$

96. (a) For solid cylinder $\frac{K^2}{R^2} = \frac{1}{2}$,

and for hollow cylinder $\frac{K^2}{R^2} = 1$.

Acceleration down the inclined plane $\propto \frac{1}{K^2/R^2}$.

Solid cylinder has greater acceleration, so it reaches the bottom first.

97. (a) For solid sphere, $\frac{K^2}{R^2} = \frac{2}{5}$

For disc and solid cylinder, $\frac{K^2}{R^2} = \frac{1}{2}$

As $\frac{K^2}{R^2}$ is smallest for solid sphere, it takes minimum time to reach the bottom of the incline, disc and cylinder reach together later.

98. (a) P.E. = total K.E.

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$gh = \frac{v^2}{2} + \frac{v^2}{5} = \frac{7v^2}{10}$$

$$v = \sqrt{\frac{10}{7}gh}$$

99. (d) Total kinetic energy $= E_{trans} + E_{rot} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$$= \frac{1}{2}mv^2 + \frac{1}{2} \times \left(\frac{2}{5}mr^2\right)\omega^2$$

$$= \frac{1}{2}mv^2 + \frac{1}{5}mv^2 = \frac{7}{10}mv^2$$

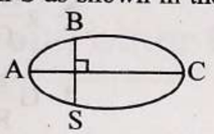
$$\therefore \frac{E_{trans}}{E_{total}} = \frac{\frac{1}{2}mv^2}{\frac{7}{10}mv^2} = \frac{5}{7}$$

CHAPTER 7

Gravitation

Kelper's Laws and Dynamics of Planetary Motion

1. The kinetic energies of a planet in an elliptical orbit about the Sun at positions A, B and C are K_A, K_B and K_C respectively. AC is the major axis and SB is perpendicular to AC at the position of the Sun S as shown in the figure. Then *(2018)*



a. $K_B < K_A < K_C$ b. $K_A > K_B > K_C$
c. $K_A < K_B < K_C$ d. $K_B > K_A > K_C$

2. Kepler's third law states that square of period of revolution (T) of a planet around the sun, is proportional to third power of average distance r between sun and planet, i.e., $T^2 = Kr^3$ here K is constant. If the masses of sun and planet are M and m respectively then as per Newton's law of gravitation force of attraction between them is $F = \frac{GMm}{r^2}$ here G is gravitational constant. The relation between G and K is described as: *(2015)*

a. $GMK = 4\pi^2$ b. $K = G$
c. $K = \frac{1}{G}$ d. $GM = 4\pi^2$

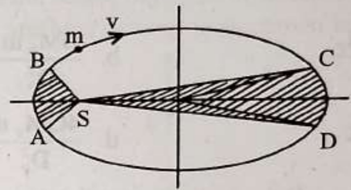
3. A geostationary satellite is orbiting the earth at a height of 5R above that surface of the earth, R being the radius of the earth. The time period of another satellite in hours at a height of 2R from the surface of the earth is: *(2012 Pre)*

a. 5   b. 10
c. $6\sqrt{2}$ d. $\sqrt{2}$

4. A planet moving along an elliptical orbit is closest to the sun at a distance r_1 and farthest away at a distance of r_2. If v_1 and v_2 are the linear velocities at these points respectively, then the ratio is *(2011 Mains)*

a. $\left(\frac{r_1}{r_2}\right)^2$ b. $\frac{r_2}{r_1}$
c. $\left(\frac{r_2}{r_1}\right)^2$ d. $\frac{r_1}{r_2}$

5. The figure shows elliptical orbit of a planet m about the sun S. The shaded area SCD is twice the shaded area SAB. If t_1 is the time for the planet to move from C to D and t_2 is the time to move from A to B then: *(2009)*



a. $t_1 = 4t_2$ b. $t_1 = 2t_2$
c. $t_1 = t_2$ d. $t_1 > t_2$

6. The period of revolution of planet A around the sun is 8 times that of B. The distance of A from the sun is how many times greater than that of B from the sun? *(1997)*

a. 4   b. 5
c. 2   d. 3

7. The distance of two planets from the sun are 10^{13} m and 10^{12} m respectively. The ratio of time periods of the planets is: *(1994, 88)*

a. $\sqrt{10}$ b. $10\sqrt{10}$
c. 10   d. $1/\sqrt{10}$

8. A satellite A of mass m is at a distance of r from the surface of the earth. Another satellite B of mass 2m is at a distance of 2r from the earth's centre. Their time periods are in the ratio of: *(1993)*

a. 1:2   b. 1:16
c. 1:32   d. $1:2\sqrt{2}$

9. The largest and the shortest distance of the earth from the sun are r_1 and r_2. Its distance from the sun when it is at perpendicular to the major axis of the orbit drawn from the sun is: *(1988)*

a. $\frac{r_1 + r_2}{4}$

b. $\frac{r_1 + r_2}{r_1 - r_2}$

c. $\frac{2r_1r_2}{r_1 + r_2}$

d. $\frac{r_1 + r_2}{3}$

Newton's Law of Gravitation & Acceleration Due to Gravity

10. If the mass of the Sun were ten times smaller and the universal gravitational constant were ten times larger in magnitude, which of the following is not correct? *(2018)*

a. Time period of a simple pendulum on the Earth would decrease

b. Walking on the ground would become more difficult

c. Raindrops will fall faster

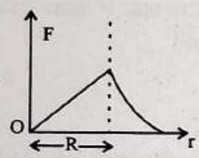
d. 'g' on the Earth will not change

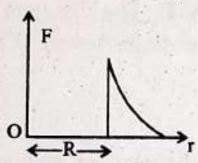
11. A spherical planet has a mass M_P and diameter D_P. A particle of mass m falling freely near the surface of this planet will experience an acceleration due to gravity, equal to: *(2012 Pre)*

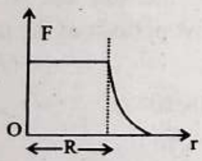
a. $\frac{4GM_P}{D_P^2}$ b. $\frac{GM_P m}{D_P^2}$

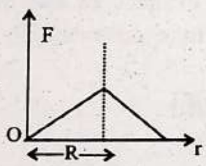
c. $\frac{GM_P}{D_P^2}$ d. $\frac{4GM_P m}{D_P^2}$

12. Which one of the following plots represents the variation of gravitational field on a particle with distance r due to a thin spherical shell of radius R? (r is measured from the center of the spherical shell) *(2012 Mains)*

a.


b.


c.


d.


13. Imagine a new planet having the same density as that of earth but it is 3 times bigger than the earth in size. If the acceleration due to gravity on the surface of earth is g and that on the surface of the new planet is g′, then: *(2005)*

a. $g' = 3g$ b. $g' = \frac{g}{9}$

c. $g' = 9g$ d. $g' = \frac{g}{3}$

14. The density of a newly discovered planet is twice that of earth. The acceleration due to gravity at the surface of the planet is equal to that at the surface of the earth. If the radius of the earth is R, the radius of the planet would be: *(2004)*

a. 4R b. $\frac{1}{4}R$

c. $\frac{1}{2}R$ d. 2R

15. Two spheres of masses m and M are situated in air and the gravitational force between them is F. The space around the masses is now filled with a liquid of specific density c. The gravitational force will now be: *(2003)*

a. 3F b. F

c. $\frac{F}{3}$ d. $\frac{F}{9}$

16. The acceleration due to gravity on the planet A is 9 times the acceleration due to gravity on planet B. A man jumps to a height of 2 m on the surface of A. What is the height of jump by the same person on the planet B: *(2003)*

a. $\frac{2}{9}$m b. 18 m

c. 6 m d. $\frac{2}{3}$m

17. For moon, its mass is $\frac{1}{81}$ of earth mass and its diameter is $\frac{1}{3.7}$ of earth diameter. If acceleration due to gravity at earth surface is 9.8 m/s^2 then at moon its value is: *(1999)*

a. 2.86 m/s^2 b. 1.65 m/s^2

c. 8.65 m/s^2 d. 5.16 m/s^2

18. What will be the formula of mass of the earth in terms of g, R and G? *(1996)*

a. $G\frac{R}{g}$ b. $g\frac{R^2}{G}$

c. $g^2\frac{R}{G}$ d. $G\frac{g}{R}$

19. The acceleration due to gravity g and mean density of the earth ρ are related by which of the following relations? (where G is the gravitational constant and R is the radius of the earth.): *(1995)*

a. $\rho = \frac{3g}{4\pi GR}$ b. $\rho = \frac{3g}{4\pi GR^3}$

c. $\rho = \frac{4\pi gR^2}{3G}\infty$ d. $\rho = \frac{4\pi gR^3}{3G}$

20. Two particles of equal mass go around a circle of radius R under the action of their mutual gravitational attraction. The speed v of each particle is: *(1995)*

a. $\frac{1}{2}\sqrt{\frac{Gm}{R}}$ b. $\sqrt{\frac{4Gm}{R}}$

c. $\frac{1}{2R}\sqrt{\frac{1}{Gm}}$ d. $\sqrt{\frac{Gm}{R}}$

21. The earth (mass = 6×10^{24} kg) revolves around the sun with an angular velocity of 2×10^{-7} rad/s in a circular orbit of radius 1.5×10^8 km. The force exerted by the sun on the earth, in newton, is: *(1995)*

a. 36×10^{21} b. 27×10^{39}

c. Zero d. 18×10^{25}

22. The radius of earth is about 6400 km and that of planet mars is 3200 km. The mass of the earth is about 10 times mass of planet mars. An object weighs 200 N on the surface of earth. Its weight on the surface of planet mars will be: *(1994)*

a. 20 N b. 8 N

c. 80 N d. 40 N

23. If the gravitational force between two objects were proportional to $\frac{1}{R}$ (and not as $\frac{1}{R^2}$), where R is the distance between them, then a particle in a circular path (under such a force) would have its orbital speed v, proportional to: *(1994, 89)*

a. R
b. R^0 (independent of R)
c. $\frac{1}{R^2}$
d. $\frac{1}{R}$

24. The mean radius of earth is R, its angular speed on its own axis is ω and the acceleration due to gravity at earth's surface is g. What will be the radius of the orbit of a geostationary satellite? *(1992)*

a. $\left(\frac{R^2g}{\omega^2}\right)^{\frac{1}{3}}$
b. $\left(\frac{Rg}{\omega^2}\right)^{\frac{1}{3}}$
c. $\left(\frac{R^2\omega^2}{g}\right)^{\frac{1}{3}}$
d. $\left(\frac{R^2g}{\omega}\right)^{\frac{1}{3}}$

Variation in g Due to Altitude, Depth and other Factors

25. A body weighs 72 N on the surface of the earth. What is the gravitation force on it, at a height equal to half the radius of the earth? *(2020)*

a. 32 N
b. 30 N
c. 24 N
d. 48 N

26. What is the depth at which the value of acceleration due to gravity becomes $\frac{1}{n}$ times the value that at the surface of earth? (radius of earth = R) *(2020-Covid)*

a. $\frac{R(n-1)}{n}$
b. $\frac{Rn}{(n-1)}$
c. $\frac{R}{n}$
d. $\frac{R}{n^2}$

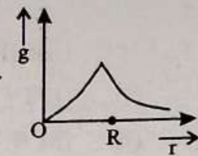
27. A body weighs 200 N on the surface of the earth. How much will it weigh half way down to the centre of the earth? *(2019)*

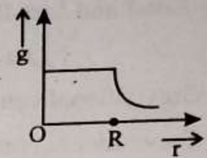
a. 150 N
b. 200 N
c. 250 N
d. 100 N

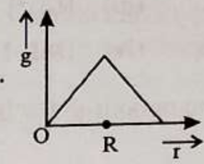
28. The acceleration due to gravity at a height 1 km above the earth is the same as at a depth d below the surface of earth. Then: *(2017-Delhi)*

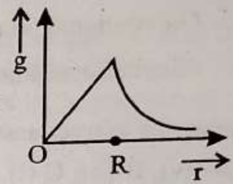
a. d = 1 km
b. $d = \frac{3}{2}$ km
c. d = 2 km
d. $d = \frac{1}{2}$ km

29. Starting from the center of the earth having radius R, the variation of g (acceleration due to gravity) is shown by: *(2016 - II)*

a.


b.


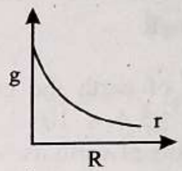
c.


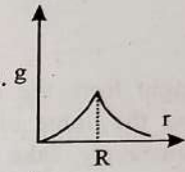
d.


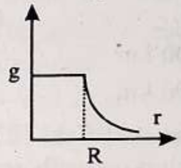
30. The height at which the weight of a body becomes $1/16^{th}$, its weight on the surface of earth (radius R), is: *(2012 Pre)*

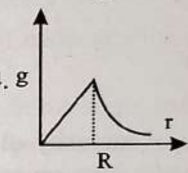
a. 5R
b. 15R
c. 3R
d. 4R

31. The dependence of acceleration due to gravity g on the distance r from the center of the earth, assumed to be a sphere of radius R of uniform density is as shown in figures below

1.


2.


3.


4.


The correct figure is: *(2010 Mains)*

a. 4
b. 1
c. 2
d. 3

32. A body of weight 72 N moves from the surface of earth to a height half of the radius of the earth, then gravitational force exerted on it will be: *(2000)*

a. 36 N
b. 32 N
c. 144 N
d. 50 N

33. If radius of earth shrinks by 1% then for acceleration due to gravity: *(1999)*

a. No change at poles
b. No change at equator
c. Max. change at equator
d. Equal change at all locations

Gravitational Intensity, Potential and Potential Energy

34. A body of mass 60 g experiences a gravitational force of 3.0 N, when placed at a particular point. The magnitude of the gravitational field intensity at that point is : *(2022)*

a. 180 N/kg
b. 0.05 N/kg
c. 50 N/kg
d. 20 N/kg

35. Match List-I and List-II (2022)

	List-I		List-II
A.	Gravitational constant (G)	(i)	$[L^2T^{-2}]$
B.	Gravitational potential Energy	(ii)	$[M^{-1}L^3T^{-2}]$
C.	Gravitational potential	(iii)	$[LT^{-2}]$
D.	Gravitational Intensity	(iv)	$[ML^2T^{-2}]$

Choose the correct answer from the options given below:

a. A-(iv), B-(ii), C-(i), D-(iii)
b. A-(ii), B-(i), C-(iv), D-(iii)
c. A-(ii), B-(iv), C-(i), D-(iii)
d. A-(ii), B-(iv), C-(iii), D-(i)

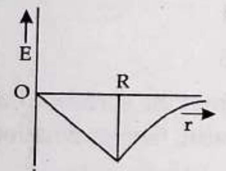
36. The work done to raise a mass m from the surface of the earth to a height h, which is equal to the radius of the earth, is: (2019)

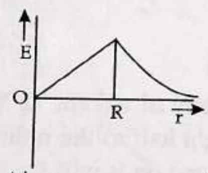
a. mgR b. 2mgR
c. $\frac{1}{2}$mgR d. $\frac{3}{2}$mgR

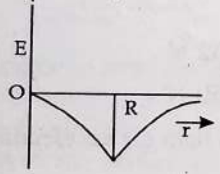
37. At what height from the surface of earth the gravitation potential and the value of g are -5.4×10^7 J kg^{-1} and 6.0 ms^{-2} respectively. Take the radius of earth as 6400 km: (2016 - I)

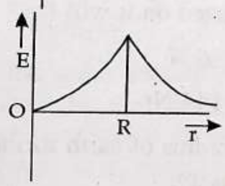
a. 2600 km b. 1600 km
c. 1400 km d. 2000 km

38. Dependence of intensity of gravitational field (E) of earth with distance (r) from center of earth is correctly represented by: (2014)

a. 

b. 

c. 

d. 

39. Infinite number of bodies, each of mass 2 kg are situated on x-axis at distances 1 m, 2 m, 4 m, 8m, ... respectively, from the origin. The resulting gravitational potential due to this system at the origin will be: (2013)

a. –4G b. –G
c. $-\frac{8}{3}G$ d. $-\frac{4}{3}G$

40. A body of mass 'm' taken from the earth's surface to the height equal to twice the radius (R) of the earth. The change in potential energy of body will be: (2013)

a. $\frac{1}{3}$mgR b. 2 mgR
c. $\frac{2}{3}$mgR d. 3 mgR

41. A particle of mass M is situated at the center of a spheri shell of same mass and radius a. The magnitude of the gravitational potential at a point situated at $\frac{a}{2}$ distance fro the center, will be: (2011 Mains)

a. $\frac{2GM}{a}$ b. $\frac{3GM}{a}$
c. $\frac{4GM}{a}$ d. $\frac{GM}{a}$

42. A particle of mass M is situated at the center of a spheric shell of same mass and radius a. The gravitational potenti at a point situated at $\frac{a}{2}$ distance from the center, will be (2010 Pre)

a. $-\frac{4GM}{a}$ b. $-\frac{3GM}{a}$
c. $-\frac{2GM}{a}$ d. $-\frac{GM}{a}$

43. For a satellite moving in an orbit around the earth, the ratio of kinetic energy to potential energy is (2005)

a. $\frac{1}{2}$ b. $\frac{1}{\sqrt{2}}$
c. 2 d. $\sqrt{2}$

44. A body of mass m is placed on earth surface which is taken from earth surface to a height of h = 3R then change in gravitational potential energy is: (2002)

a. $\frac{mgR}{4}$ b. $\frac{2}{3}$mgR
c. $\frac{3}{4}$mgR d. $\frac{mgR}{2}$

45. A planet is moving in an elliptical orbit around the sun. If T, V, E and L stand respectively for its kinetic energy, gravitational potential energy, total energy and magnitude of angular momentum about the centre of force, which of the following is correct? (1990)

a. T is conserved
b. V is always positive
c. E is always negative
d. L is conserved but direction of vector L changes continuously

Satellite, Orbital Velocity and Escape Velocity

46. The escape velocity from the Earth's surface is v. The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is: (2021)

a. 2v b. 3v
c. 4v d. v

47. A particle of mass 'm' is projected with a velocity v = kV, (k < 1) from the surface of the earth. The maximum height above the surface reached by the particle is: (2021)

a. $R\left(\frac{k}{1+k}\right)^2$
b. $\frac{R^2k}{1+k}$
c. $\frac{Rk^2}{1-k^2}$
d. $R\left(\frac{k}{1-k}\right)^2$

48. The ratio of escape velocity at earth (v_e) to the escape velocity at a planet (v_p) whose radius and mean density are twice as that of earth is: *(2016 - I)*

a. 1 : 2
b. $1:2\sqrt{2}$
c. 1 : 4
d. 1: 2

49. A satellite of mass m is orbiting the earth (of radius R) at a height h from its surface. The total energy of the satellite in terms of g_0, the value of acceleration due to gravity at the earth's surface, is: *(2016 - II)*

a. $\frac{2mg_0R^2}{R+h}$
b. $-\frac{2mg_0R^2}{R+h}$
c. $\frac{mg_0R^2}{2(R+h)}$
d. $-\frac{mg_0R^2}{2(R+h)}$

50. A remote-sensing satellite of earth revolves in a circular orbit at a height of 0.25×10^6 m above the surface of earth. If earth's radius is 6.38×10^6 m and g = 9.8 m/s², then the orbital speed of the satellite is: *(2015 Re)*

a. 6.67 km/s
b. 7.76 km/s
c. 8.56 km/s
d. 9.13 km/s

51. A satellite S is moving in an elliptical orbit around the earth. The mass of the satellite is very small compared to the mass of the earth. Then: *(2015 Re)*

a. The acceleration of S is always directed towards the center of the earth.
b. The angular momentum of S about the center of the earth changes in direction, but its magnitude remains constant.
c. The total mechanical energy of S varies periodically with time.
d. The linear momentum of S remains constant in magnitude.

52. A black hole is an object whose gravitational field is so strong that even light cannot escape from it. To what approximate radius would earth (mass = 5.98×10^{24} kg) have to be compressed to be a black hole? *(2014)*

a. 10^{-2} m
b. 10^{-6} m
c. 10 m
d. 100 m

53. If v_e is escape velocity and v_o is orbital velocity of a satellite for orbit close to the earth's surface, then these are related by: *(2012 Mains)*

a. $v_0 = \sqrt{2}v_e$
b. $v_0 = v_e$
c. $v_e = \sqrt{2v_0}$
d. $v_e = \sqrt{2}v_0$

54. A particle of mass m is thrown upwards from the surface of the earth, with a velocity u. The mass and the radius of the earth are, respectively, M and R. G is gravitational constant and g is acceleration due to gravity on the surface of the earth. The minimum value of u so that the particle does not return back to earth, is: *(2011 Mains)*

a. $\sqrt{\frac{2GM}{R}}$
b. $\sqrt{\frac{2gM}{R^2}}$
c. $\sqrt{2gR^2}$
d. $\sqrt{\frac{2GM}{R^2}}$

55. The additional kinetic energy to be provided to a satellite of mass m revolving around a planet of mass M, to transfer it from a circular orbit of radius R_1 to another of radius R_2 ($R_2 > R_1$) is: *(2010 Mains)*

a. $GmM\left(\frac{1}{R_1^2}-\frac{1}{R_2^2}\right)$
b. $GmM\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$
c. $2GmM\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$
d. $\frac{1}{2}GmM\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

56. The radii of circular orbits of two satellites A and B of the earth, are 4R and R, respectively. If the speed of satellite A is 3 V, then the speed of satellite B will be: *(2010 Pre)*

a. $\frac{3V}{2}$
b. $\frac{3V}{4}$
c. 6V
d. 12V

57. Two satellites of earth, S_1 and S_2 are moving in the same orbit. The mass of S_1 is four times the mass of S_2. Which one of the following statements is true? *(2007)*

a. The potential energies of earth satellites in the two cases are equal.
b. S_1 and S_2 are moving with the same speed.
c. The kinetic energies of the two satellites are equal.
d. The time period of S_1 is four times that of S_2.

58. The earth is assumed to be a sphere of radius R. A platform is arranged at a height R from the surface of the earth. The escape velocity of a body from this platform is *f*v, where v is its escape velocity from the surface of the Earth. The value of *f* is *(2006)*

a. $\frac{1}{2}$
b. $\sqrt{2}$
c. $\frac{1}{\sqrt{2}}$
d. $\frac{1}{3}$

59. With what velocity should a particle be projected so that its height becomes equal to radius of earth: *(2001)*

a. $\left(\frac{GM}{R}\right)^{1/2}$
b. $\left(\frac{8GM}{R}\right)^{1/2}$
c. $\left(\frac{2GM}{R}\right)^{1/2}$
d. $\left(\frac{4GM}{R}\right)^{1/2}$

60. For a planet having mass equal to mass of the earth but radius is one fourth of radius of the earth, Then escape velocity for this planet will be: *(2000)*

a. 11.2 km/s
b. 22.4 km/s
c. 5.6 km/s
d. 44.8 km/s

61. Rohini satellite is at a height of 500 km and Insat-B is at a height of 3600 km from surface of earth then relation between their orbital velocity (V_R, V_I) is: *(1999)*

a. $V_R > V_I$
b. $V_R < V_I$
c. $V_R = V_I$
d. No relation

62. The escape velocity of a body on the surface of the earth is 11.2 km/s. If the earth's mass increases to twice its present value and radius of the earth becomes half, the escape velocity becomes: *(1997)*

a. 22.4 km/s
b. 44.8 km/s
c. 5.6 km/s
d. 11.2 km/s

63. A ball is dropped from a spacecraft revolving around the earth at a height of 120 km. What will happen to the ball? *(1996)*

a. It will fell down to the earth gradually
b. It will go very far in the space
c. It will continue to move with the same speed along the original orbit of spacecraft
d. It will move with the same speed, tangentially to the spacecraft.

64. The escape velocity from earth is 11.2 km/s. If a body is to be projected in a direction making an angle 45° to the vertical, then the escape velocity is: *(1993)*

a. 11.2 × 2 km/s
b. 11.2 km/s
c. $\frac{11.2}{\sqrt{2}}$ km/s
d. $11.2\sqrt{2}$ km/s

65. The satellite of mass m is orbiting around the earth in a circular orbit with a velocity v. What will be its total energy? *(1991)*

a. $\left(\frac{3}{4}\right)mv^2$
b. $\left(\frac{1}{2}\right)mv^2$
c. mv^2
d. $-\left(\frac{1}{2}\right)mv^2$

66. For a satellite escape velocity is 11 km/s. If the satellite is launched at an angle of 60° with the vertical, then escape velocity will be: *(1989)*

a. 11 km/s
b. $11\sqrt{3}$ km/s
c. $\frac{11}{\sqrt{3}}$ km/s
d. 33 km/s

Weightlessness

67. Two astronauts are floating in gravitational free space after having lost contact with their spaceship. The two will *(2017-Delhi)*

a. Move towards each other
b. Move away from each other
c. Will become stationary
d. Keep floating at the same distance between them

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	a	c	b	b	a	b	d	c	d	a	b	a	c	b	b	b
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
b	a	a	a	c	b	a	a	a	d	c	d	c	a	b	d	c
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
c	c	a	a	a	c	b	b	a	c	c	c	c	b	d	b	a
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	
a	d	a	d	c	b	c	a	b	a	a	c	b	d	a	a	

Explanations

1. (b) When a planet revolves around the sun its angular momentum is conserved

$\Rightarrow mvr = \text{constant}$

$\therefore \quad v \propto \frac{1}{r}$ [$\because$ mass is constant]

As $r_A < r_B < r_C$

$\Rightarrow v_A > v_B > v_C$

hence

$K_A > K_B > K_C$

2. (a) $T_p^2 = Kr^3$ given ...(1)

As the planet revolves in nearly circular orbit

Time Period $= \frac{\text{Circumference of the orbit}}{\text{Orbital speed}}$

$T_{planet} = \frac{2\pi r}{V_p}$ $\left[\because \text{orbital speed } V_p = \sqrt{\frac{GM}{r}}\right]$

$= \frac{2\pi r . r^{1/2}}{\sqrt{GM}}$

$$T_p = \frac{2\pi r^{3/2}}{\sqrt{GM}} \qquad ...(2)$$

Squaring Eq. (2), we have

$$T_P^2 = \frac{4\pi^2 r^3}{GM}$$

$$Kr^3 = \frac{4\pi^2 r^3}{GM}$$

$$\Rightarrow K = \frac{4\pi^2}{GM}$$

3. (c) Time period $T \propto r^{3/2}$

Height of the satellites from surface of earth are 5R and 2R

$\Rightarrow$ Distance of the satellites from center of the earth are 6R and 3R

$\because$ Time period of geostationary satellite = 24 hrs

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2} \Rightarrow T_2 = (24\text{hrs})\left(\frac{3R}{6R}\right)^{\frac{3}{2}} = 6\sqrt{2}\text{ hrs}$$

4. (b) According to the law of conservation of angular momentum

$L_1 = L_2$

$mv_1r_1 = mv_2r_2 \Rightarrow v_1r_1 = v_2r_2$

$$\frac{v_1}{v_2} = \frac{r_2}{r_1}$$

5. (b) As we know that areal velocity of planet is constant

$A(SCD) = 2A(SAB)$

$$\Rightarrow \frac{\text{Area covered in time } t_1\,(SCD)}{t_1} = A(SCD)$$

$$\Rightarrow \frac{\text{Area covered in time } t_2\,(SAB)}{t_2} = A(SAB)$$

$$\Rightarrow \frac{2SAB}{t_1} = \frac{SAB}{t_2} \Rightarrow t_1 = 2t_2$$

6. (a) Period of revolution of planet A $(T_A) = 8T_B$. According to Kepler's III law of planetary motion, $T^2 \propto r^3$

$$\text{Therefore } \left(\frac{r_A}{r_B}\right)^3 = \left(\frac{T_A}{T_B}\right)^2 = \left(\frac{8T_B}{T_B}\right)^2 = 64$$

$\Rightarrow r_A = 4r_B$.

7. (b) Distance of two planets from sun, $r_1 = 10^{13}$ m and $r_2 = 10^{12}$ m.

$\because T^2 \propto r^3$

Therefore

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2} = \left(\frac{10^{13}}{10^{12}}\right)^{3/2} = 10^{3/2} = 10\sqrt{10}.$$

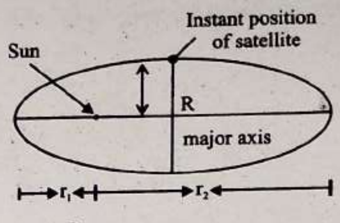
8. (d) Time period does not depend on the mass.

As $T^2 \propto r^3$

$$\Rightarrow \frac{T_A}{T_B} = \frac{r^{3/2}}{2^{3/2}r^{3/2}} = 1 : 2\sqrt{2}$$

9. (c) From the properties of ellipse, we have

$$\frac{2}{R} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{r_1 + r_2}{r_1 r_2}$$



$$R = \frac{2r_1r_2}{r_1 + r_2}$$

10. (d) $\because g = \frac{GM}{R^2}$

If universal constant becomes 10 times, then

$G' = 10G$

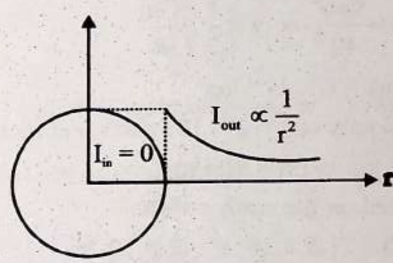
New acceleration due to gravity is $g' = \frac{G'M}{R^2}$

$$g' = \frac{10\,GM}{R^2} \Rightarrow g' = 10\,g$$

So g on earth surface will change

11. (a) $g = \frac{GM}{R^2} = \frac{GM_P}{\left(\frac{D_P}{2}\right)^2} = \frac{4GM_P}{D_P^2}$ $\left[\because \text{Radius} = \frac{\text{diameter}}{2} \quad \therefore R = \frac{D_P}{2}\right]$

12. (b) At any point inside the shell gravitational field remains zero.



13. (a) $g = \frac{GM}{R^2} = \frac{G(V \times \rho)}{R^2}$ $\left[\therefore \rho = \frac{M}{V}\right]$

$$= \frac{G\left(\frac{4}{3}\pi R^3 \times \rho\right)}{R^2} = \frac{4}{3}G\pi R\rho$$

$\because R' = 3R$

So, $g' = \frac{4}{3}G\pi(3R)\rho = 3g$

14. (c) $g = \frac{GM}{R^2} = \frac{G\frac{4}{3}\pi R^3\rho}{R^2} = \frac{4}{3}\pi G\rho R$

Now according to question $g_{planet} = g_{earth}$

$$\Rightarrow \frac{4}{3}\pi G\rho_{planet} \times R_{planet} = \frac{4}{3}\pi G\rho_{earth} \times R_{earth}$$

$$\Rightarrow R_{planet} = \frac{R_{earth}}{2} \; (\because \rho_{planet} = 2\rho_{earth})$$

15. (b) Gravitational force between two masses does not depend on the medium between the masses, thus it remains the same.

16. (b) $\because \quad mg_A h_A = mg_B h_B$

$\Rightarrow h_B = \dfrac{g_A}{g_B} \times h_A$

$= 9 \times 2$

$= 18$ m

17. (b) $g = \dfrac{GM}{R^2}$

$\Rightarrow g \propto \dfrac{M}{R^2}$

$$g_M = \frac{M_M}{M_E} \times \left(\frac{R_E}{R_m}\right)^2 \times g_E = \frac{1}{81} \times (3.7)^2 \times 9.8 = 1.65 \text{ m/s}^2$$

18. (b) The gravitational force $(F) = \dfrac{GMm}{R^2}$ and F = mg.

Equating both the values of gravitational force, $\dfrac{GMm}{R^2} = mg$

$\Rightarrow M = g\dfrac{R^2}{G}$, where M is the mass of the earth.

19. (a) Acceleration due to gravity $(g) = G \times \dfrac{M}{R^2}$

$= G\dfrac{(4/3)\pi R^3 \times \rho}{R^2} = G \times \dfrac{4}{3}\pi R \times \rho$

$\Rightarrow \rho = \dfrac{3g}{4\pi GR}$

20. (a) $m \times \dfrac{v^2}{R} = \dfrac{Gm^2}{4R^2} \Rightarrow v = \dfrac{1}{2}\sqrt{\dfrac{Gm}{R}}$

21. (a) Mass (m) = 6×10^{24} kg;

Angular velocity $(\omega) = 2 \times 10^{-7}$ rad/s and radius (r)

$= 1.5 \times 10^8$ km $= 1.5 \times 10^{11}$ m.

Force exerted on the earth $= mR\omega^2$

$= (6 \times 10^{24}) \times (1.5 \times 10^{11}) \times (2 \times 10^{-7})^2$

$= 36 \times 10^{21}$ N.

22. (c) Radius of earth (R_e) = 6400 km; Radius of mars (R_m) = 3200 km; Mass of earth $(M_e) = 10\ M_m$ and weight of the object on earth (W_e) = 200 N.

$$\frac{W_m}{W_e} = \frac{mg_m}{mg_e} = \frac{M_m}{M_e} \times \left(\frac{R_e}{R_m}\right)^2 = \frac{1}{10} \times (2)^2$$

$\Rightarrow W_m = W_e \times \dfrac{2}{5} = 200 \times 0.4 = 80$ N

23. (b) Centripetal force $(F) = \dfrac{mv^2}{R}$ and the gravitational force

$(F) = \dfrac{GMm}{R}$ [Given in the question $F \propto \dfrac{1}{R}$]

$\Rightarrow \dfrac{mv^2}{R} = \dfrac{GMm}{R}$

$\Rightarrow v = \sqrt{GM}$

Thus velocity v is independent of R.

24. (a) For a geostationary satellite,

$\dfrac{GMm}{r^2} = m\omega^2 r \Rightarrow r^3 = \dfrac{GM}{\omega^2} = \dfrac{gR^2}{\omega^2}$

$\therefore \quad r = \left(\dfrac{gR^2}{\omega^2}\right)^{\frac{1}{3}}$

25. (a) Initial weight on the surface of earth

$W_1 = mg = 72$N

weight of the body at a height $h = \dfrac{R}{2}$

$$W_2 = mg' = \frac{mg}{\left(1+\frac{h}{R}\right)^2} = \frac{72N}{\left(1+\frac{R/2}{R}\right)^2} = \frac{72}{9/4} \quad \left[\because g' = \frac{g}{\left(1+\frac{h}{R}\right)^2}\right]$$

$W_2 = 32$N

26. (a) Inside the earth at depth d from the surface

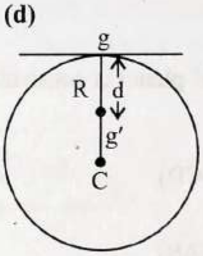
$g_{eff} = g\left(1 - \dfrac{d}{R}\right)$

As the new acceleration due to gravity becomes $\dfrac{1}{n}$ times,

$\therefore g_{eff} = \dfrac{g}{n}$

$\Rightarrow \dfrac{g}{n} = g\left(1 - \dfrac{d}{R}\right) \Rightarrow d = \dfrac{(n-1)R}{n}$

27. (d)



Acceleration due to gravity at a depth d from surface of earth

$g' = g\left(1 - \dfrac{d}{R}\right)$ ...(I)

Here g = acceleration due to gravity at earth's surface

Multiplying by mass 'm' on both sides of a.

$mg' = mg\left(1 - \dfrac{d}{R}\right)$; $\left(\because d = \dfrac{R}{2}\right)$

$= 200\left(1 - \dfrac{R}{2R}\right) = \dfrac{200}{2} = 100$ N

28. (c) Acceleration due to gravity at a height 1 km above earth's surface

$a_h = g\left[1 - \dfrac{2h}{R}\right] = g\left[1 - \dfrac{2 \times 1}{R}\right] = g\left[1 - \dfrac{2}{R}\right]$

Acceleration due to gravity at depth 'd' below earth surface

$a_d = g\left[1 - \dfrac{d}{R}\right]$

$\because \quad a_h = a_d$

$\Rightarrow g\left[1 - \dfrac{2}{R}\right] = g\left[1 - \dfrac{d}{R}\right]$

d = 2 km

29. (d) For $0 < r \le R_e$

$$g = \left(\frac{GM_e}{R_e^3}\right)r \Rightarrow g \propto r$$

For $r > R_e$

$$g = \frac{GM_e}{r^2} \Rightarrow g \propto \frac{1}{r^2}$$

30. (c) $mg_h = \dfrac{mg}{\left(1+\dfrac{h}{R}\right)^2} \Rightarrow \dfrac{mg}{16} = \dfrac{mg}{\left(1+\dfrac{h}{R}\right)^2} \Rightarrow h = 3R$

31. (a) Inside the earth $g \propto r$ and outside the earth

$$g \propto \frac{1}{r^2}$$

32. (b) $F = mg = 72$ N

$$g' = g\left(\frac{R_e}{R_e+h}\right)^2 = g\left(\frac{R_e}{R_e+R_e/2}\right)^2$$

$$= g\left[\frac{2R_e}{3R_e}\right]^2 = \frac{4}{9}g$$

$$F' = mg' = mg \times \frac{4}{9} = 72 \times \frac{4}{9} = 32\text{ N}$$

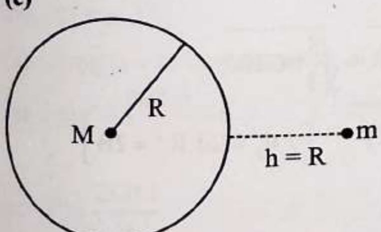
33. (d) If the radius of earth shrinks then acceleration due to gravity get affected equally at all locations.

34. (c) Gravitational field intensity is

$$I_g = \frac{F}{m} = \frac{3}{60\times10^{-3}} = 50\text{ N/kg} \quad [\because m = 60g = 60 \times 10^{-3}\text{kg}]$$

35. (c) Gravitational constant = $[M^{-1}L^3T^{-2}]$
Gravitational potential energy = $[ML^2T^{-2}]$
Gravitational potential = $[L^2T^{-2}]$
Gravitational intensity = $[LT^{-2}]$

36. (c)



Initial gravitational potential energy

$$U_i = \frac{-GMm}{R}$$

Final gravitational potential energy at height h = R

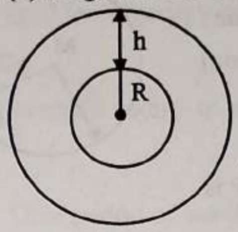
$$U_f = \frac{-GMm}{(R+R)}$$

As work done = Change in PE

$$\therefore W = U_f - U_i$$

$$= \frac{GMm}{2R} = \frac{gR^2m}{2R} = \frac{mgR}{2} \quad (\because GM = gR^2)$$

37. (a) Magnitude of Gravitational Potential at a height h



$$= \frac{GM}{R+h} = 5.4\times10^7\text{ JKg}^{-1} \qquad ...(1)$$

$$g = \frac{GM}{(R+h)^2} = 6\text{ m/s}^2 \qquad ...(2)$$

Dividing eqn. (1) by (2)

$$(R+h) = \frac{5.4\times10^7}{6} = 0.9\times10^7\text{ m} = 9000\text{ km}$$

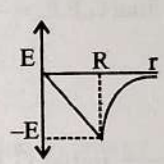
$h = 9000$ km $- 6400$ km $= 2600$ km

38. (a) Gravitational field intensity

For a point inside the earth $E = \dfrac{-GMr}{R^3}$

For a point outside the earth $E = \dfrac{-GM}{r^2}$

Where –ve sign indicates that the gravitational field is attractive



Accurate graph to show variation of E with r

39. (a) Total Gravitation Potential = $V = \dfrac{-GM}{r_1} + \left(\dfrac{-GM}{r_2}\right) + ...$

$$\Rightarrow V = -G(2)\left[\frac{1}{1}+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+...\right]$$

Because it forms geometric progression and for a geometric progression

$$S_{GP} = \frac{r^n-1}{r-1}, \text{ where } r = \frac{1}{2}$$

$$S_{GP} = \frac{\left(\frac{1}{2}\right)^n - 1}{-1/2} = \frac{-1}{-1/2} = 2$$

$\therefore V = -2G \times S_{GP}$
$= -2G \times 2 = -4G$

40. (c) Change in gravitational P.E. = $U_f - U_i$

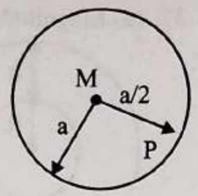
$$= -\frac{GMm}{3R} - \left(-\frac{GMm}{R}\right)$$

$$= \frac{2}{3}\frac{GMm}{R} = \frac{2}{3}mgR$$

41. (b) Total potential = magnitude of potential due to shell + magnitude of potential due to the mass

$$= \frac{GM}{a} + \frac{GM}{a/2} = \frac{3GM}{a}$$

42. (b) In a spherical shell, potential throughout the shell remains constant



Potential at surface [General equation]

$$V_{surface} = -\frac{GM}{r}$$

Total gravitation potential at point P is

$$V_p = V_{spherical\ shell} + V_{[particle]}$$

$$V_p = -\frac{GM}{a} - \frac{GM}{a/2}$$

$$\Rightarrow V_p = \frac{-3GM}{a}$$

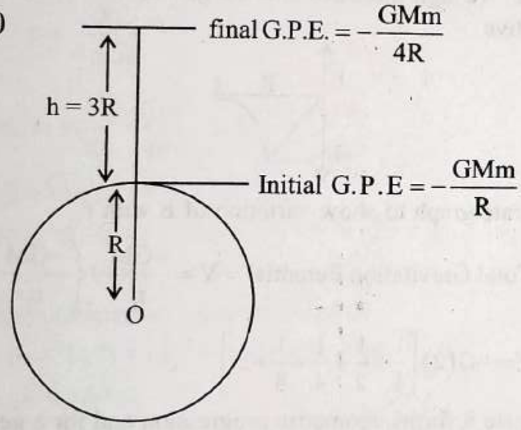
43. (a) $\because$ $K.E. = \frac{GMm}{2R}$

$$P.E. = -\frac{GMm}{R}$$

$$\therefore K.E. = \frac{|P.E.|}{2}$$

$$\therefore \frac{K.E.}{|P.E.|} = \frac{1}{2}$$

44. (c)



Change in G.P.E. = final energy – initial energy

$$= \frac{3}{4}\frac{GMm}{R} = \frac{3}{4}\frac{GM}{R^2}mR$$

$$= \frac{3}{4}gmR$$

45. (c) Total energy E of the planet is always negative. In attractive field, potential energy is negative. Kinetic energy changes as velocity increase when distance is less. Moreover, if the motion is in a plane, the direction of L does not change.

46. (c) Escape velocity from the earth's surface is,

$$V = \sqrt{\frac{2GM}{R}} \Rightarrow V = \sqrt{\frac{2G}{R} \times \frac{4}{3}\pi R^3 \rho}$$

$$\begin{bmatrix} \because \text{Mass} = \text{Volume} \times \text{density} \\ \therefore M = \frac{4}{3}\pi R^3 \times \rho \end{bmatrix}$$

$$\Rightarrow V \propto R$$

$$\Rightarrow \frac{V}{V_e} = \frac{R}{4R} \Rightarrow V_e = 4V$$

47. (c) Given,

$v = Kv_e$

where,

$$v_e = \text{escape velocity} = \sqrt{\frac{2GM}{R}}$$

Let maximum height reached above the surface of earth be 'h'.

Total energy remains conserved at each point.

$\therefore$ Total energy at the surface of earth = Total energy at height h

$$\Rightarrow \frac{1}{2}mv^2 - \frac{GmM}{R} = 0 - \frac{GmM}{(R+h)}$$

[$\because$ at max^{-m} height, velocity becomes zero.]

$$\Rightarrow \frac{1}{2}mv^2 - \frac{GmM}{R} = -\frac{GmM}{(R+h)}$$

$$\Rightarrow \frac{1}{2}m(KV_e)^2 = -\frac{GmM}{(R+h)} + \frac{GmM}{R}$$

$$\Rightarrow \frac{1}{2}K^2\left(\sqrt{\frac{2GM}{R}}\right)^2 = GM\left[\frac{R+h-R}{R(R+h)}\right]$$

$$\Rightarrow \frac{1}{2}K^2\left(\frac{2GM}{R}\right) = GM\left[\frac{h}{R(R+h)}\right]$$

$$\Rightarrow K^2 = \frac{h}{(R+h)} \Rightarrow RK^2 + hK^2 = h$$

$$\Rightarrow h = \frac{RK^2}{1-K^2}$$

48. (b) $V_e \quad \sqrt{2gR}$

$$\because g = \frac{GM}{R^2} = \frac{G}{R^2} \times \frac{4}{3}\pi R^3 d, \text{ Where d is density}$$

$$\Rightarrow g = \frac{4}{3}\pi GdR$$

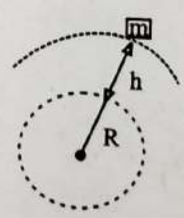
$$\Rightarrow V_e = \sqrt{2 \times \frac{4}{3}\pi GdR.R} = \sqrt{\frac{8}{3}\pi GdR^2}$$

$$V_p = \sqrt{\frac{8}{3}\pi G\, 2d\,(2R)^2} \quad [\because d_p = 2d, R_p = 2R]$$

$$\frac{V_e}{V_p} = \frac{1}{\sqrt{8}} = \frac{1}{2\sqrt{2}}$$

49. (d) PE of the satellite at a height h $= \frac{-GMm}{R+h}$

$$KE = \frac{1}{2}m\left[\sqrt{\frac{GM}{R+h}}\right]^2 \quad \left[\because \text{Orbital velocity} = \sqrt{\frac{GM}{R+h}}\right]$$



$$KE = \frac{GMm}{2(R+h)}$$

$$TE = \frac{-GMm}{R+h} + \frac{GMm}{2[R+h]}$$

$$= \frac{-GMm}{2(R+h)}\frac{R^2}{R^2} \qquad \left[\because g = \frac{GM}{R^2}\right]$$

$$= \frac{-g_0 mR^2}{2[R+h]}$$

50. **(b)** For the satellite revolving around earth

$$v_0 = \sqrt{\frac{GM_e}{(R_e+h)}} = \sqrt{\frac{GM_e}{R_e\left(1+\frac{h}{R_e}\right)}} = \sqrt{\frac{gR_e}{1+\frac{h}{R_e}}}$$

Substituting the values

$v_0 = \sqrt{60\times10^6}$ m/s

$v_0 = 7.76 \times 10^3$ m/s = 7.76 km/s

51. **(a)** Acceleration due to earth to the satellite is centripetal, hence directed towards centre.

Angular momentum conservation holds good for comparable masses but

$M_{earth} >> M_{satellite}$

52. **(a)** Escape velocity $(v_e) = \sqrt{\frac{2GM}{R}} = c =$ speed of light

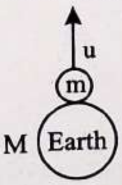
$$\Rightarrow R = \frac{2GM}{c^2} = \frac{2\times6.6\times10^{-11}\times5.98\times10^{24}}{(3\times10^8)^2}\,m = 10^{-2}\,m$$

53. **(d)** Escape velocity $v_e = \sqrt{\frac{2GM}{R}}$

Orbital velocity (near earth's surface) $v_0 = \sqrt{\frac{GM}{R}}$

So, $v_e = \sqrt{2}\,v_0$

54. **(a)** Using law of conservation of mechanical energy



$$\Rightarrow \frac{1}{2}mu^2 - \frac{GMm}{R} = 0$$

$$\Rightarrow u^2 = \frac{2GM}{R}$$

$$u = \sqrt{\frac{2GM}{R}}$$

55. **(d)** Additional kinetic energy $= TE_2 - TE_1$

$$= -\frac{GMm}{2R_2} - \left(-\frac{GMm}{2R_1}\right) = \frac{1}{2}GmM\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

56. **(c)** Speed of satellite $V = \sqrt{\frac{GM}{r}}$

$$\Rightarrow \frac{V_B}{V_A} = \sqrt{\frac{r_A}{r_B}} = \sqrt{\frac{4R}{R}} = 2 \Rightarrow V_B = (3V)(2) = 6V$$

57. **(b)** Kinetic energy of satellite $= \frac{GMm}{2(R+h)}$

Potential energy of satellite $= \frac{-GMm}{(R+h)}$

Time period of satellite $= \frac{2\pi(R+h)^{3/2}}{\sqrt{GM}}$

Velocity of satellite $= \sqrt{\frac{GM}{(R+h)}}$

Since, orbital speed is independent of the mass of satellite, V_1 and V_2 are equal

58. **(c)** For a platform at a height h,

$$v' = \sqrt{\frac{2GM}{R+h}} = \sqrt{\frac{2GM}{2R}} \qquad (\because h = R)$$

But at the surface of earth, $v = \sqrt{\frac{2GM}{R}}$

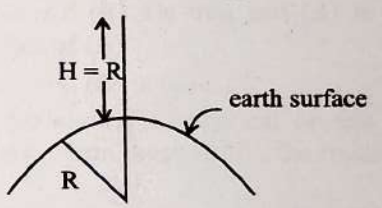
According to the question

$v' = fv$

$$\sqrt{\frac{2GM}{2R}} = f\sqrt{\frac{2GM}{R}}$$

Solving, $f = \frac{1}{\sqrt{2}}$

59. **(a)** Apply conservation of energy



$TE_{surface} = TE_{height}$

$KE + PE_1 = KE_2 + PE_2$

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0 - \frac{GMm}{R+h}$$

$$\frac{1}{2}mv^2 = \frac{GMm}{R} - \frac{GMm}{2R} \qquad [\because h = R]$$

$$\frac{mv^2}{2} = \frac{GMm}{2R}$$

$$v = \sqrt{\frac{GM}{R}}$$

60. **(b)** V_{es} for earth is 11.2 km/sec.

$$V_{es} = \sqrt{\frac{2GM_e}{R_e}} = 11.2 \text{ km/sec}$$

$$v'_{es} = \sqrt{\frac{2GM_e\times4}{R_e}} = 2\sqrt{\frac{2GM_e}{R_e}}$$

$= 2 \times 11.2 = 22.4$ km/sec.

61. (a) $V_0 = \sqrt{\frac{GM}{r}}$; M = mass of earth

$V_0 \propto \frac{1}{\sqrt{r}}$ hence, $V_R > V_1$

62. (a) Escape velocity of a body (v_e) = 11.2 km/s; New mass of the earth $M'_e = 2\ M_e$ and new radius of the earth $R'_e = 0.5\ R_e$.

Escape velocity $(v_e) = \sqrt{\frac{2GM_e}{R_e}} \propto \sqrt{\frac{M_e}{R_e}}$.

Therefore $\frac{v_e}{v'_e} = \sqrt{\frac{M_e}{R_e} \times \frac{0.5R_e}{2M_e}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$

Or, $v'_e = 2v_e$ = 22.4 km/sec.

63. (c) The ball will continue to move with the same angular velocity along the original orbit of the spacecraft, as no external torque is applied.

64. (b) Escape velocity will remain same as it does not depend on the angle of projection.

65. (d) Total energy = – K.E = $-\frac{1}{2}mv^2$

$K.E = \frac{|P.E.|}{2} \Rightarrow K.E = \frac{1}{2}mv^2$

66. (a) Since escape velocity $\left(v_e = \sqrt{2gR_e}\right)$ is independent of angle of projection, so it will not change.

67. (a) Since two astronauts are floating in gravitational free space, the only force acting on the two astronauts is the gravitational pull of their masses, $F = \frac{Gm_1m_2}{r^2}$ which is attractive in nature.

Hence they move towards each other.

CHAPTER 8 Mechanical Properties of Solids

Stress, Strain and Young's Modulus

1. A wire of length L area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is: *(2020)*

   a. $\frac{Mg(L_1 - L)}{AL}$ b. $\frac{MgL}{AL_1}$

   c. $\frac{MgL}{A(L_1 - L)}$ d. $\frac{MgL_1}{AL}$

2. Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area 3 A. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount? *(2018)*

   a. 4 F b. 6 F

   c. 9 F d. F

3. The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of: *(2015 Re)*

   a. 1 : 1 b. 1 : 2

   c. 2 : 1 d. 4 : 1

4. Copper of fixed volume V is drawn into wire of length ℓ. When this wire is subjected to a constant force F, the extension produced in the wire is D_l. Which of the following graphs is a straight line? *(2014)*

   a. $\Delta\ell$ versus $1/\ell$ b. $\Delta\ell$ versus ℓ^2

   c. $\Delta\ell$ versus $1/\ell^2$ d. $\Delta\ell$ versus ℓ^3

5. The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied? *(2013)*

   a. Length = 300 cm, diameter = 3 mm

   b. Length = 50 cm, diameter = 0.5 mm

   c. Length = 100 cm, diameter = 1 mm

   d. Length = 200 cm, diameter = 2 mm

Bulk Modulus and Shear Modulus

6. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R)

   Assertion (A): The stretching of a spring is determined by the shear modulus of the material of the spring

   Reason (R): A coil spring of copper has more tensile strength than a steel spring of same dimensions.

   In the light of the above statements, choose the most appropriate answer from the options given below : *(2022)*

   a. (A) is false but (R) is true

   b. Both (A) and (R) are true and (R) is the correct explanation of (A)

   c. Both (A) and (R) are true and (R) is not the correct explanation of (A)

   d. (A) is true but (R) is false

7. The bulk modulus of a spherical objects is 'B'. If it is subjected to uniform pressure 'P', the fractional decrease in radius is: *(2017-Delhi)*

   a. $\frac{B}{3p}$ b. $\frac{3P}{B}$

   c. $\frac{P}{3B}$ d. $\frac{P}{B}$

8. The approximate depth of an ocean is 2700 m. The compressibility of water is $45.4 \times 10^{-11}\ Pa^{-1}$ and density of water is $10^3\ kg/m^3$. What fractional compression of water will be obtained at the bottom of the ocean? *(2015)*

   a. 1.0×10^{-2} b. 1.2×10^{-2}

   c. 1.4×10^{-2} d. 0.8×10^{-2}

Elastic Potential Energy

9. When a block of mass M is suspended by a long wire of length L, the length of the wire becomes $(L + l)$. The elastic potential energy stored in the extended wire is : *(2019)*

   a. Mgl b. MgL

   c. $\frac{1}{2}Mgl$ d. $\frac{1}{2}MgL$

Answer Key

1	2	3	4	5	6	7	8	9
c	c	c	b	b	d	c	b	c

Explanations

1. (c) When the new mass is hanged to the wire, the force exerted on the wire is: $F = Mg$

The initial length of the wire is L and the new length is L_1.

Young's Modulus is given as:

$$Y = \frac{FL}{A\Delta L}$$ (Where ΔL is the change in length of the wire)

Now, put all the values in the above formula, we get,

$$Y = \frac{MgL}{A(L_1 - L)}$$

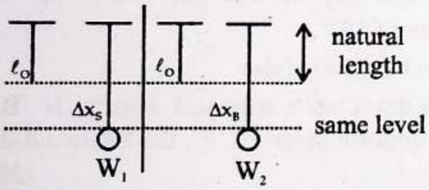
2. (c) $Y = \frac{F\ell}{A\Delta\ell}$

$\because V = A\ell$ so $\ell = \frac{V}{A}$

$$F = \frac{YA\Delta\ell}{\ell} = \frac{YA^2\Delta\ell}{V}$$

$$\frac{F_1}{F_2} = \left(\frac{A_1}{A_2}\right)^2 \Rightarrow \frac{F}{F_2} = \left(\frac{A}{3A}\right)^2 = \frac{1}{9} \Rightarrow F_2 = 9F$$

3. (c)



$\Delta x_S = \Delta x_B$

$$\frac{F_S L_O}{AY_S} = \frac{F_B L_O}{AY_B}$$

$$\frac{F_S}{F_B} = \frac{Y_S}{Y_B} = \frac{2Y_B}{Y_B} \quad (\because Y_S = 2Y_B)$$

$$\frac{F_S}{F_B} = \frac{2}{1}$$

4. (b) We know that, Young's modulus (Y)

$$Y = \frac{FL}{A\Delta L} \quad \text{...(i)}$$

Volume, $V = AL$ ...(ii)

Eq. (i) can be written as,

$$\Rightarrow Y = \frac{FL^2}{AL\Delta L} = \frac{FL^2}{V\Delta L}$$

$$\Rightarrow \Delta L = \frac{FL^2}{VY} \quad \text{...(iii)}$$

[Here, F, V and Y are constant, therefore $\frac{F}{VY}$ is constant]

$\Rightarrow \Delta L \propto L^2$

On comparing Eq. (iii) with the equation of a straight line $Y = mx + c$, $y = \Delta L$, $c = O$

Slope, $m = \frac{F}{VY}$ = constant & $x = L^2$

$\Rightarrow \Delta L$ versus L^2 graph is of the form $y = mx$.

Hence, ΔL versus L^2 graph is a straight line passing through origin.

5. (b) $Y = \frac{F/A}{\Delta\ell/\ell} \Rightarrow \Delta\ell = \frac{F\ell}{YA} = \frac{F\ell}{Y\pi r^2} \Rightarrow \Delta\ell \propto \frac{\ell}{r^2}$

Therefore maximum for $\ell = 50$ cm & diameter $= 0.5$ mm

6. (d) When we stretch a spring, the length of the wire does not change but the coil experiences an angular twist. Hence shear modulus is used to determine the stretching of a spring.

Therefore, the assertion is true.

Also, for a given dimension, $Y_{steel} > Y_{copper}$

Hence, the reason is false.

7. (c) $B = \frac{P}{-\frac{\Delta v}{v}}$ ...(i) (where B = bulk modulus)

Volume of spherical body $V = \frac{4}{3}\pi R^3$

Fractional change in volume

$$\frac{\Delta V}{V} = \frac{3\Delta R}{R}$$

Substituting the value of $\frac{\Delta V}{V}$ in Equation (i)

$$B = \frac{P}{-\frac{3\Delta R}{R}} \Rightarrow \frac{\Delta R}{R} = (-)\frac{P}{3B}$$

where (– ve) sign is decrease in radius.

8. (b) As we know

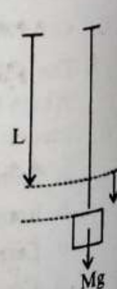
$$B = \frac{P}{\frac{\Delta V}{V}} \Rightarrow \frac{\Delta V}{V} = \frac{P}{B}$$

Now $P = \rho gh$ and compressibility 'K' $= \frac{1}{B}$

so $\frac{\Delta V}{V} = \rho gh(K)$

$$= 10^3 \times 9.8 \times 2700 \times 45.4 \times 10^{-11} = 1.201 \times 10^{-2}$$

9. (c)



$$U = \frac{1}{2}Mgl$$

$$U = \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{vol}^m$$

$$= \frac{1}{2} \times \frac{F}{A} \times \frac{\Delta l}{l} \times Al = \frac{1}{2} \times Mg \times \Delta l$$

CHAPTER 9

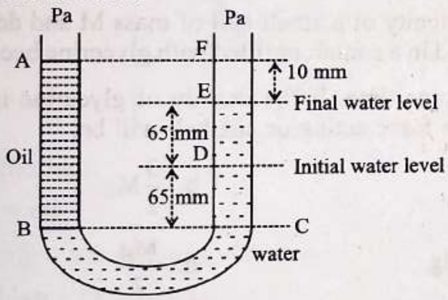
Mechanical Properties of Fluids

Variation of Pressure with Depth and Pascal's Law

1. A barometer is constructed using a liquid (density = 760 kg/m^3). What would be the height of the liquid column, when a mercury barometer reads 76 cm? (density of mercury = 13600 kg/m^3) *(2020-Covid)*

a. 13.6 m
b. 136 m
c. 0.76 m
d. 1.36 m

2. A U tube with both ends open to the atmosphere, is partially filled with water. Oil, which is immiscible with water, is poured into one side until it stands at a distance of 10 mm above the water level on the other side. Meanwhile the water rises by 65 mm from its original level (see diagram). The density of the oil is: *(2017-Delhi)*



a. 425 kg m^{-3}
b. 800 kg m^{-3}
c. 928 kg m^{-3}
d. 650 kg m^{-3}

3. Two non-mixing liquids of densities ρ and $n\rho$ $(n > 1)$ are put in a container. The height of each liquid is h. A solid cylinder of length L and density d is put in this container. The cylinder floats with its axis vertical and length pL $(p < 1)$ in the denser liquid. The density d is equal to: *(2016 - I)*

a. $\{1 + (n + 1)p\}\rho$
b. $\{2 + (n + 1)p\}\rho$
c. $\{2 + (n - 1)p\}\rho$
d. $\{1 + (n - 1)p\}\rho$

Bernoulli's Theorem and its Application

4. A small hole of area of cross-section 2 mm^2 is present near the bottom of a fully filled open tank of height 2 m. Taking g = 10 m/s^2, the rate of flow of water through the open hole would be nearly *(2019)*

a. 12.6×10^{-6} m^3/s
b. 8.9×10^{-6} m^3/s
c. 2.23×10^{-6} m^3/s
d. 6.4×10^{-6} m^3/s

5. A wind with speed 40 m/s blows parallel to the roof of a house. The area of the roof is 250 m^2. Assuming that the pressure inside the house is atmospheric pressure, the force exerted by the wind on the roof and the direction of the force will be (P_{air} = 1.2 kg/m^3): *(2015)*

a. 4.8×10^5 N, upwards
b. 2.4×10^5 N, upwards
c. 2.4×10^5 N, downwards
d. 4.8×10^5 N, downwards

Surface Tension and Surface Energy

6. If a soap bubble expands, the pressure inside the bubble : *(2022)*

a. Is equal to the atmospheric pressure
b. Decreases
c. Increases
d. Remains the same

7. A soap bubble, having radius of 1 mm, is blown from a detergent solution having a surface tension of 2.5×10^{-2} N/m. The pressure inside the bubble equals at a point Z_0 below the free surface of water in a container. Taking g = 10 m/s^2, density of water = 10^3 kg/m^3, the value of Z_0 is : *(2019)*

a. 100 cm
b. 10 cm
c. 1 cm
d. 0.5 cm

8. A rectangular film of liquid is extended from (4 cm × 2 cm) to (5 cm × 4 cm). If the work done is 3×10^{-4} J, the value of the surface tension of the liquid is: *(2016 - II)*

a. 0.2 Nm^{-1}
b. 8.0 Nm^{-1}
c. 0.250 Nm^{-1}
d. 0.125 Nm^{-1}

9. A certain number of spherical drops of a liquid of radius r coalesce to form a single drop of radius R and volume V. If 'T' is the surface tension of the liquid, then: *(2014)*

a. Energy $= 4VT\left(\frac{1}{r}-\frac{1}{R}\right)$ is released

b. Energy $= 3VT\left(\frac{1}{r}+\frac{1}{R}\right)$ is absorbed

c. Energy $= 3VT\left(\frac{1}{r}-\frac{1}{R}\right)$ is released

d. Energy is neither released nor absorbed

Angle of Contacts and Ascent/Descent Formula

10. A liquid does not wet the solid surface if angle of contact is: *(2020-Covid)*

a. Equal to 60°
b. Greater than 90°
c. Zero
d. Equal to 45°

11. A capillary tube of radius r is immersed in water and water rises in it to a height h. The mass of the water in the capillary is 5g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is: *(2020)*

a. 5.0 g
b. 10.0 g
c. 20.0 g
d. 2.5 g

12. Three liquids of densities ρ_1, ρ_2 and ρ_3 (with $\rho_1 > \rho_2 > \rho_3$) having the same value of surface tension T, rise to the same height in three identical capillaries. The angles of contact obey: *(2016 - II)*

a. $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$

b. $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$

c. $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geq 0$

d. $0 \leq \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$

13. Water rises to height 'h' in capillary tube. If the length of capillary tube above the surface of water is made less than 'h', then: *(2015 Re)*

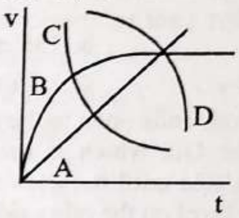
a. Water does not rise at all.
b. Water rises up to the tip of capillary tube and then starts overflowing like a fountain.
c. Water rises up to the top of capillary tube and stays there without overflowing.
d. Water rises up to a point a little below the top and stays there.

14. The wettability of a surface by a liquid depends primarily on: *(2013)*

a. Angle of contact between the surface and the liquid
b. Viscosity
c. Surface tension
d. Density

Viscosity, Stoke's Law and Terminal Velocity

15. A spherical ball is dropped in a long column of a highly viscous liquid. The curve in the graph shown which represents the speed of the ball (v) as a function of time (t) is: *(2022)*



a. D
b. A
c. B
d. C

16. The velocity of a small ball of mass M and density d, when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be: *(2021)*

a. Mg
b. $\frac{3}{2}$Mg
c. 2 Mg
d. $\frac{Mg}{2}$

17. A small sphere of radius 'r' falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to :- *(2018)*

a. r^5
b. r^2
c. r^3
d. r^4

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	c	d	a	b	b	c	d	c	b	b	d	c	a	c	d	a

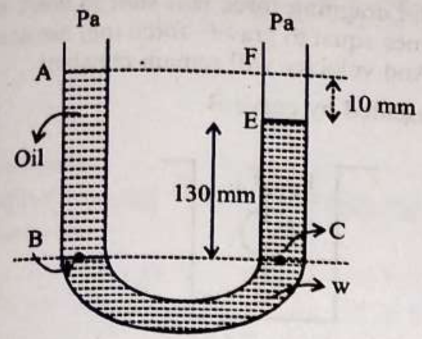
Explanations

1. (a) According to the question

$76\text{ cm} \times \rho_{Hg} \times g = h \times \rho_L \times g$

$$h = 76\text{cm} \times \frac{\rho_{Hg}}{\rho_L} = 76\text{ cm} \times \frac{13600}{760} = 13.6\text{ m}$$

2. (c)



$P_B = P_C$ (pressure at same level is same) [Pascal law]

$H\rho_{oil}\, g = h\rho_W g$

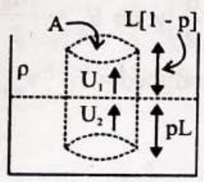
$(65 + 65 + 10) \times 10^{-3} \times \rho_0 = (65 + 65) \times 10^{-3} \times 10^3$

$140\rho_0 = 130 \times 10^3$

$$\rho_0 = \frac{130}{140} \times 10^3$$

$\rho_0 = 928\text{ kgm}^{-3}$ (approx)

3. (d)



For equilibrium.

$U_1 + U_2 = mg$

$AL[1 - p]g \times \rho + ApL\, n\rho g = d\, ALg$

$ALg\rho\,[1 - p + np] = d\, ALg$

$d = \rho[1 + (n - 1)p]$

4. (a) Rate of flow liquid

$Q = Au = A\sqrt{2gh}$

$= 2 \times 10^{-6}\text{m}^2 \times \sqrt{2 \times 10 \times 2}\text{m/s}$

$= 2 \times 2 \times 3.16 \times 10^{-6}\text{ m}^3/\text{s}$

$= 12.6 \times 10^{-6}\text{ m}^3/\text{s}$

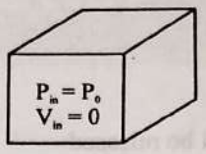
5 (b) Applying Bernoulli's theorem between two points just inside the house and on the roof,

$$P + \frac{\rho v^2}{2} = P_0$$

$$P_0 - P = \frac{\rho v^2}{2}$$

$F = (P_0 - P)A$

$$F = \frac{\rho v^2 A}{2} = \frac{1}{2} \times 1.2 \times (40)^2 \times 250 = 2.4 \times 10^5$$



Force is in upward direction pressure goes from higher to lower pressure area.

6. (b) Total pressure inside a soap bubble is given by

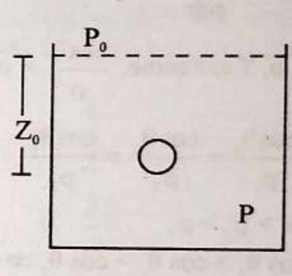
$$P = P_0 + \frac{4T}{R}$$

$\Rightarrow$ As the bubble expands R increases and P decreases

7. (c) Excess pressure $= \frac{4T}{R}$,

Gauge pressure $= \rho g Z_0$

Pressure at depth Z_0 = pressure inside soap bubble



$$P_0 + \frac{4T}{R} = P_0 + \rho g Z_0$$

$$Z_0 = \frac{4T}{R \times \rho g}$$

$$Z_0 = \frac{4 \times 2.5 \times 10^{-2}}{10^{-3} \times 1000 \times 10}\text{m}$$

$Z_0 = 1\text{ cm}$

8. (d) Initial area $(A_1) = 4\text{ cm} \times 2\text{ cm} = 8\text{ cm}^2$

Final area $(A_2) = 5\text{ cm} \times 4\text{ cm} = 20\text{ cm}^2$

Change in area $(\Delta A) = 20\text{ cm}^2 \times 8\text{ cm}^2 = 12\text{ cm}^2$

The required work done, $W = T \times 2 \times \Delta A$

Where, T is the surface tension,

$$\Rightarrow T = \frac{W}{2\Delta A} = \frac{3 \times 10^{-4}}{2 \times 12 \times 10^{-4}} = 0.125\text{ N/m}$$

9. (c) Volume of n smaller drops = volume of single bigger drop

$$\Rightarrow n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3 = V$$

$$\Rightarrow R = n^{\left(\frac{1}{3}\right)} r$$

We know that surface energy = surface tension × change in surface area

$\Rightarrow E = T[A_{final} - A_{intial}]$

$$\Rightarrow E = T[A_{final} - A_{initial}]$$

$$= T[4\pi R^2 - n4\pi r^2]$$

$$= 4\pi R^2 T\left[1 - n^{\left(\frac{1}{3}\right)}\right]$$

$$= 4\pi R^3 T\left[\frac{1}{R} - \frac{1}{r}\right] = 3VT\left[\frac{1}{R} - \frac{1}{r}\right]$$

As $R > r \Rightarrow \frac{1}{R} < \frac{1}{r}$

$\therefore$ E is negative

Hence, $3VT\left[\frac{1}{r} - \frac{1}{R}\right]$ energy will be released

10. **(b)** When angle of contact $\geq 90°$ then liquid doesn't wet solid.

11. **(b)** In a capillary tube force of surface tension balances the weight of water in capillary tube

$F_s = 2\pi r T\cos\theta = mg$

Here, T and θ are constant

So, $m \propto r$

Hence, $\frac{m_2}{5.0} = \frac{2r}{r} \Rightarrow m_2 = 10.0g$

12. **(d)** $h = \frac{2T\cos\theta}{\rho g r}$

As r, h, T are same, $\frac{\cos\theta}{\rho}$ = constant

$$\Rightarrow \frac{\cos\theta_1}{\rho_1} = \frac{\cos\theta_2}{\rho_2} = \frac{\cos\theta_3}{\rho_3}$$

As $\rho_1 > \rho_2 > \rho_3$

$\Rightarrow \cos\theta_1 > \cos\theta_2 > \cos\theta_3 \Rightarrow \theta_1 < \theta_2 < \theta_3$

As water rises so θ must be acute

So, $0 \leq \theta_1 < \theta_2 < \theta_3 < \pi/2$

13. **(c)** At normal condition water height h is given as $h = \frac{2s}{\rho g R}$

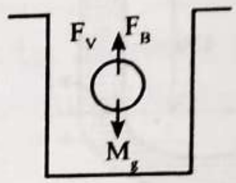
It is clear that, $h \propto \frac{1}{R}$ So, when height above the surface is decreased, there won't be any motion as the water rises till height h and stays there. Overflowing of water is not possible because there is no external force on molecules that causes it to overflow from the capillary tube.

14. **(a)** The wettability of a surface by a liquid depends primarily on angle of contact between the surface and the liquid.

15. **(c)** When a ball is dropped in a liquid that is highly viscous in nature then initially gravitational force acts in it. As it goes inside the liquid dragging force will start to work and when its value becomes equal to gravity force mg, net acceleration will be zero. And velocity will remain constant.

It is best represented by curve B.

16. **(d)**



$$F_B = V_s\rho_l g = V \times \frac{d}{2} \times g = \frac{Vdg}{2}$$

$F_g = M_g = Vdg$ $\quad$ [$\because$ M = Vd, V is the volume of ball]

When velocity of moving body becomes constant, then $F_{net} = 0$.

$\Rightarrow F_B + F_V = Mg$

$$\Rightarrow \frac{Vdg}{2} + F_V = Vdg \Rightarrow F_V = Vdg - \frac{Vdg}{2} = \frac{(2-1)Vdg}{2}$$

$$\Rightarrow F_V = \frac{Vdg}{2} = V\left(\frac{d}{2}\right)g \Rightarrow F_V = \frac{Mg}{2}$$

17. **(a)** Rate of heat produced $= F_v \cdot v_{terminal}$

$$\frac{dQ}{dT} = F_V.v_T$$

$$= 6\pi\eta r v_T.v_T$$

$$\therefore \frac{dQ}{dt} \propto r.v_T^2$$

$$\propto r.r^4 \quad [\because v_T \propto r^2]$$

$$\propto r^5$$

CHAPTER 10

Thermal Properties of Matter

Heat and Temperature Scales

1. On a new scale of temperature (which is linear) and called the W scale, the freezing and boiling points of water are 39° W and 239° W respectively. What will be the temperature on the new scale, corresponding to a temperature of 39°C on the Celsius scale? *(2008)*
   a. 139° W
   b. 78° W
   c. 117° W
   d. 200° W

2. Mercury thermometer can be used to measure temperature upto: *(1992)*
   a. 260°C
   b. 100°C
   c. 360°C
   d. 500°C

3. A Centigrade and a Fahrenheit thermometer are dipped in boiling water. The water temperature is lowered until the Fahrenheit thermometer registers 140°F. What is the fall in temperature as registered by the centigrade thermometer? *(1990)*
   a. 80°C
   b. 60°C
   c. 40°C
   d. 30°C

Thermal Expansion

4. A copper rod of 88 cm and an aluminium rod of unknown length have their increase in length independent of increase in temperature. The length of aluminium rod is : *(2019)*
   ($\alpha_{Cu} = 1.7 \times 10^{-5}\ K^{-1}$ and $\alpha_{Al} = 2.2 \times 10^{-5}\ K^{-1}$)
   a. 6.8 cm
   b. 113.9 cm
   c. 88 cm
   d. 68 cm

5. Coefficient of linear expansion of brass and steel rods are α_1 and α_2. Lengths of brass and steel rods are l_1 and l_2 respectively. If $(l_2 - l_1)$ is maintained same at all temperatures, which one of the following relations holds good? *(2016 - I)*
   a. $\alpha_1 l_2 = \alpha_2 l_1$
   b. $\alpha_1 l_2^2 = \alpha_2 l_1^2$
   c. $\alpha_1^2 l_2 = \alpha_2^2 l_1$
   d. $\alpha_1 l_1 = \alpha_2 l_2$

6. The value of coefficient of volume expansion of glycerin is 5×10^{-4}/K. The fractional change in the density of glycerin for a rise of 40°C in its temperature, is: *(2015 Re)*
   a. 0.010
   b. 0.015
   c. 0.020
   d. 0.025

7. Which of the following rods, (given radius r and length l) each made of the same material and whose ends are maintained at the same temperature will conduct most heat? *(2005)*
   a. $r = r_0,\ l = l_0$
   b. $r = 2r_0,\ l = l_0$
   c. $r = r_0,\ l = 2l_0$
   d. $r = 2r_0,\ l = 2l_0$

Specific Heat, Latent Heat and Calorimetry

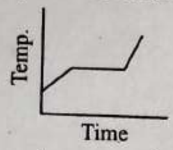
8. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5\ r_2$) through 1 K are in the ratio: *(2020)*
   a. $\frac{9}{4}$
   b. $\frac{3}{2}$
   c. $\frac{5}{3}$
   d. $\frac{27}{8}$

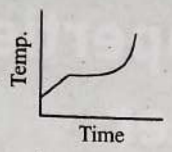
9. A piece of ice falls from a height h so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h is [Latent heat of ice is 3.4×10^5 J/kg and g = 10 N/kg]: *(2016 - I)*
   a. 34 km
   b. 544 km
   c. 136 km
   d. 68 km

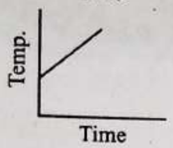
10. Two identical bodies are made of a material for which the heat capacity increases with temperature. One of these is at 100°C, while the other one is at 0°C. If the two bodies are brought into contact, then, assuming no heat loss, the final common temperature is: *(2016 - II)*
    a. Less than 50°C but greater than 0°C
    b. 0°C
    c. 50°C
    d. More than 50°C

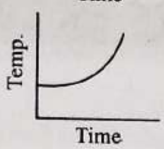
11. Steam at 100°C is passed into 20 g of water at 10°C. When water acquires a temperature of 80°C, the mass of water present will be: [Take specific heat of water = 1 cal /g /°C and latent heat of steam = 540 cal g^{-1}]: *(2014)*
    a. 24 g
    b. 31.5 g
    c. 42.5 g
    d. 22.5 g

12. Liquid oxygen at 50 K is heated to 300 K at constant pressure of 1 atm. The rate of heating is constant. Which one of the following graphs represents the variation of temperature with time? *(2012 Pre)*

a. 

b. 

c. 

d. 

13. When 1 kg of ice at 0°C melts to water at 0°C, the resulting change in its entropy, taking latent heat of ice to be 80 Cal/°C, is: *(2011 Pre)*

a. 273 cal/K
b. 8×10^4 cal/K
c. 80 cal/K
d. 293 cal/K

14. Thermal capacity of 40 g of aluminum (s = 0.2 cal/g K) is: *(1990)*

a. 168 J/K
b. 672 J/K
c. 840 J/K
d. 33.6 J/K

15. 10 gm of ice cubes at 0°C are released in a tumbler (water equivalent 55 g) at 40°C. Assuming that negligible heat is taken from the surroundings the temperature of water in the tumbler becomes nearly (L = 80 cal/g): *(1988)*

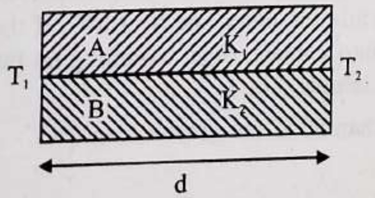
a. 31°C
b. 22°C
c. 19°C
d. 15°C

Heat Transfer and Thermal Conductivity

16. The unit of thermal conductivity is : *(2019)*

a. J m K^{-1}
b. J m^{-1} K^{-1}
c. W m K^{-1}
d. W m^{-1} K^{-1}

17. Two rods A and B of different material are welded together as shown in figure. Their thermal conductivities are K_1 and K_2. The thermal conductivity of the composite rod will be: *(2017-Delhi)*



a. $\frac{3(K_1+K_2)}{2}$
b. $K_1 + K_2$
c. $2(K_1 + K_2)$
d. $\frac{K_1+K_2}{2}$

18. The two ends of a metal rod are maintained at temperatures 100°C and 110°C. The rate of heat flow in the rod is found to be 4.0 J/s. If the ends are maintained at temperatures 200°C and 210°C, the rate of heat flow will be: *(2015)*

a. 16.8 J/s
b. 8.0 J/s
c. 4.0 J/s
d. 44.0 J/s

19. A slab of stone of area 0.36 m^2 and thickness 0.1 m is exposed on the lower surface to steam at 100°C. A block of ice at 0°C rests on the upper surface of the slab. In one hour 4.8 kg of ice is melted. The thermal conductivity of slab is (Given latent heat of fusion of ice = 3.36×10^5 J kg^{-1}) *(2012 Mains)*

a. 1.24 J/m/s/° C
b. 1.29 J/m/s/° C
c. 2.05 J/m/s/° C
d. 1.02 J/m/s/° C

20. A cylindrical metallic rod in thermal contact with two reservoirs of heat at its two ends conducts an amount of heat Q in time t. The metallic rod is melted and the material is formed into a rod of half the radius of the original rod. What is the amount of heat conducted by the new rod, when placed in thermal contact with the two reservoirs in time t? *(2010 Pre)*

a. Q/2
b. Q/4
c. Q/16
d. 2Q

21. The two ends of a rod of length L and a uniform cross-sectional area A are kept at two temperatures T_1 and T_2 ($T_1 > T_2$). The rate of heat transfer, $\frac{dQ}{dt}$ through the rod in a steady state is given by: *(2009)*

a. $\frac{dQ}{dt}=\frac{k(T_1-T_2)}{LA}$
b. $\frac{dQ}{dt}=kLA(T_1-T_2)$
c. $\frac{dQ}{dt}=\frac{kA(T_1-T_2)}{L}$
d. $\frac{dQ}{dt}=\frac{kL(T_1-T_2)}{A}$

22. Consider a compound slab consisting of two different materials having equal thicknesses and thermal conductivities K and 2K, respectively. The equivalent thermal conductivity of the slab is: *(2003)*

a. $-K$
b. $\sqrt{2}K$
c. 3K
d. $\frac{4}{3}K$

23. Consider two rods of same length and different specific heats (S_1, S_2), conductivities (K_1 K_2) and area of cross-sections (A_1, A_2) and both having temperature T_1 and T_2 at their ends. If rate of loss of heat due to conduction is equal, then *(2002)*

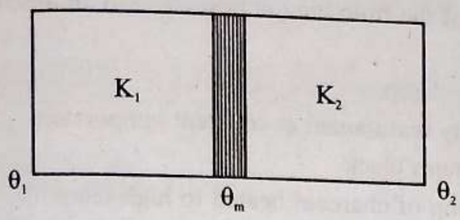
a. $K_1A_1 = K_2A_2$
b. $\frac{K_1A_1}{S_1}=\frac{K_2A_2}{S_2}$
c. $K_2A_1 = K_1A_2$
d. $\frac{K_2A_1}{S_2}=\frac{K_1A_2}{S_1}$

24. A cylindrical rod having temperature T_1 and T_2 at its end. The rate of flow of heat Q_1 cal/sec. If all the linear dimension are doubled keeping temperature constant, then rate of flow of heat Q_2 will be: *(2001)*

a. $4Q_1$
b. $2Q_1$
c. $\frac{Q_1}{4}$
d. $\frac{Q_1}{2}$

25. Gravitational force is required for: *(2000)*
a. Stirring of liquid
b. Convection
c. Conduction
d. Radiation

26. Two conducting slabs of heat conductivity K_1 and K_2 are joined as shown in figure. The temperature at ends of the slabs are θ_1 and θ_2 ($\theta_1 > \theta_2$) the, final temperature (θ_m) of junction is: *(1999)*



a. $\dfrac{K_1\theta_1 + K_2\theta_2}{K_1 + K_2}$
b. $\dfrac{K_1\theta_2 + K_2\theta_1}{K_1 + K_2}$
c. $\dfrac{K\theta_2 - K_2\theta_1}{K_1 + K_2}$
d. None

27. Heat is flowing through two cylindrical rods of the same material. The diameters of the rods are in the ratio 1 : 2 and the lengths in the ratio 2 : 1. If the temperature difference between the ends is same, then ratio of the rate of flow of heat through them will be: *(1995)*
a. 2 :1
b. 8 : 1
c. 1: 1
d. 1 : 8

Newton's Law of Cooling

28. A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C. The time taken by a similar cup of coffee to cool from 80° C to 60°C at a room temperature same at 20°C is: *(2021)*
a. $\dfrac{13}{5}t$
b. $\dfrac{10}{13}t$
c. $\dfrac{5}{13}t$
d. $\dfrac{13}{10}t$

29. A body cools from a temperature 3T to 2T in 10 minutes. The room temperature is T. Assume that Newton's law of cooling is applicable. The temperature of the body at the end of next 10 minutes will be: *(2016 - II)*
a. $\dfrac{4}{3}T$
b. T
c. $\dfrac{7}{4}T$
d. $\dfrac{3}{2}T$

30. Certain quantity of water cools from 70°C to 60°C in the first 5 minutes and to 54°C in the next 5 minutes. The temperature of the surroundings is: *(2014)*
a. 45°C
b. 20°C
c. 42°C
d. 10°C

31. A beaker full of hot water is kept in a room. If it cools from 80°C to 75°C in t_1 minutes, from 75°C to 70°C in t_2 minutes and from 70°C to 65°C in t_3 minutes, then: *(1995)*
a. $t_1 < t_2 < t_3$
b. $t_1 > t_2 > t_3$
c. $t_1 = t_2 = t_3$
d. $t_1 < t_2 = t_3$

Stefan's law, Wien's Displacement Law, Kirchhoff's Law and Black Body

32. Three stars A, B, C have surface temperatures T_A, T_B, T_C respectively. Star A appears bluish, star B appears reddish and star C yellowish. Hence, *(2020-Covid)*
a. $T_B > T_C > T_A$
b. $T_C > T_B > T_A$
c. $T_A > T_C > T_B$
d. $T_A > T_B > T_C$

33. The power radiated by a black body is P and it radiates maximum energy at wavelength, λ_0. If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it beco .. : nP. The value of n is: *(2018)*
a. $\dfrac{256}{81}$
b. $\dfrac{4}{3}$
c. $\dfrac{3}{4}$
d. $\dfrac{81}{256}$

34. A spherical black body with a radius of 12 cm radiates 450 watt power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be: *(2017-Delhi)*
a. 450
b. 1000
c. 1800
d. 225

35. A black body is at a temperature of 5760 K. The energy of radiation emitted by the body at wavelength 250 nm is U_1, at wavelength 500 nm is U_2 and that at 1000 nm is U_3. Wien's constant, b = 2.88×10^6 nmK. Which of the following is correct? *(2016 - I)*
a. $U_1 = 0$
b. $U_3 = 0$
c. $U_1 > U_2$
d. $U_2 > U_1$

36. On observing light from three different stars P, Q and R, it was found that intensity of violet color is maximum in the spectrum of P, the intensity of green color is maximum in the spectrum of R and the intensity of red color is maximum in the spectrum of Q. If T_P, T_Q and T_R are the respective absolute temperatures of P, Q and R then it can be concluded from the above observations that: *(2015)*
a. $T_P > T_R > T_Q$
b. $T_P < T_R < T_Q$
c. $T_P < T_Q < T_R$
d. $T_P > T_Q > T_R$

37. A piece of iron is heated in a flame. It first becomes dull red then becomes reddish yellow and finally turns to white hot. The correct explanation for the above observation is possible by using: *(2013)*
a. Newton's Law of cooling
b. Stefan's Law
c. Wien's displacement Law
d. Kirchoff's Law

38. If the radius of a star is R and it acts as a black body, what would be the temperature of the star, in which the rate of energy production is Q? *(2012 Pre)*

a. $Q/4\pi R^2\sigma$
b. $(Q/4\pi R^2\sigma)^{-1/2}$
c. $(4\pi R^2Q/\sigma)^{1/4}$
d. $(Q/4\pi R^2\sigma)^{1/4}$

(σ stands for Stefan's constant)

39. The total radiant energy per unit area, normal to the direction of incidence, received at a distance R from the center of a star of radius r, whose outer surface radiates as a black body at a temperature T K is given by: *(2010 Pre)*

a. $\frac{4\pi\sigma r^2T^4}{R^2}$
b. $\frac{\sigma r^2T^4}{R^2}$
c. $\frac{\sigma r^2T^4}{4\pi r^2}$
d. $\frac{\sigma r^4T^4}{r^2}$

(Where σ is Stefan's Constant)

40. A black body at 227°C radiates heat at the rate of 7 cal/cm² s. At a temperature of 727°C, the rate of heat radiated in the same units will be *(2009)*

a. 50
b. 112
c. 80
d. 60

41. Assuming the sun to have a spherical outer surface of radius r, radiating like a black body at temperature t°C, the power received by a unit surface, (normal to the incident rays) at a distance R from the center of the sun is where σ is the Stefan's constant: *(2007)*

a. $\frac{r^2\sigma(t+273)^4}{4\pi R^2}$
b. $\frac{16\pi^2r^2\sigma t^4}{R^2}$
c. $\frac{r^2\sigma(t+273)^4}{R^2}$
d. $\frac{4\pi r^2\sigma t^4}{R^2}$

42. A black body is at 727°C. It emits energy at a rate which is proportional to *(2007)*

a. $(1000)^4$
b. $(1000)^2$
c. $(727)^4$
d. $(727)^2$

43. A block body at 1227°C emits radiations with maximum intensity at a wavelength of 5000Å. If the temperature of the body is increased by 1000°C, the maximum intensity will be observed at *(2006)*

a. 3000Å
b. 4000Å
c. 5000Å
d. 6000Å

44. If λ_m denotes the wavelength at which the radiative emission from a black body at a temperature T K is maximum, then: *(2004)*

a. λ_m is independent of T
b. $\lambda_m \propto T$
c. $\lambda_m \propto T^{-1}$
d. $\lambda_m \propto T^4$

45. We consider the radiation emitted by the human, body. Which of the following statements is true: *(2003)*

a. The radiation emitted is in the infrared regions
b. The radiation is emitted only during the day.
c. The radiation is emitted during the summers and absorbed during the winters.
d. The radiation emitted lies in the ultraviolet region and hence is not visible

46. The Wien's displacement law express relation between: *(2002)*

a. Wavelength corresponding to maximum energy and temperature.
b. Radiation energy and wavelength
c. Temperature and wavelength
d. Colour of light and temperature

47. Which of the following is best close to an ideal black body: *(2002)*

a. Black lamp
b. Cavity maintained at constant temperature
c. Platinum black
d. A lump of charcoal heated to high temp.

48. For a black body at temperature 727°C, its radiating power is 60 watt and temperature of surrounding is 227°C. If temperature of black body is changed to 1227°C then its radiating power will be: *(2002)*

a. 304 W
b. 320 W
c. 240 W
d. 120 W

49. Unit of Stefan's constant is: *(2002)*

a. Watt-m²-K⁴
b. Watt-m²/K⁴
c. Watt/m²-K
d. Watt/m²K⁴

50. A black body has wavelength λ_m corresponding to maximum energy at 2000 K. Its wavelength corresponding to maximum energy at 3000 K will be: *(2001)*

a. $\frac{3}{2}\lambda_m$
b. $\frac{2}{3}\lambda_m$
c. $\frac{16}{81}\lambda_m$
d. $\frac{81}{16}\lambda_m$

51. A sphere maintained at temperature 600 K, has cooling rate R in an external environment of 200 K temperature If its temperature, falls to 400 K then its cooling rate will be: *(1999)*

a. $\frac{3}{16}R$
b. $\frac{16}{3}R$
c. $\frac{9}{27}R$
d. None

52. Radiation energy corresponding to the temperature T of the sun is E. If its temperature is doubled, then its radiation energy will be: *(1998)*

a. 32E
b. 16E
c. 8E
d. 4E

53. A black body is at a temperature of 500 K. It emits energy at a rate which is proportional to: *(1997)*

a. $(500)^3$
b. $(500)^4$
c. 500
d. $(500)^2$

54. If the temperature of the sun is doubled, the rate of energy received on earth will be increased by a factor of: *(1993)*

a. 2
b. 4
c. 8
d. 16

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	c	c	d	d	c	b	d	c	d	d	a	d	d	b	d	d
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
c	a	c	c	d	a	b	b	a	d	a	d	a	a	c	a	c
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
d	a	c	d	b	b	c	a	a	c	a	a	b	b	d	b	a
52	53	54														
b	b	d														

Explanations

1. (c) The freezing point and boiling point of unknown scale are given. We have to find the given celsius in terms of unknown scale. So use the formula-

$$\frac{\text{Temp. of one scale} - \text{Lower fixed point}}{\text{Upper fixed point} - \text{Lower fixed point}} = \text{Constant}$$

$$\Rightarrow \frac{T^\circ - 39^\circ}{239^\circ - 39^\circ} = \frac{39^\circ - 0^\circ}{100^\circ - 0^\circ}$$

$$\Rightarrow \frac{T^\circ - 39^\circ}{200^\circ} = \frac{39^\circ}{100^\circ} \Rightarrow T^\circ = 78^\circ + 39^\circ = 117^\circ W$$

2. (c) Volume of the mercury thermometer changes with rise of temperature and can measure temperatures ranging from $-30°C$ to $357°C$ ($\approx 360°$ C).

3. (c) Using $\frac{F-32}{180} = \frac{C}{100}$,

$$\Rightarrow \frac{140-32}{180} = \frac{C}{100} \Rightarrow C = 60°C$$

As the temperature of boiling water = 100°C

fall in temperature = 100 – 60 = 40°C

4. (d) Increase in length of a metallic rod = $L\,\alpha\,\Delta T$

As the increase in length is independent of temp

$\therefore L\,\alpha$ = constant

$\Rightarrow \alpha_{Cu} L_{Cu} = \alpha_{Al} L_{Al}$

$1.7 \times 10^{-5} \times 88 \text{ cm} = 2.2 \times 10^{-5} \times L_{Al}$

$$L_{Al} = \frac{1.7 \times 88}{2.2} = 68\,\text{cm}$$

5. (d) As increase in length are equal $\Rightarrow \Delta l_1 = \Delta l_2$

$l_1 \alpha_1 \Delta T = l_2 \alpha_2 \Delta T$

$\Rightarrow l_1 \alpha_1 = l_2 \alpha_2$

6. (c) We know that, for volumetric expansion:

$V = V_0 (1 + \gamma \Delta T)$

$$\frac{M}{d} = \frac{M}{d_0}(1 + \gamma \Delta T) \qquad \left\{ \because \text{Volume}(V) = \frac{\text{Mass}(M)}{\text{Density}(d)} \right\}$$

$d = d_0 (1 - \gamma \Delta T)$

$\Rightarrow \quad d = d_0 - d_0 \gamma \Delta T$

$$\Rightarrow \quad \frac{d_0 - d}{d_0} = \gamma \Delta T$$

Fractional change in density $= \frac{\Delta d}{d_0} = 5 \times 10^{-4} \times 40 = 0.020$

7. (b) Heat conducted

$$= \frac{KA(T_1 - T_2)}{l} = \frac{K\pi r^2 (T_1 - T_2)}{l}$$

The rod with the maximum ratio of A/l will conduct most. Here the rod with $r = 2r_0$ and $l = l_0$ will conduct most heat.

8. (d) $\Delta Q = MS\Delta T$

Where S is specific heat capacity and is same for same material

Since, $\Delta Q \propto M$ and $M = \frac{4}{3}\pi r^3 \rho$

$\Delta Q \propto r^3$

$$\frac{\Delta Q_1}{\Delta Q_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{1.5}{1}\right)^3 = \frac{27}{8}$$

9. (c) The potential energy gets converted into heat and only one quarter of it is absorbed by ice.

$$\frac{mgh}{4} = mL$$

$$h = \frac{4L}{g} = \frac{4 \times 3.4 \times 10^5}{10} = 136\,\text{km}$$

10. (d) Let θ be the final common temperature. Further, let s_c and s_h be the average heat capacities of the cold and hot bodies respectively (where $s_c < s_h$ given).

Using, principle of calorimetry,

Heat lost = Heat gained

$s_h (100 - \theta) = s_c \theta$

$$\therefore\ \theta = \frac{s_h}{(s_h + s_c)} \times 100°C = \frac{100°C}{\left(1 + \frac{s_c}{s_h}\right)}$$

$\because\ s_c/s_h < 1$

$\Rightarrow\ \theta > 50°C$

$\therefore$ As the body at 100°C has more heat capacity than body at 0°C so final temperature must be greater than 50°C.

11. (d) Heat lost = Heat gained

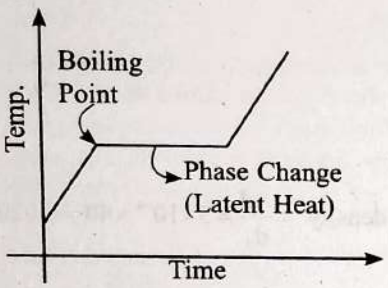
$mL_v + ms_w\Delta\theta_1 = m_W s_W \Delta\theta_2$

$\Rightarrow m \times 540 + m \times 1 \times (100 - 80) = 20 \times 1 \times (80 - 10)$

$\Rightarrow m = 2.5$ g

Total mass of water = (20 + 2.5) g = 22.5 g

12. (a) Initially liquid oxygen will gain the temperature upto its boiling temperature, then it changes its state of gas (during phase change temperature is constant). After this again its temperature will increase, so corresponding graph will be option a.



13. (d) Heat required to melt 1 kg ice at 0°C to water at 0°C

$Q = M_{ice} L_{ice} = (1\text{ kg})(80\text{ cal/g})$

$= 8 \times 10^4$ cal $\qquad [\because 1\text{kg} = 1000\text{g}]$

$$\Delta S = \frac{Q}{T} = \frac{8\times10^4\text{cal}}{273\text{ K}} = 293\text{ cal/K}$$

14. (d) Thermal capacity = ms = 40 × 0.2 = 8 cal/K = 33.6 joule/K.

($\because$ 1 cal = 4.2 J)

15. (b) Let the final temperature be T

Heat required by ice = mL + m × s × (T – 0)

= 10 × 80 + 10 × 1 × T

Heat lost by water = 55 × 1 × (40 – T)

$\because$ Heat gained = Heat lost

800 + 10 T = 55 × (40 – T)

$\Rightarrow T = 21.54°\text{ C} \approx 22°C$

16. (d) Thermal conductivity is the measure of a material's ability to conduct heat.

$$\frac{dQ}{dt} = \frac{kA}{l}\Delta T \quad (k = \text{coefficient of thermal conductivity})$$

$$\therefore k = \frac{l\,dQ}{A\,dt\,\Delta T} \qquad \left[\text{unit of } k = \frac{m \times J}{m^2 \times t \times K}\right]$$

Unit of k = $Wm^{-1}K^{-1}$

17. (d) $\because$ Rods are in parallel combination

$$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$$

$$\frac{l}{K_{eq}(2A)} = \frac{\frac{l}{K_1A}\cdot\frac{l}{K_2A}}{\frac{l}{K_1A} + \frac{l}{K_2A}} = \frac{\frac{1}{K_1K_2}\frac{l}{A}}{\frac{1}{K_1} + \frac{1}{K_2}}$$

$$= \frac{\frac{1}{K_1K_2}\frac{l}{A}}{\frac{K_1+K_2}{K_1K_2}} = \frac{1}{(K_1+K_2)}\frac{l}{A} \Rightarrow K_{eq} = \frac{K_1+K_2}{2}$$

18. (c) We know that the rate of flow of heat is directly proportional to temperature difference i.e.

$$\frac{\Delta Q}{\Delta t} = \frac{KA\Delta T}{l}$$

$$4\text{ J/s} = \frac{KA[110-100]}{l} = \frac{10KA}{l}$$

Similarly,

$$\frac{\Delta Q}{\Delta t} = \frac{kA[210-200]}{l}$$

$$\Rightarrow \frac{\Delta Q}{\Delta t} = 4\text{ J/s}$$

$\because$ Temperature difference is same so rate of heat flow will also same.

19. (a) $\frac{\Delta Q}{\Delta t} = KA\frac{\Delta T}{\Delta x} = \frac{mL}{\Delta t}$

$K = \frac{mL \times \Delta x}{\Delta t \times \Delta T \times A}$ (K: coefficient of thermal conductivity)

$$K = \frac{4.8\times3.36\times10^5\times0.1}{3600\times100\times0.36} = 1.24\text{ J/m/s/°C}$$

20. (c) We know that, $\frac{Q}{t} = \frac{kA(T_1 - T_2)}{l}$

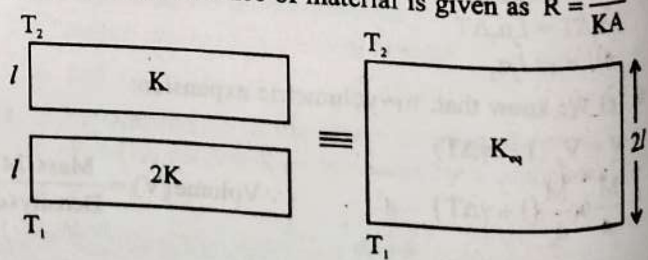
$$\frac{Q'}{t} = \frac{k\left(\frac{A}{4}\right)(T_1 - T_2)}{4l} = \frac{1}{16}\frac{kA(T_1 - T_2)}{l}$$

$$\Rightarrow Q' = \frac{Q}{16}$$

21. (c) Rate of heat transfer is directly proportional to surface area, temperature difference and thermal conductivity.

$$\frac{dQ}{dt} = \frac{kA(T_1 - T_2)}{L}$$

22. (d) Thermal resistance of material is given as $R = \frac{l}{KA}$



Since the slabs are in series:

$$\frac{2l}{K_{eq}(A)}=\frac{l}{2KA}+\frac{l}{KA} \quad \text{(series connection } R_{eq}=R_1+R_2\text{)}$$

$$\Rightarrow K_{eq}=\frac{4}{3}K$$

23. **(a)** We know that,

$$H=\frac{dQ}{dt}=\frac{KA}{L}(T_1-T_2)$$

$$\therefore H_1=\frac{K_1A_1}{L_1}(T_1-T_2) \text{ and } H_2=\frac{K_2A_2}{L_2}(T_1-T_2)$$

Given, $H_1 = H_2$ ($\because$ Rate of loss due to conduction is same)

$\therefore K_1A_1(T_1-T_2)=K_2A_2(T_1-T_2)$ [$\because L_1 = L_2$]

$\Rightarrow K_1A_1 = K_2A_2$

24. **(b)** Heat flow rate $=\frac{KA(T_1-T_2)}{L}=Q$

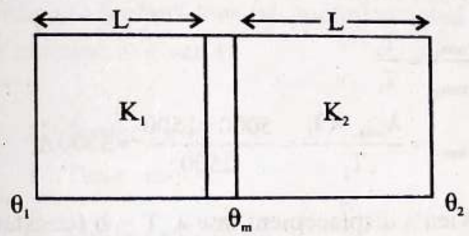
when linear dimensions are doubled.

$A_1 \propto r_1^2, \quad L_1 = L$

$A_2 \propto 4r_1^2, \quad L_2 = 2L_1 \quad$ so $Q_2 = 2Q_1$

25. **(b)** When liquid is heated in a flask, the particles of the liquid at the bottom, get heated, become lighter and actually rises up. The cold particles from above at lower temperature being heavy come down due to gravity and receive heat.

26. **(a)** Heat flow rate through uniform cross section area is constant.



$$\Delta Q_1 = \Delta Q_2$$

$$\frac{K_1(\theta_1-\theta_m)A}{L}=\frac{K_2(\theta_m-\theta_2)A}{L}$$

$$K_1\theta_1 - K_1\theta_m = K_2\theta_m - K_2\theta_2$$

$$K_1\theta_1 + K_2\theta_2 = \theta_m[K_2 + K_1]$$

$$\theta_m=\frac{K_1\theta_1+K_2\theta_2}{K_1+K_2}$$

27. **(d)** Ratio of diameters of rod = 1 : 2 and ratio of their lengths = 2 : 1.

The rate of flow of heat, $(R)=\frac{KA\Delta T}{l}\propto\frac{A}{l}$.

Therefore $\frac{R_1}{R_2}=\frac{A_1}{A_2}\times\frac{l_2}{l_1}=\left(\frac{1}{2}\right)^2\times\frac{1}{2}=\frac{1}{8}$

$\Rightarrow R_1 : R_2 = 1 : 8$

28. **(a)** According to Newton's law of cooling.

$$\frac{T_1-T_2}{t}=K\left[\frac{T_1+T_2}{2}-T_0\right]$$

Case-I:

$$\Rightarrow\frac{90-80}{t}=K\left[\frac{90+80}{2}-20\right] \quad \text{...(i)}$$

Case-II:

$$\Rightarrow\frac{80-60}{t_1}=K\left[\frac{80+60}{2}-20\right] \quad \text{...(ii)}$$

Dividing (i) by (ii):

$$\Rightarrow\frac{\left(\frac{10}{t}\right)}{\left(\frac{20}{t_1}\right)}=\frac{K[65]}{K[50]}\Rightarrow\frac{t_1}{2t}=\frac{13}{10}\Rightarrow t_1=\frac{13}{5}t$$

29. **(d)** Newton's law of cooling

$$\frac{T_1-T_2}{t}=k\left(\frac{T_1+T_2}{2}-T\right)$$

$$\frac{3T-2T}{10}=k\left(\frac{5T-2T}{2}\right)\Rightarrow\frac{T}{10}=k\left(\frac{3T}{2}\right) \quad \text{...(i)}$$

$$\frac{2T-T'}{10}=k\left(\frac{2T+T'}{2}-T\right)\Rightarrow\frac{2T-T'}{10}=k\left(\frac{T'}{2}\right) \quad \text{...(ii)}$$

By solving Eqs. (i) and (ii)

$$T'=\frac{3}{2}T$$

30. **(a)** According to Newton's law of cooling

$$\frac{\theta_1-\theta_2}{t}=k\left[\frac{\theta_1+\theta_2}{2}-\theta_0\right]\Rightarrow\frac{70-60}{5}=k\left[\frac{70+60}{2}-\theta_0\right]$$

$$2 = k[65-\theta_0] \quad \text{...(i)}$$

and

$$\frac{60-54}{5}=k\left[\frac{60+54}{2}-\theta_0\right]$$

$$\Rightarrow\frac{6}{5}=k[57-\theta_0] \quad \text{...(ii)}$$

By dividing Eqs. (i) by (ii) we have

$$\frac{10}{6}=\frac{65-\theta_0}{57-\theta_0}\Rightarrow\theta_0=45°C$$

31. **(a)** The rate of cooling is directly proportional to the difference in temperature of the body and the surroundings. So, cooling will be fastest in the first case and slowest in the third case.

32. **(c)** From Wien's displacement law

$$\lambda=\frac{b}{T}\Rightarrow\lambda\propto\frac{1}{T}$$

as $\lambda_A < \lambda_C < \lambda_B$

$\therefore T_A > T_C > T_B$

33. **(a)** As λT = const. (Wien's displacement law)

$$T_1=\frac{C}{\lambda_0}, T_2=\frac{4C}{3\lambda_0} \quad \text{(C is constant)}$$

Power radiated (P) $\propto T^4$

$$P_1=\frac{C^4}{\lambda_0^4}$$

$$P_2=\frac{256}{81}\frac{C^4}{\lambda_0^4}=nP_1$$

Comparing $n=\frac{256}{81}$

34. (c) Power radiated = $\sigma A \varepsilon_0 T^4$

$P \propto r^2 T^4$

$$\frac{P_1}{P_2} = \frac{r_1^2}{r_2^2}\left[\frac{T_1}{T_2}\right]^4$$

$$\frac{450}{P_2} = \frac{(12\times10^{-2})^2}{\left(\frac{12}{2}\times10^{-2}\right)^2}\times\left[\frac{500}{1000}\right]^4$$

$P_2 = 1800$ W

35. (d) Maximum amount of emitted radiation corresponding to $\lambda_m = \frac{b}{T}$

$$\lambda_m = \frac{2.88\times10^6\,\text{nmK}}{5760\text{K}} = 500\text{ nm}$$

$\therefore$ U is maximum at 500 nm. Hence, $U_2 > U_1$

36. (a) From Wein's displacement law

$$\lambda_m \propto \frac{1}{T}$$

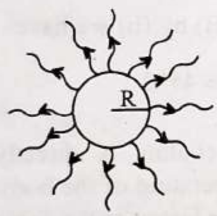
Now from sequence 'VIBGYOR'

$(\lambda_m)_P < (\lambda_m)_R < (\lambda_m)_Q$

$\Rightarrow T_P > T_R > T_Q$

37. (c) We can explain this observation by using Wien's displacement law, $\lambda_m T = b$, where b is Wien's constant.

38. (d) Heat radiated by star of radius R



$Q = A\sigma\varepsilon T^4$

$\varepsilon = 1$ $\quad\therefore$ acts as black body

$Q = 4\pi R^2\sigma T^4$ $\quad$ Area of star = $4\pi R^2$

$$T = \left[\frac{Q}{4\pi R^2\sigma}\right]^{1/4}$$

39. (b) Total radiant energy per unit area

$$= \frac{\sigma(4\pi r^2)T^4}{4\pi R^2} = \frac{\sigma r^2 T^4}{R^2}$$

40. (b) Rate of heat radiated at (227 + 273) K = 7 cal/(cm²s)

Rate of heat radiated at (727 + 273) K = x

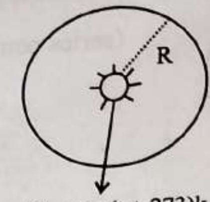
By Stefan's law, $7 \propto (500)^4$ ...(1)

$x \propto (1000)^4$ ...(2)

Dividing eqn. (2) by (1)

$$\therefore \frac{x}{7} = 2^4 \Rightarrow x = 7\times2^4 = 112\text{ cal/(cm}^2\text{s)}$$

41. (c)



Temp = (t° + 273) K

Power received by a unit surface

$$= \frac{\text{Intensity}}{\text{Surface Area}}$$

$$= \frac{\sigma(4\pi r^2)(t+273)^4}{4\pi R^2}$$

$$= \frac{\sigma r^2(t+273)^4}{R^2}$$

42. (a) According to Stefan's law

Rate of energy radiated $E \propto T^4$

where T is the absolute temperature of a black body.

$\therefore E \propto (727 + 273)^4$ or $E \propto [1000]^4$

43. (a) According to Wein's displacement law,

$\lambda_{max} T$ = constant

$$\therefore \frac{\lambda_{max_1}}{\lambda_{max_2}} = \frac{T_2}{T_1}$$

$$\text{or } \lambda_{max_2} = \frac{\lambda_{max_1}\times T_1}{T_2} = \frac{5000\times1500}{2500} = 3000\,\text{Å}$$

44. (c) Wien's displacement law $\lambda_m T = b$ (constant)

$\therefore \lambda_m \propto T^{-1}$

45. (a) As we know that the wavelength of the emitted radiations is inversely proportional to the temperature of the body.

Wavelength, $\lambda = \frac{b}{T}$ $\quad\left[\begin{array}{l}\text{Where } b = \text{constant} \\ T = \text{Temperature of body}\end{array}\right]$

We know that, temperature of human body is:-

T = 37°C = 310K

$\therefore$ Wavelength $\lambda = \frac{b}{T}$

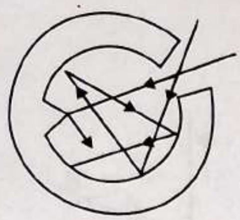
$$\lambda = \frac{2898}{10} \qquad (\because b = 2898\ \mu\text{mK})$$

$\lambda = 9.35\ \mu$m (corresponds to infrared region)

So, the infrared radiations are emitted by human body.

46. (a) Wien's displacement law expresses relation between wavelength corresponding to maximum energy and temperature. It states that the product of absolute temperature and the wavelength at which the emissive power is maximum is constant, i.e. λ_{max} T = constant.

47. **(b)** An ideal black body absorbs all the incident radiation without reflecting or transmitting any part of it. An ideal black body can be realized in practice by a small hole in the wall of a hollow body (as shown in figure) which is at uniform temperature. Any radiation entering the hollow body through the holes suffers a number of reflections and ultimately gets completely absorbed. This can be facilitated by coating the interior surface with black so that about 96% of the radiation is absorbed at each reflection. The portion of the interior surface opposite to the hole is made conical to avoid the escape of the reflected ray after one reflection.



48. **(b)** We know that, Radiated power,

$$\therefore \; P \propto (T^4 - T_0^{\,4})$$

$$\frac{P_2}{P_1} = \frac{(1500)^4 - (500)^4}{(1000)^4 - (500)^4} = \frac{500^4(3^4-1)}{500^4(2^4-1)}$$

$$\frac{P_2}{60} = \frac{80}{15} \Rightarrow P_2 = 320\,\text{W}$$

49. **(d)** According to Stefan's law, the energy radiated per second or power radiated is given by:

$P = \sigma A T^4$

Where A : Surface area

T : Temperature

σ : Stefans's Constant $= \dfrac{P}{AT^4}$

Unit of $\sigma = \dfrac{W}{m^2K^4}$

50. **(b)** According to Wien's displacement law,

$$\lambda_m T = b \quad \Rightarrow \quad \lambda_1 T_1 = \lambda_2 T_2$$

$$\because T_1 = 2000\text{ K}, T_2 = 3000\text{ K}$$

$$\therefore \lambda_m \times 2000 = \lambda_2 \times 3000$$

$$\lambda_2 = \frac{2}{3}\lambda_m$$

51. **(a)** According to Stefan's law:

Rate of cooling $\propto T^4 - T_0^4$

$$\frac{R'}{R} = \frac{(400)^4 - (200)^4}{(600)^4 - (200)^4} = \frac{4^4 - 2^4}{6^4 - 2^4} \Rightarrow R' = \frac{3}{16}R$$

52. **(b)** According to Stefan's law

$E \propto T^4$

$$\frac{E}{E_2} = \left(\frac{T}{2T}\right)^4$$

$$\Rightarrow E_2 = 16E$$

53. **(b)** Temperature of black body (T) = 500 K. Therefore total energy emitted by the black body $(E) \propto T^4 \propto (500)^4$.

54. **(d)** Amount of energy radiated $\propto T^4$.

$$\frac{E_1}{E_2} = \left(\frac{T_1}{T_2}\right)^4 = \left(\frac{T}{2T}\right)^4$$

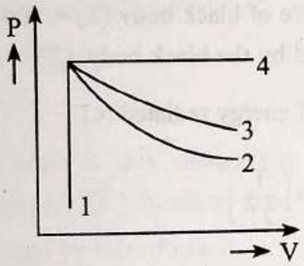
$$E_2 = 16E_1$$

$\therefore$ Energy increased by factor 16.

CHAPTER 11 Thermodynamics

Thermodynamic Processes and First Law of Thermodynamics

1. An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal isobaric and isochoric. The curve which represents the adiabatic process among 1, 2, 3 and 4 is: *(2022)*

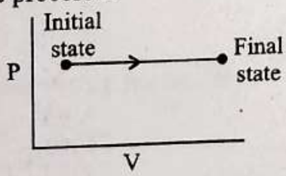


a. 4 b. 1
c. 2 d. 3

2. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is: *(2020)*

a. Adiabatic b. Isochoric
c. Isobaric d. Isothermal

3. The P-V diagram for an ideal gas in a piston cylinder assembly undergoing a thermodynamic process is shown in the figure. The process is *(2020-Covid)*

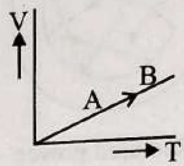


a. Isochoric b. Isobaric
c. Isothermal d. Adiabatic

4. In which of the following processes, heat is neither absorbed nor released by a system? *(2019)*

a. Isothermal b. Adiabatic
c. Isobaric d. Isochoric

5. The volume (V) of a monoatomic gas varies with its temperature (T), as shown in the graph. The ratio of the work done by the gas, to the heat absorbed by it, when it undergoes a change from state A to State B, is *(2018)*

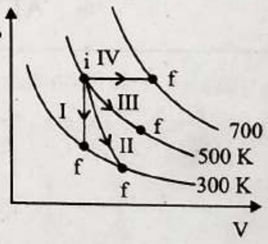


a. $\frac{1}{3}$ b. $\frac{2}{3}$
c. $\frac{2}{5}$ d. $\frac{2}{7}$

6. A sample of 0.1 g of water at 100°C and normal pressure (1.013×10^5 Nm^{-2}) requires 54 cal of heat energy to convert to steam at 100°C. If the volume of the steam produced is 167.1 cc, the change in internal energy of the sample, is: *(2018)*

a. 42.2 J b. 208.7 J
c. 104.3 J d. 84.5 J

7. Thermodynamic processes are indicated in the following diagram. *(2017-Delhi)*



Match the following:

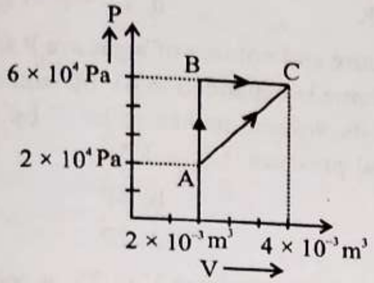
	Column-1		Column-2
P.	Process I	a.	Adiabatic
Q.	Process II	b.	Isobaric
R.	Process III	c.	Isochoric
S.	Process IV	d.	Isothermal

a. P → c, Q → a, R → d, S → b
b. P → c, Q → d, R → b, S → a
c. P → d, Q → b, R → a, S → c
d. P → a, Q → c, R → d, S → b

8. A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then: *(2016 - I)*

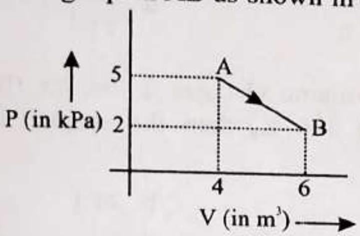
a. Compressing the gas isothermally will require more work to be done
b. Compressing the gas through adiabatic process will require more work to be done
c. Compressing the gas isothermally or adiabatically will require the same amount of work
d. Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas

9. Figure below shows two paths that may be taken by a gas to go from a state A to a state C. In process AB, 400 J of heat is added to the system and in process BC, 100 J of heat is added to the system. The heat absorbed by the system in the process AC will be: *(2015)*



a. 500 J
b. 460 J
c. 300 J
d. 380 J

10. One mole of an ideal diatomic gas undergoes a transition from A to B along a path AB as shown in the figure,



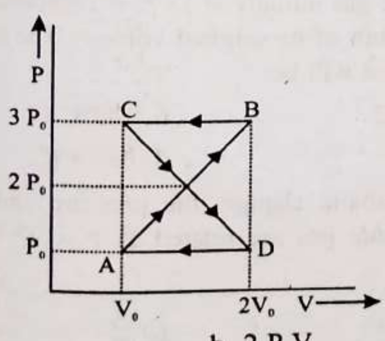
The change in internal energy of the gas during the transition is: *(2015)*

a. –20 kJ
b. 20 J
c. –12 kJ
d. 20 kJ

11. An ideal gas is compressed to half its initial volume by means of several processes. Which of the process results in the maximum work done on the gas? *(2015 Re)*

a. Isothermal
b. Adiabatic
c. Isobaric
d. Isochoric

12. A thermodynamic system undergoes cyclic process ABCDA as shown in figure. The work done by the system in the cycle is: *(2014)*



a. P_0V_0
b. $2 P_0V_0$
c. $\frac{P_0V_0}{2}$
d. Zero

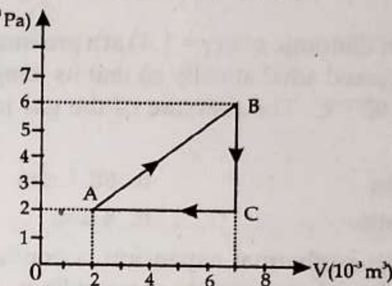
13. A monoatomic gas at a pressure P, having a volume V expands isothermally to a volume 2 V and then adiabatically to a volume 16 V. The final pressure of the gas is (take $\gamma = 5/3$): *(2014)*

a. 64 P
b. 32 P
c. P/64
d. 16 P

14. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its temperature. The ratio of $\frac{C_P}{C}$ for the gas is: *(2013)*

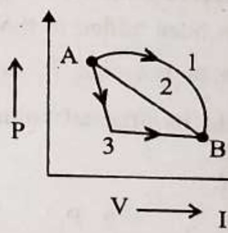
a. 3/2
b. 2
c. 4/3
d. 5/3

15. A gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$, as shown. What is the net work done by the gas? *(2013)*



a. –2000 J
b. 2000 J
c. 1000 J
d. Zero

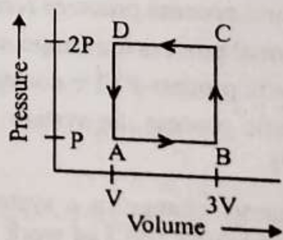
16. An ideal gas goes from state A to state B via three different processes as indicated in the P-V diagram



If Q_1, Q_2, Q_3 indicate the heat absorbed by the gas along the three processes and ΔU_1, ΔU_2, ΔU_3 indicate the change in internal energy along the three processes respectively, then: *(2012 Mains)*

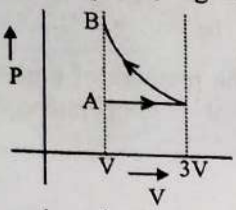
a. $Q_1 > Q_2 > Q_3$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$
b. $Q_3 > Q_2 > Q_1$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$
c. $Q_1 = Q_2 = Q_3$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$
d. $Q_3 > Q_2 > Q_1$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$

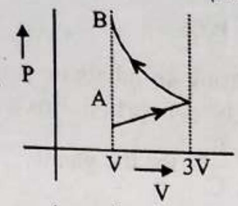
17. A thermodynamic system is taken through the cycle ABCD as shown in figure. Heat rejected by the gas during the cycle is: *(2012 Pre)*

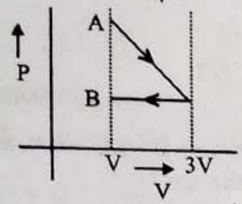


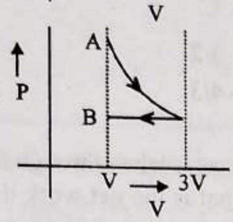
a. 2 PV
b. 4 PV
c. 1/2 PV
d. PV

18. One mole of an ideal gas goes from an initial state A to final state B via two processes: It first undergoes isothermal expansion from volume V to 3V and then its volume is reduced from 3V to V at constant pressure. The correct P-V diagram representing the two processes is: *(2012 Pre)*

a. 

b. 

c. 

d. 

19. A mass of diatomic gas ($\gamma = 1.4$) at a pressure of 2 atmospheres is compressed adiabatically so that its temperature rises from 27°C to 927°C. The pressure of the gas in the final state is: *(2011 Mains)*

a. 28 atm
b. 68.7 atm
c. 256 atm
d. 8 atm

20. During an isothermal expansion, a confined ideal gas does 150 J of work against its surroundings. This implies that: *(2011 Pre)*

a. 150 J heat has been removed from the gas
b. 300 J of heat has been added to the gas
c. No heat is transferred because the process is isothermal
d. 150 J of heat has been added to the gas

21. A monoatomic gas at pressure P_1 and V_1 is compressed adiabatically to $\frac{1}{8}$th its original volume. What is the final pressure of the gas? *(2010 Mains)*

a. $64\ P_1$
b. P_1
c. $16\ P_1$
d. $32\ P_1$

22. ΔU and ΔW represent the increase in internal energy and work done by the system respectively in a thermodynamical process, which of the following is true? *(2010 Pre)*

a. $\Delta U = -\Delta W$, in a isothermal process
b. $\Delta U = -\Delta W$, in a adiabatic process
c. $\Delta U = \Delta W$, in a isothermal process
d. $\Delta U = \Delta W$, in a adiabatic process

23. In thermodynamic processes which of the following statements is not true? *(2009)*

a. In an isochoric process pressure remains constant
b. In an isothermal process the temperature remains constant
c. In an adiabatic process PV^γ = constant
d. In an adiabatic process the system is insulated from the surroundings

24. The internal energy change in a system that has absorbed 2 kcal of heat and done 500 J of work is: *(2009)*

a. 6400 J
b. 5400 J
c. 7900 J
d. 8900 J

25. If Q, E and W denote respectively the heat added, change in internal energy and the work done in a closed cycle process, then: *(2008)*

a. $Q = 0$
b. $W = 0$
c. $Q = W = 0$
d. $E = 0$

26. One mole of an ideal gas at an initial temperatures of T K does 6 R joules of work adiabatically. If the ratio of specific heats of this gas at constant pressure and at constant volume is $\frac{5}{3}$, the final temperatures of gas will be: *(2004)*

a. $(T - 0.4)$ K
b. $(T + 14)$ K
c. $(T - 4)$ K
d. $(T + 0.4)$ K

27. Initial pressure and volume of a gas are P and V respectively. First its volume is expanded to 4V by isothermal process and then again its volume makes to be V by adiabatic process then its final pressure is ($\gamma = 1.5$): *(1999)*

a. 8P
b. 4P
c. P
d. 2P

28. When volume changes from V to 2V at constant pressure(P) then the change in internal energy will be: *(1998)*

a. PV
b. 3PV
c. $\frac{PV}{\gamma - 1}$
d. $\frac{RV}{\gamma - 1}$

29. A gas of volume changes 2 litre to 10 litre at constant temperature 300 K, then the change in internal energy will be: *(1998)*

a. 12 J
b. 24 J
c. 36 J
d. 0 J

30. A sample of gas expands from volume V_1 to V_2. The amount of work done by the gas is greatest, when the expansion is: *(1997)*

a. Adiabatic
b. Equal in all cases
c. Isothermal
d. Isobaric

31. An ideal gas, undergoing adiabatic change, has which of the following pressure temperature relationship? *(1996)*

a. $P^\gamma T^{1-\gamma}$ = constant
b. $P^{1-\gamma} T^\gamma$ = constant
c. $P^{\gamma-1} T^\gamma$ = constant
d. $P^\gamma T^{\gamma-1}$ = constant

32. A diatomic gas initially at 18°C is compressed adiabatically to one eighth of its original volume. The temperature after compression will be: *(1996)*

a. 395.4°C
b. 144°C
d. 18°C
d. 887.4°C

33. In an adiabatic change, the pressure and temperature of a monatomic gas are related as $P \propto T^C$ where C equals: *(1994)*

a. $\frac{3}{5}$
b. $\frac{5}{3}$
c. $\frac{2}{5}$
d. $\frac{5}{2}$

34. 110 joule of heat is added to a gaseous system whose internal energy is 40 J, then the amount of external work done is: *(1993)*

a. 150 J b. 70 J
c. 110 J d. 40 J

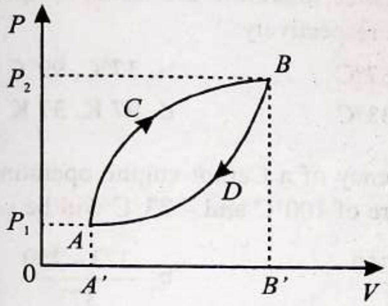
35. An ideal gas A and a real gas B have their volumes increased from V to 2V under isothermal conditions. The increase in internal energy: *(1993)*

a. Will be same in both A and B
b. Will be zero in both the gases
c. Of B will be more than that of A
d. Of A will be more than that of B

36. Which of the following is not thermodynamical function? *(1993)*

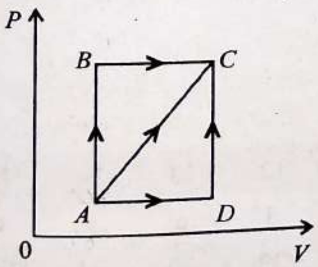
a. Enthalpy b. Work done
c. Gibb's energy d. Internal energy

37. A thermodynamic system is taken from state A to B along ACB and is brought back to A along BDA as shown in the PV diagram. The net work done during the complete cycle is given by the area: *(1992)*



a. $P_1ACBP_2P_1$ b. ACBB'A'A
c. ACBDA d. ADBB'A'A

38. A thermodynamic process is shown in the figure. The pressure and volumes corresponding to some points in the figure are
$P_A = 3 \times 10^4$ Pa; $V_A = 2 \times 10^{-3}$ m³.
$P_B = 8 \times 10^4$ Pa; $V_D = 5 \times 10^{-3}$ m³.
In the process AB, 600 J of heat is added to the system and in process BC, 200 J of heat is added to the system. The change in internal energy of the system in process AC would be: *(1991)*



a. 560 J b. 800 J
c. 600 J d. 640 J

39. One mole of an ideal gas requires 207 J heat to rise the temperature by 10 K when heated at constant pressure. If the same gas is heated at constant volume to raise the temperature by the same 10 K, the heat required is (Given the gas constant R = 8.3 J/mole K): *(1990)*

a. 198.7 J b. 29 J
c. 215.3 J d. 124 J

40. First law of thermodynamics is consequence of conservation of: *(1988)*

a. Work b. Energy
c. Heat d. All of these

Second Law of Thermodynamics & Refrigerator

41. The temperature inside a refrigerator is t_2°C and the room temperature is t_1°C. The amount of heat delivered to the room for each joule of electrical energy consumed ideally will be: *(2016 - II)*

a. $\frac{t_1° + 273}{t_1° + t_2°}$ b. $\frac{t_1° + t_2°}{t_1° + 273}$
c. $\frac{t_1°}{t_1° + t_2°}$ d. $\frac{t_1° + 273}{t_1° - t_2°}$

42. A refrigerator works between 4°C and 30°C. It is required to remove 600 calories of heat every second in order to keep the temperature of the refrigerated space constant. The power required is
(Take 1 cal = 4.2 joules): *(2016 - I)*

a. 2.365 W b. 23.65 W
c. 236.5 W d. 2365 W

43. The coefficient of performance of a refrigerator is 5. If the temperature inside freezer is –20°C, the temperature of the surroundings to which it rejects heat is: *(2015 Re)*

a. 21°C b. 31°C
c. 41°C d. 11°C

44. Which of the following processes is reversible? *(2005)*

a. Transfer of heat by radiation
b. Electrical heating of a nichrome wire
c. Transfer of heat by conduction
d. Isothermal compression

Carnot Engine

45. The efficiency of a carnot engine depends upon *(2020-Covid)*

a. The temperatures of the source and sink
b. The volume of the cylinder of the engine
c. The temperature of the source only
d. The temperature of the sink only

46. The efficiency of an ideal heat engine working between the freezing point and boiling point of water, is: *(2018)*

a. 6.25% b. 20%
c. 26.8% d. 12.5%

47. A carnot engine having an efficiency of 1/10 as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is: *(2017-Delhi)*

a. 90 J
b. 99 J
c. 100 J
d. 1 J

48. Carnot engine, having an efficiency of $\eta = 1/10$. As heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is: *(2015)*

a. 99 J
b. 90 J
c. 1 J
d. 100 J

49. The efficiency of Carnot engine is 50% and temperature of sink is 500 K. If temperature of source is kept constant and its efficiency raised to 60%, then the required temperature of sink will be: *(2007)*

a. 100 K
b. 600 K
c. 400 K
d. 500 K

50. An engine has an efficiency of 1/6. When the temperature of sink is reduced by 62°C, its efficiency is doubled. Temperature of the source is *(2007)*

a. 37°C
b. 62°C
c. 99°C
d. 124°C

51. A Carnot engine whose sink is at 300 K has an efficiency of 40%. By how much should the temperature of source be increased so as to increase its efficiency by 50% of original efficiency? *(2006)*

a. 275 K
b. 175 K
c. 250 K
d. 225 K

52. An ideal gas heat engine operates in Carnot cycle between 227°C and 127°C. It absorbs 6×10^4 cal of heat at higher temperature. Amount of heat converted to work is: *(2005)*

a. 2.4×10^4 cal
b. 3.6×10^4 cal
c. 1.2×10^4 cal
d. 6.4×10^4 cal

53. An ideal gas heat engine operates in a Carnot cycle. Between 227°C and 127°C. It absorbs 6 kcal at the higher temperature. The amount of heat (in kcal) converted into work is equal to: *(2003)*

a. 4.8
b. 3.5
c. 1.6
d. 1.2

54. The efficiency of carnot engine is 50% and temperature of sink is 500 K. If temperature of source is kept constant and its efficiency raised to 60%, then the required temperature of the sink will be: *(2002)*

a. 100 K
b. 600 K
c. 400 K
d. 500 K

55. A scientist says that the efficiency of his heat engine which work at source temperature 127°C and sink temperature 27°C is 26%, then: *(2001)*

a. It is impossible
b. It is possible but less probable
c. It is quite probable
d. Data are incomplete

56. The ratio (W/Q) for a carnot-engine is $\frac{1}{6}$. Now the temperature, of sink is reduced by 62°C, then this ratio becomes twice, therefore the initial temp, of the sink and source are respectively: *(2000)*

a. 33°C, 67°C
b. 37°C, 99°C
c. 67°C, 33°C
d. 97 K, 37 K

57. The efficiency of a Carnot engine operating with reservoir temperature of 100°C and – 23°C will be: *(1997)*

a. $\frac{373+250}{373}$
b. $\frac{373-250}{373}$
c. $\frac{100-23}{100}$
d. $\frac{100+23}{100}$

58. An ideal Carnot engine, whose efficiency is 40%, receives heat at 500 K. If its efficiency is 50%, then the intake temperature for the same exhaust temperature is: *(1995)*

a. 800 K
b. 900 K
c. 600 K
d. 700 K

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	a	b	b	c	b	a	b	b	a	b	d	c	a	c	a	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
d	c	d	d	b	a	c	d	c	d	c	d	d	b	a	d	b
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
b	b	c	a	d	b	d	c	b	d	a	c	a	b	c	c	c
52	53	54	55	56	57	58										
c	d	c	a	b	b	c										

Explanations

1. (c) Curve 1: As the volume remains constant, so the process is Isochoric
Curve 2: Adiabatic curve is steeper than isothermal.
Curve 3: Curve 3 represents isothermal process.
Curve 4: Isobaric process occurs at constant pressure.

2. (a) Entire system is thermally insulated. So, no heat exchange will take place. Hence process will be adiabatic.

3. (b) From the given p-V diagram, pressure p= constant
$\therefore$ The given process is isobaric.

4. (b) An adiabatic process occurs without exchange of heat or mass between a thermodynamic system and its surroundings. Therefore, the given process is adiabatic.

5. (c) In the given diagram, volume V changes with temperature T and pressure P remains constant, $\Rightarrow P = \text{constant}$

$W = \text{Work done} = P(V_2 - V_1)$

$= nR(T_2 - T_1) = nR\,\Delta T \qquad \left[\begin{array}{l}\because PV = nRT \\ \therefore P\Delta V = nR\Delta T\end{array}\right]$

As the pressure remains constant, so specific heat capacity used is C_p & for monoatomic gas $C_p = \frac{5R}{2}$

$\Rightarrow Q = nC_P\Delta T \Rightarrow Q = n\left(\frac{5}{2}R\right)\Delta T$

Ratio of $\frac{W}{Q} = \frac{2}{5}$

6. (b) Using first law of thermodynamics

$\Delta Q = W + \Delta U \Rightarrow \Delta U = \Delta Q - W$

$W = P(V_f - V_i)$

$V_f - V_i = 167$ cc

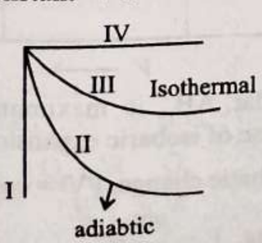
$= \frac{167}{10^6}\,m^3$

$\Rightarrow W = 1.013\times10^5 \times \frac{167}{10^6} = 16.9$ J

$\Rightarrow \Delta U = 54 \times 4.18 - 16.9$

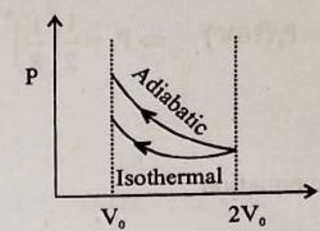
$\Delta U = 208.7$ J

7. (a) Isochoric = V constant $\rightarrow$ I
Isobaric = P constant $\rightarrow$ IV
Isothermal = T constant $\rightarrow$ III
Adiabtic = Q constant $\rightarrow$ II



In an expansion adiabtic curve is more steeper than that of isothermal.

8. (b)



Area under the adiabatic curve is more than the area under the isothermal curve.

$\therefore W_{(\text{adiabatic})} > W_{(\text{Isothermal})}$

9. (b) In cyclic process ABCA,

$\Delta U_{\text{cyclic}} = 0 \Rightarrow Q_{\text{cyclic}} = W_{\text{cyclic}}$

$Q_{AB} = +400$ J is the heat absorbed in process AB

$Q_{BC} = +100$ J is the heat absorbed in process BC

Area under loop = +W (clockwise)

$Q_{AB} + Q_{BC} - Q_{CA} =$ closed loop area.

$400 + 100 - Q_{CA} = \frac{1}{2}\times(2\times10^{-3})\times4\times10^4$

$400 + 100 - Q_{AC} = 40$

$Q_{AC} = 460$ J

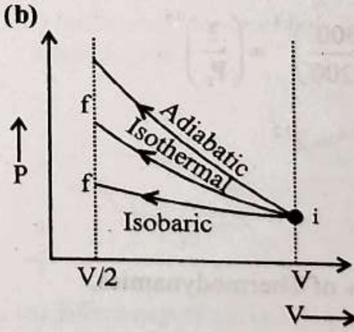
10. (a) $\Delta U = nC_v\Delta T$ & $T = \frac{PV}{nR}$

so $\Delta T = T_2 - T_1 = \frac{P_2V_2 - P_1V_1}{nR}$

so $\Delta U = \frac{nR}{\gamma - 1}\left(\frac{P_2V_2 - P_1V_1}{nR}\right) = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$

$\Rightarrow \Delta U = \frac{-8\times10^3}{2/5} = -20$ kJ

11. (b)



work done on the gas

$W_{\text{isochoric}} = 0$

and $W_{\text{adiabatic}} > W_{\text{Isothermal}} > W_{\text{Isobaric}}$

12. (d) Work done by the system in the cycle
= Area under P – V curve

$= \frac{1}{2}(2P_0 - P_0)(2V_0 - V_0) + \left[-\left(\frac{1}{2}\right)(3P_0 - 2P_0)(2V_0 - V_0)\right]$

$= \frac{P_0V_0}{2} - \frac{P_0V_0}{2} = 0$

13. (c) For isothermal process $P_1V_1 = P_2V_2$

$$\Rightarrow PV = P_2(2V) \Rightarrow P_2 = \frac{P}{2}$$

For adiabatic process $P_2V_2^\gamma = P_3V_3^\gamma$

$$\Rightarrow \left(\frac{P}{2}\right)(2V)^\gamma = P_3(16V)^\gamma \quad \Rightarrow P_3 = \frac{1}{2}\left(\frac{1}{8}\right)^{5/3} = \frac{P}{64}$$

14. (a) $P \propto T^3$

$P = kT^3$ ...(1)

Using adiabatic equation

$P^{1-\gamma}T^\gamma = k$

$P = kT^{-\gamma/1-\gamma}$

Compare this with equation (1)

$$\frac{-\gamma}{1-\gamma} = 3 \quad \Rightarrow \quad \gamma = \frac{3}{2}$$

15. (c) Net work done = Area of triangle ABC

$$= \frac{1}{2}\times[7-2]\times10^{-3}\times\left[(6-2)\times10^5\right] = 1000 \text{ J}$$

16. (a) For three paths initial and final positions are same hence $\Delta U_A = \Delta U_B = \Delta U_C$ and for path (1) area is maximum hence $Q_1 > Q_2 > Q_3$ because $Q = W + \Delta U$

17. (a) W = –ve As the cycle is Anticlockwise

In cyclic process $\Delta U = 0$

So $Q = W = -(2P - P)(3V - V) = -2PV$

Heat absorbed by the system = –2PV

Heat rejected = 2PV

18. (d) As the expansion is isothermal and compression is isobaric, therefore only graph given in option (d) satisfy the given requirements.

19. (c) $T_i = 27°C = 300$ K

$T_f = 927°C = 1200$ K

$$\therefore \left(\frac{T_i}{T_f}\right)^\gamma = \left(\frac{P_i}{P_f}\right)^{\gamma-1} \Rightarrow \left(\frac{300}{1200}\right)^{1.4} = \left(\frac{2}{P_f}\right)^{0.4}$$

$$\Rightarrow \frac{1}{4^{1.4}\times2^{0.4}} = \frac{1}{P_f^{0.4}} \Rightarrow P^{0.4} = 2^{3.2}$$

$$P_f = 2^{\frac{3.2}{0.4}} = 2^8 = 256 \text{ atm}$$

20. (d) According to first law of Thermodynamics,

$\Delta Q = \Delta U + W$

Since it is isothermal expansion $\Delta Q = 0$

$\Delta Q = 0 - (-150) \Rightarrow 150$ J

21. (d) PV^γ = Constant $\Rightarrow P_2 = P_1\left(\frac{V_1}{V_2}\right)^\gamma$

$$\Rightarrow P_2 = P_1\left(\frac{V}{V/8}\right)^{\frac{5}{3}} = P_1(8)^{5/3} = 32P_1$$

22. (b) $\Delta Q = \Delta U + \Delta W$. In adiabatic process $\Delta Q = 0$

$\therefore \Delta U = -\Delta W$

23. (a) In an isochoric process, volume remains constant. All other statements are correct

24. (c) According to 1st Law of Thermodynamics

$Q = \Delta U + W$

2000 cal = ΔU + 500 J

(2000 × 4.2) J = ΔU + 500 J

$\Delta U = 8400 - 500 = 7900$ J

25. (d) Since E which is change in internal energy is a state function. Therefore, in a closed cyclic process,

E = 0

26. (c) $\Delta U = \mu C_v \Delta T$ and $0 = W + \Delta U$

$\Rightarrow \Delta U = -6R \quad (\because W = 6R)$

Therefore $-6R = 1\left(\frac{R}{\gamma-1}\right)\Delta T = \frac{3}{2}R\Delta T$

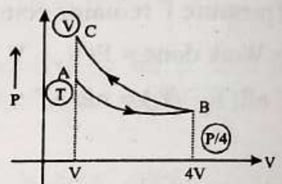
$\Rightarrow \Delta T = -4 \Rightarrow T_{final} = (T - 4)$ K

27. (d) For isothermal process

$P_AV_A = P_BV_B$

$PV = P_B(4V)$

$P_B = \frac{P}{4}$



For adiabatic process

$P_BV_B^\gamma = P_CV_C^\gamma$

$$P_c = \frac{P}{4}\left(\frac{4V}{V}\right)^{1.5} = \frac{P}{4}\times8 = 2P$$

28. (c) Change in internal energy is given by

$$dU = \mu C_V dT = \frac{\mu R dT}{\gamma-1} = \frac{P(2V-V)}{\gamma-1} = \frac{PV}{\gamma-1}$$

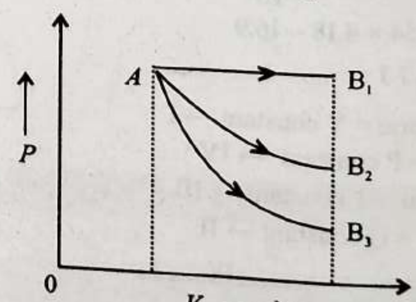
29. (d) $\because \Delta U = \mu C_0 \Delta T$

at constant temperature

$\Delta T = 0$

$\Rightarrow \Delta U = 0$

30. (d) During expansion, work is performed by the gas. The isobaric expansion is represented by the horizontal straight line AB_1, since the adiabatic curve is steeper than the isothermal curve, the adiabatic expansion curve (AB_3) must lie below the isothermal curve (AB_2) as shown in the figure below



Since area under AB_1, is maximum, the work done is maximum in case of isobaric expansion.

31. (b) For the adiabatic change, PV^γ = constant.

And for ideal gas, $V = \frac{RT}{P} \propto \frac{T}{P}$.

$\Rightarrow P^{1-\gamma}T^\gamma$ = constant.

32. (a) Initial temperature $(T_1) = 18°C$.
$= (273 + 18) = 291$ K and $V_2 = V_1/8$.
For adiabatic compression, $TV^{\gamma-1}$ = constant
$\Rightarrow T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$.
Therefore $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$
$= 291 \times (8)^{1.4-1} = 291 \times (8)^{0.4}$
$= 291 \times 2.297 = 668.4$ K. $= 395.4°C$

33. (d) For adiabatic change, PV^{γ} = constant
$\Rightarrow P\left(\frac{RT}{\rho}\right)^{\gamma} = \text{constant} \Rightarrow P^{1-\gamma}T^{\gamma} = \text{constant}$
$\Rightarrow P \propto T^{\frac{-\gamma}{1-\gamma}}$.
Therefore, the value of constant $C = \frac{\gamma}{(\gamma-1)}$. For monoatomic gas, $\gamma = \frac{5}{3}$.
Therefore $C = \frac{5/3}{(5/3)-1} = \frac{5/3}{2/3} = \frac{5}{2}$.

34. (b) According to first law of thermodynamics
$\Delta Q = \Delta U + \Delta W$
$\Rightarrow \Delta W = \Delta Q - \Delta U = 110 - 40 = 70$ J.

35. (b) Under isothermal conditions, there is no change in internal energy.

36. (b) Work done is not a thermodynamical functions.

37. (c) Work done during the complete cycle = Area under curve ACBDA

38. (a) Since AB is a isochoric process. So no work is done. BC is isobaric process
$W = P_B \times (V_D - V_A) = 240$ J
Therefore $\Delta Q = 600 + 200 = 800$ J
Using $\Delta Q = \Delta U + \Delta W$
$\Rightarrow \Delta U = \Delta Q - \Delta W = 800 - 240 = 560$ J

39. (d) Using $C_P - C_V = R$
C_P is heat needed for raising by 10 K
$\therefore$ $C_P = 20.7$ J/mole K
Given R = 8.3 J /mole K
$\therefore$ $C_V = 20.7 - 8.3 = 12.4$ J/mole K
$\therefore$ Heat required to raise temperature by 10 K = 124 J

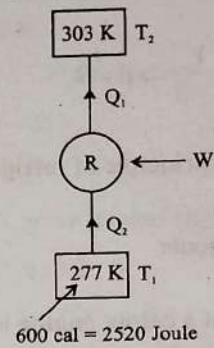
40. (b) First law of thermodynamics is a consequence of Conservation of energy.

41. (d) Let Heat delivered = Q_1
And heat absorbed from the sink = Q_2

$$\text{C.O.P } (\beta) = \frac{Q_2}{W} = \frac{Q_1 - W}{W} = \frac{Q_1}{W} - 1 = \frac{T_2}{T_1 - T_2}$$

$$\Rightarrow \frac{Q_1}{W} = 1 + \frac{t_2° + 273}{t_1° - t_2°} = \frac{t_1° + 273}{t_1° - t_2°}$$

42. (c)



$\because$ Coefficient of performance of the refrigerator β is given by,

$$\beta = \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2}$$

$$\frac{2520}{W} = \frac{277}{303 - 277}$$

$\Rightarrow$ W = 236.5 joule

$$\text{Power} = \frac{W}{t} = \frac{236.5}{1 \text{ sec}} \text{ joule} = 236.5 \text{ watt}$$

43. (b) Coefficient of performance of refrigerator

$$\text{C.O.P} = \frac{T_L}{T_H - T_L}$$

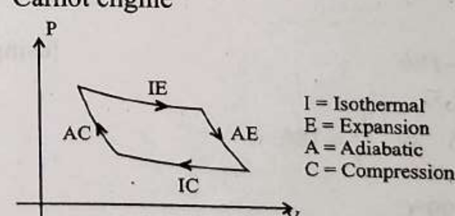
where $T_L \rightarrow$ lower Temperature
and $T_H \rightarrow$ Higher Temperature

$$\text{So, } 5 = \frac{T_L}{T_H - T_L}$$

$$\Rightarrow T_H = \frac{6}{5}T_L = \frac{6}{5}(253) = 303.6 \text{ K}$$

$\Rightarrow T_H = 303.6 - 273 = 30.6°C$
$= 31°C$

44. (d) Isothermal compression is reversible phenomenon. E.g. Carnot engine



45. (a) Efficiency of carnot engine (η)

$$= 1 - \frac{T_2}{T_1} \quad \begin{pmatrix} T_1 = \text{Temperature of source} \\ T_2 = \text{Temperature of sink} \end{pmatrix}$$

46. (c) $\% \eta_{\text{carnot}} = \left(1 - \frac{T_L}{T_H}\right) \times 100 = \left(1 - \frac{273}{373}\right) \times 100 = 26.8\%$

47. (a) $\beta = \frac{Q_2}{W} = \frac{1-\eta}{\eta}$

$$\frac{Q_2}{10} = \frac{1-0.1}{0.1} \Rightarrow Q_2 = 90 \text{ joule}$$

48. **(b)** For engine and refrigerators operating between two same temperatures

$$\eta=\frac{1}{1+\beta}\Rightarrow\frac{1}{10}=\frac{1}{1+\beta}\Rightarrow\beta=9$$

$\beta=\frac{Q_2}{10}$ (From the principle of refrigerator)

$$9=\frac{Q_2}{10}\Rightarrow Q_2=90\,\text{joule}$$

49. **(c)** Efficiency (η) of a carnot engine is given by

$\eta=1-\frac{T_2}{T_1}$ where T_1 is the temperature of the source and T_2 is the temperature of the sink.

Here, $T_2 = 500$ K.

$$\therefore\ 0.5=1-\frac{500}{T_1}\Rightarrow T_1=1000\,\text{K}$$

Now, $\eta=0.6=1-\frac{T_2'}{1000}$

(T'_2 is the new sink temperature)

$= T'_2 = 400$ K

50. **(c)** Efficiency of an engine, $\eta=1-\frac{T_2}{T_1}$

Where T_1 is the temperature of the source and T_2 is the temperature of the sink.

$$\therefore\ \frac{1}{6}=1-\frac{T_2}{T_1}\quad\text{or,}\ \frac{T_2}{T_1}=\frac{5}{6}\qquad\text{...(i)}$$

When the temperature of the sink is decreased by 62°C, its efficiency becomes doubled.

Since, the temperature of the source remain unchanged

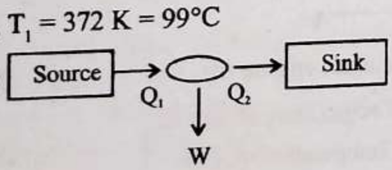
$$\therefore\ 2\times\frac{1}{6}=1-\frac{(T_2-62)}{T_1}\quad\text{or,}\ \frac{1}{3}=1-\frac{(T_2-62)}{T_1}$$

$$\Rightarrow 2T_1=3T_2-186$$

$$\text{or } 2T_1=3\left[\frac{5}{6}\right]T_1-186\qquad\text{[using (i)]}$$

$$\therefore\left[\frac{5}{2}-2\right]T_1=186\quad\text{or,}\ \frac{T_1}{2}=186$$

or, $T_1 = 372\text{ K} = 99°\text{C}$

51. **(c)** 

$T_2 = 300$K

Efficiency $(\eta_1) = 40\% = 0.4$

Let temperature of source be T_1

$$\eta_1=1-\frac{T_2}{T_1}\Rightarrow 0.4=1-\frac{300}{T_1}$$

$\Rightarrow T_1 = 500$ K

Now, new efficiency is 50% more than the original efficiency

$$\Rightarrow\eta_2=0.4+0.4\times\frac{50}{100}=0.6$$

$$\because\ \eta_2=1-\frac{T_2}{T_1}$$

$$\Rightarrow 0.6=1-\frac{T_2}{T_1}$$

Since T_2 is fixed and T_1 is changed,

$$0.6=1-\frac{300}{T_1}$$

$$T_1=750\,\text{K}$$

Increase in temperature of source $= 750 - 500 = 250$ K

52. **(c)**

$$\frac{T_2}{T_1}=\frac{Q_2}{Q_1}$$

Here, $T_1 = 227 + 273 = 500$ K

$T_2 = 127 + 273 = 400$ K

$$\Rightarrow\ \frac{400}{500}=\frac{Q_2}{6\times10^4}$$

$$\Rightarrow\ Q_2=4.8\times10^4\,\text{cal}$$

So, amount of heat converted to work

$= (6 - 4.8) \times 10^4 = 1.2 \times 10^4$ cal

53. **(d)** $\eta=1-\frac{T_2}{T_1}=\frac{W}{Q}$

$T_2(\text{sink}) = 127 + 273 = 400$ K

$T_1(\text{source}) = 227 + 273 = 500$ K

$Q = 6$ kcal

$$1-\frac{400}{500}=\frac{W}{6}$$

$$\frac{1}{5}=\frac{W}{6}$$

$$W=\frac{6}{5}=1.2\,\text{kcal}$$

54. **(c)** $\%\eta=\left(1-\frac{T_2}{T_1}\right)\times100$

For 50% efficiency

$$\frac{50}{100}=1-\frac{500}{T_1}\Rightarrow T_1=1000\,\text{K}$$

For 60% efficiency

$$\frac{60}{100}=1-\frac{T_2}{1000}\Rightarrow T_2=400\,\text{K}$$

55. **(a)** Carnot engine is an Ideal engine so its efficiency will be maximum

$$\therefore\ \eta_{max.}=\frac{400-300}{400}\times100\%=25\%$$

therefore 26% efficient engine is impossible.

56. (b) $\frac{W}{Q}=\frac{1}{6}$

$1-\frac{T_L}{T_H}=\frac{1}{6} \Rightarrow \frac{T_L}{T_H}=\frac{5}{6}$

If sink temp. decrease by 62°C then

$1-\frac{T_L-62}{T_H}=\frac{2}{6} \Rightarrow \frac{T_L-62}{T_H}=\frac{2}{3}$

$2T_H=3T_L-186 \Rightarrow 2T_H=3\times\frac{5}{6}T_H-186$

$2T_H-\frac{5}{2}T_H=-186 \Rightarrow \frac{5-4}{2}T_H=186$

$T_H=186\times 2=372\text{ K}=99°C$

$T_L=\frac{5}{6}\times 372=310\text{K}=37°C$

57. (b) Reservoir temperature (T_1) = 100°C = 373 K and $T_2=-23°C=250$ K.

The efficiency of a Carnot engine

$$\eta=\frac{T_1-T_2}{T_1}=\frac{373-250}{373}.$$

58. (c) Efficiency of Carnot engine $(\eta_1)=40\%=0.4$;

Heat intake = 500 K and

New efficiency $(\eta_2)=50\%=0.5$

The efficiency $(\eta)=1-\frac{T_2}{T_1}$ *or* $\frac{T_2}{T_1}=1-\eta$

For first case, $\frac{T_2}{500}=1-0.4 \Rightarrow T_2=300\text{K}$

For second case, $\frac{300}{T_1}=1-0.5 \Rightarrow T_1=600\text{K}.$

CHAPTER 12

Kinetic Theory

Ideal Gas Equations and Vander Waals Relation

1. The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces vanish away is: *(2022)*

a. $5.6\ m^3$ b. $5.6 \times 10^6\ m^3$
c. $5.6 \times 10^3\ m^3$ d. $5.6 \times 10^{-3}\ m^3$

2. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27° C.
Its density is : ($R = 8.3\ J\ mol^{-1}\ K^{-1}$) *(2020)*

a. $0.2\ kg/m^3$ b. $0.1\ kg/m^3$
c. $0.02\ kg/m^3$ d. $0.5\ kg/m^3$

3. An ideal gas equation can be written as $P = \frac{\rho RT}{M_0}$ where ρ and M_0 are respectively, *(2020-Covid)*

a. Number density, molar mass
b. Mass density, molar mass
c. Number density, mass of the gas
d. Mass density, mass of the gas

4. Increase in temperature of a gas filled in a container would lead to: *(2019)*

a. Increase in its mass
b. Increase in its kinetic energy
c. Decrease in its pressure
d. Decrease in intermolecular distance

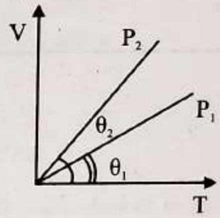
5. A given sample, of an ideal gas occupies a volume V at a pressure P and absolute temperature T. The mass of each molecule of the gas is m. Which of the following gives the density of the gas? *(2016 - II)*

a. P/(kTV) b. mkT
c. P/(kT) d. Pm/(kT)

6. Two vessels separately contain two ideal gases A and B at the same temperature, the pressure of A being twice that of B. Under such conditions, the density of A is found to be 1.5 times the density of B. The ratio of molecular weight of A and B is: *(2015 Re)*

a. 1/2 b. 2/3
c. 3/4 d. 2

7. In the given (V-T) diagram, what is the relation between pressures P_1 and P_2? *(2013)*



a. Cannot be predicted b. $P_2 = P_1$
c. $P_2 > P_1$ d. $P_2 < P_1$

8. At 10°C the value of the density of a fixed mass of an ideal gas divided by its pressure is x. At 110°C this ratio is: *(2008)*

a. $\frac{283}{383}x$ b. x
c. $\frac{383}{283}x$ d. $\frac{10}{100}x$

9. The equation of state for 5 g of oxygen at a pressure P and temperature T, when occupying a volume V, will be: *(2004)*

a. PV = 5 RT b. PV = (5/2) RT
c. PV = (5/16) RT d. PV = (5/32)RT
where R is the gas constant.

10. The value of critical temperature in terms of Van der Waals' constant a and b is given by: *(1996)*

a. $T_C = \frac{8a}{27Rb}$ b. $T_C = \frac{27a}{8Rb}$
c. $T_C = \frac{a}{2Rb}$ d. $T_C = \frac{a}{27Rb}$

11. According to kinetic theory of gases, at absolute zero temperature *(1990)*

a. Water freezes b. Liquid helium freezes
c. Molecular motion stops d. Liquid hydrogen freezes

12. At constant volume temperature is increased then: *(1989)*

a. Collision on walls will be less
b. Number of collisions per unit time will increase
c. Collisions will be in straight lines
d. Collisions will not change

Pressure Exerted by an Ideal Gas and its Kinetic Energy

13. At 0 K which of the following properties of a gas will be zero? *(1996)*
a. Vibrational energy
b. Density
c. Kinetic energy
d. Potential energy

14. Three containers of the same volume contain three different gases. The masses of the molecules are m_1, m_2 and m_3 and the number of molecules in their respective containers are N_1, N_2 and N_3. The gas pressure in the containers are P_1, P_2 and P_3 respectively. All the gases are now mixed and put in one of these containers. The pressure P of the mixture will be: *(1991)*
a. $P < (P_1 + P_2 + P_3)$
b. $P = \frac{P_1 + P_2 + P_3}{3}$
c. $P = P_1 + P_2 + P_3$
d. $P > (P_1 + P_2 + P_3)$

15. Relation between pressure (P) and energy (E) of a gas of *(1991)*
a. $P = \frac{2}{3}E$
b. $P = \frac{1}{3}E$
c. $P = E$
d. $P = 3E$

16. According to kinetic theory of gases, at absolute zero of temperature: *(1990)*
a. Water freezes
b. Liquid helium freezes
c. Molecular motion stops
d. Liquid hydrogen freezes

17. Two containers A and B are partly filled with water and closed. The volume of A is twice that of B and it contains half the amount of water in B. If both are at the same temperature, the water vapour in the containers will have pressure in the ratio of: *(1988)*
a. 1: 2
b. 1 : 1
c. 2 : 1
d. 4 : 1

Speed of Gas Molecules

18. Match Column-I and Column-II and choose the correct match from the given choices. *(2021)*

Column-I		Column-II	
(A)	Root mean square speed of gas molecules	(P)	$\frac{1}{3}nm\bar{v}^2$
(B)	Pressure exerted by ideal gas	(Q)	$\sqrt{\frac{3RT}{M}}$
(C)	Average kinetic energy of a molecule	(R)	$\frac{5}{2}RT$
(D)	Total internal energy of 1 mole of a diatomic gas	(S)	$\frac{3}{2}k_BT$

a. (A) - (Q), (B) - (R), (C) - (S), (D) - (P)
b. (A) - (Q), (B) - (P), (C) - (S), (D) - (R)
c. (A) - (R), (B) - (Q), (C) - (P), (D) - (S)
d. (A) - (R), (B) - (P), (C) - (S), (D) - (Q)

19. At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere? (Given: Mass of oxygen molecule (m) = 2.76×10^{-26} kg, Boltzmann's constant $k_B = 1.38 \times 10^{-23}$ JK^{-1}) *(2018)*
a. 5.016×10^4 K
b. 8.360×10^4 K
c. 2.508×10^4 K
d. 1.254×10^4 K

20. The molecules of a given mass of a gas have r.m.s velocity of 200 ms^{-1} at 27°C and 1.0×10^5 Nm^{-2} pressure. When the temperature and pressure of the gas are respectively, 127°C and 0.05×10^5 Nm^{-2}, the r.m.s. velocity of its molecules in ms^{-1} is: *(2016 - I)*
a. $100\sqrt{2}$
b. $\frac{400}{\sqrt{3}}$
c. $\frac{100\sqrt{2}}{3}$
d. $\frac{100}{3}$

Variation of Pressure with Depth and Pascal's Law

21. The average thermal energy for a mono-atomic gas is : (k_B) is Boltzmann constant and T is absolute temperature) *(2020)*
a. $\frac{3}{2}k_BT$
b. $\frac{5}{2}k_BT$
c. $\frac{7}{2}k_BT$
d. $\frac{1}{2}k_BT$

22. A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is: *(2017-Delhi)*
a. 15 RT
b. 9 RT
c. 11 RT
d. 4 RT

23. The molar specific heats of an ideal gas at constant pressure and volume are denoted by C_P and C_V respectively. If $\gamma = \frac{C_P}{C_V}$ and R is the universal gas constant, then C_V is equal to: *(2013)*
a. γR
b. $\frac{1+\gamma}{1-\gamma}$
c. $\frac{R}{(\gamma-1)}$
d. $\frac{(\gamma-1)}{R}$

24. To find out degree of freedom, the correct expression is: *(2000)*
a. $\frac{2}{\gamma-1}$
b. $\frac{\gamma+1}{2}$
c. $\frac{2}{\gamma+1}$
d. $\frac{1}{\gamma+1}$

25. The number of translational degrees of freedom for a diatomic gas is: (1993)
a. 2
b. 3
c. 5
d. 6

26. If for a gas, $\frac{R}{C_V} = 0.67$, this gas is made up of molecules which are: (1992)
a. Diatomic
b. Mixture of diatomic and polyatomic molecules
c. Monoatomic
d. Polyatomic

27. A polyatomic gas with n degrees of freedom has a mean energy per molecule given by: (1989)
a. $\frac{nkT}{N}$
b. $\frac{nkT}{2N}$
c. $\frac{nkT}{2}$
d. $\frac{3kT}{2}$

Specific and Molar Heat Capacity of Gases

28. One mole of an ideal monoatomic gas undergoes a process described by the equation PV^3 = constant. The heat capacity of the gas during this process is: (2016 - II)
a. 2 R
b. R
c. $\frac{3}{2}R$
d. $\frac{5}{3}R$

29. The ratio of the specific heats $\frac{C_P}{C_v} = \gamma$ in terms of degrees of freedom (n) is given by: (2015)
a. $\left(1+\frac{n}{3}\right)$
b. $\left(1+\frac{2}{n}\right)$
c. $\left(1+\frac{n}{2}\right)$
d. $\left(1+\frac{1}{n}\right)$

30. 4.0 g of a gas occupies 22.4 litres at NTP. The specific heat capacity of the gas at constant volume is 5.0 $JK^{-1}mol^{-1}$. If the speed of sound in this gas at NTP is 952 ms^{-1}, then the heat capacity at constant pressure is (Take gas constant R = 8.3 J/mol K): (2015 Re)
a. 8.5 J/K mol
b. 8.0 J/K mol
c. 7.5 J/K mol
d. 7.0 J/K mol

31. The amount of heat energy required to raise the temperature of 1 g of Helium at NTP, from T1 K to T2 K is: (2013)
a. $\frac{3}{4}N_A k_B\left(\frac{T_2}{T_1}\right)$
b. $\frac{3}{8}N_A k_B(T_2 - T_1)$
c. $\frac{3}{2}N_A k_B(T_2 - T_1)$
d. $\frac{3}{4}N_A k_B(T_2 - T_1)$

32. If C_p and C_v denote the specific heats (per unit mass of an ideal gas of molecular weight M then: (2010 Mains)
a. $C_P - C_v = R/M^2$
b. $C_P - C_v = R$
c. $C_P - C_v = MR$
d. $C_P - C_v = \frac{R}{M}$

where R is the molar gas constant

33. The molar specific heat at constant pressure of an ideal gas is $\frac{7}{2}R$. The ratio of specific heat at constant pressure to that at constant volume is: (2006)
a. 7/5
b. 6/7
c. 9/7
d. 4/7

34. For hydrogen gas $C_p - C_v = a$ and for oxygen gas $C_p - C_v = b$, so the relation between a and b is given by: (1991)
a. a = 16b
b. 16b = a
c. a = 4b
d. a = b

35. For a certain gas the ratio of specific heats is given to be $\gamma = 1.5$. For this gas: (1990)
a. $C_v = 3R/J$
b. $C_P = 3R/J$
c. $C_P = 5R/J$
d. $C_V = 5R/J$

Mean Free Path

36. The mean free path for a gas, with molecular diameter and number density n can be expressed as : (2020)
a. $\frac{1}{\sqrt{2}\,n\pi d^2}$
b. $\frac{1}{\sqrt{2}\,n^2\pi d^2}$
c. $\frac{1}{\sqrt{2}\,n^2\pi^2 d^2}$
d. $\frac{1}{\sqrt{2}\,n\pi d}$

37. The mean free path ℓ for a gas molecule depends upon diameter, d of the molecule as (2020-Covid)
a. $\ell \propto d$
b. $\ell \propto d^2$
c. $\ell \propto \frac{1}{d}$
d. $\ell \propto \frac{1}{d^2}$

38. The mean free path of molecules of a gas, (radius r) is inversely proportional to: (2014)
a. r^3
b. r^2
c. r
d. $\sqrt{r}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
a	a	b	b	d	c	d	a	d	a	c	b	c	c	a	c	b
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
b	b	b	a	c	c	a	b	c	c	b	b	b	b	d	a	d
35	36	37	38													
b	a	d	b													

Explanations

1. (a) Volume V = (no. of moles) (22.4 litre)

$$\Rightarrow V = \frac{\text{Total mass of the molecules}}{\text{molar mass}}(22.4\times10^{-3}\,m^3)$$

$$\Rightarrow V = \frac{4.5\times10^3}{18}\times 22.4\times10^{-3}\,m^3 = 5.6\,m^3$$

2. (a) As we know that

$$PM = \rho RT \Rightarrow \rho = \frac{PM}{RT}$$

Here, $\rho = 249 \times 10^3 N/m^2$

$M = 2 \times 10^{-3} kg$; $T = 300$ K

$$\therefore \rho = \frac{(249\times10^3)(2\times10^{-3})}{8.3\times300} = \frac{0.2kg}{m^3}$$

3. (b) From the gas equation

$$pV = nRT \Rightarrow p = \frac{1}{V}\frac{m}{M_0}RT = \left(\frac{m}{V}\right)\left(\frac{RT}{M_0}\right)$$

$$\Rightarrow p = \frac{\rho RT}{M_0} \Rightarrow \rho = \frac{m}{V} = \text{mass density}$$

M_0 = molar mass

4. (b) Increase in temperature would lead to the increase in kinetic energy of gas (assuming far as to be ideal) as increase in temperature increases the speed with which the gas molecule move. Mass will remain constant. Pressure will increase and intermolecular space will also increase.

5. (d) $\frac{P}{\rho} = \frac{RT}{M_w}$ (Ideal gas equation)

$$\Rightarrow \rho = \frac{PM_w}{RT} = \frac{P\times(mN_A)}{kN_AT} = \frac{Pm}{kT}$$

6. (c) According to ideal gas equation

$$P = \frac{\rho RT}{M} \Rightarrow M = \frac{\rho RT}{P}$$

So, $\frac{M_A}{M_B} = \frac{\rho_A T_A P_B}{\rho_B T_B P_A} = (1.5)(1)\left(\frac{1}{2}\right) \Rightarrow \frac{M_A}{M_B} = \frac{3}{4}$

7. (d) $PV = nRT \Rightarrow V = \left(\frac{nR}{P}\right)T \Rightarrow \text{slope} = \frac{nR}{P}$

As $\theta_2 > \theta_1$ so $\frac{1}{P_2} > \frac{1}{P_1} \Rightarrow P_1 > P_2$

8. (a) Temp = 10°C = 283 K

$$\frac{\text{Density of a fixed mass}}{\text{Pressure}} = x$$

$$\because PV = nRT \Rightarrow PV = \frac{m}{M}RT$$

$$\Rightarrow P = \frac{m}{V}\frac{RT}{M} \qquad \left\{\because \frac{m}{V} = \rho\right\}$$

$$\Rightarrow \frac{\rho}{P} = \frac{M}{RT} = x \Rightarrow \frac{\rho}{P} \propto \frac{1}{T}$$

$\Rightarrow \frac{x}{y} = \frac{383}{283}$ [y is ratio of $\frac{\rho}{P}$ at 383 K]

$$\Rightarrow y = \frac{283}{383}x$$

9. (d) $PV = nRT$

Where $n = \frac{m}{\text{molecular mass}} = \frac{5}{32}$ moles

10. (a) Critical temperature $T_C = \frac{8a}{27Rb}$, where a and b are Van der Waals constant.

11. (c) According to kinetic theory all motion of molecules stop at 0 K and the kinetic energy become zero.

12. (b) As the temperature increases, the average velocity increases. So, the collisions are faster.

13. (c) According to kinetic theory of gases, at zero kelvin all molecular motions of a gas stops, therefore kinetic energy will be zero.

14. (c) From Dalton's law of partial pressure, we get $P = P_1 + P_2 + P_3$

15. (a) $\frac{1}{3}Nmc^2 = \frac{2}{3}\left(\frac{1}{2}Nm\right)c^2 = \frac{2}{3}E$

16. (c) According to kinetic theory all motion of molecules stop at 0 K.

17. (b) Vapour pressure depends on the temperature alone, it does not depend on the amount of substance.

18. (b)

A. $V_{rms} = \sqrt{\frac{3RT}{M}}$

$\Rightarrow$ option (Q)

B. Pressure exerted by ideal gas,

$P = \frac{1}{3}\rho v_{rms}^2$

$\because \rho = mn$ & $v_{rms}^2 = \bar{v}^2$

$P = \frac{1}{3}mn\bar{v}^2$

$\Rightarrow$ option (P)

C. Average kinetic energy of a molecule,

$K.E = \frac{3}{2}K_B$

$\Rightarrow$ option (S)

D. Total internal energy of 1 mole of a diatomic gas,

$V = \frac{f}{2}nRT$

n = 1 (for 1 mole)

f = 5 (for diatomic)

$\Rightarrow V = \frac{5}{2} \times (1) \times RT = \frac{5}{2}RT$

$\Rightarrow$ option (R)

Therefore, (A) – (Q), (B) – (P), (C) – (S) & (D) – (R)

19. (b) $V_{rms} = \sqrt{\frac{3KT}{m}}$

V_{rms} should be equal to escape speed

$11.2 \times 10^3 = \sqrt{\frac{3KT}{m}}$

$T = 8.36 \times 10^4$ kelvin

20. (b) $V \propto \sqrt{T}$

$\Rightarrow \frac{V}{200} = \sqrt{\frac{400}{300}}$

$\Rightarrow V = \frac{200(2)}{\sqrt{3}}$

$\Rightarrow V = \frac{400}{\sqrt{3}} ms^{-1}$

21. (a) Average thermal energy $= \frac{3}{2}K_BT$

where 3 is translational degree of freedom

p = 3 (translational degree of freedom)

22. (c) $U = \frac{f}{2}nRT$

$U_T = U_{O_2} + U_{Ar}$

$U_T = \frac{5}{2} \times 2RT + \frac{3}{2} \times 4RT$

$U_T = 11RT$

23. (c) $\frac{C_p}{C_v} = \gamma$

By Mayer's equation

$C_p - C_v = R$

$\frac{C_p}{C_v} - 1 = \frac{R}{C_v}$

$\gamma - 1 = \frac{R}{C_v}$

$C_v = \frac{R}{\gamma - 1}$

24. (a) $\because \gamma = 1 + \frac{2}{f}$

$\Rightarrow \frac{2}{f} = \gamma - 1 \qquad \Rightarrow f = \frac{2}{\gamma - 1}$

25. (b) Number of translational degrees of freedom are same for all types of gas molecules, which is equal to 3.

26. (c) Since $\frac{R}{C_V} = 0.67 \Rightarrow \frac{C_P - C_V}{C_V} = 0.67$

$\Rightarrow \gamma = 1.67 = \frac{5}{3}$

Hence gas is monoatomic.

27. (c) Energy per degree of freedom $= \frac{1}{2}kT$ (law of equipartition of energy) For a polyatomic gas with n degrees of freedom, the mean energy per molecule $= \frac{1}{2}nkT$.

28. (b) PV^x = constant (Polytropic process)

Heat capacity in polytropic process is given by

$$\left[C = C_v + \frac{R}{1-x}\right] \quad ...(i)$$

Given that PV^3 = constant $\Rightarrow x = 3$ ...(ii)

Gas is monoatomic therefore $C_v = \frac{3}{2}R$ ...(iii)

Using the formula (i)

$C = \frac{3}{2}R + \frac{R}{1-3} = \frac{3}{2}R - \frac{R}{2} = R$

29. (b) $\because \gamma = 1 + \frac{2}{f}$

Where f is the degree of freedom

For degree of freedom → n

$\therefore \gamma = 1 + \frac{2}{n}$

30. (b) Molecular mass M = 4.0 g

$v_{sound} = \sqrt{\frac{\gamma RT}{M}} \Rightarrow \gamma = \frac{Mv^2}{RT} = 1.6$

So, $C_p = \gamma C_v = 1.6 \times 5$

$= 8\ JK^{-1}\ mol^{-1}$

31. (b) Number of moles in 1 g of He $= \frac{1}{4}$

As the volume of the gas remains constant

$\therefore \Delta Q = nC_v\Delta T$

Amount of heat energy required to raise its temperature from T_1K to T_2K

$= nC_v\Delta T$

$= \left(\frac{1}{4}\right)\left(\frac{3}{2}R\right)(T_2 - T_1)$

$= \frac{3}{8}k_B N_A (T_2 - T_1)\ [\because R = K_B N_A]$

32. (d) $C_p' - C_v' = R \Rightarrow \frac{C_p}{M} - \frac{C_v}{M} = \frac{R}{M} \Rightarrow C_p - C_v = \frac{R}{M}$

33. (a) Molar specific heat at constant pressure (C_p)

$= 7/2\,R$

According to Meyer's Equation

$C_p - C_v = R \Rightarrow 7/2\,R - C_v = R$

$C_V = \frac{5}{2}R$

Since, $C_p = 7/2\,R, C_V = \frac{5}{2}R$

Ratio of $C_P / C_V = \frac{7}{5}$

34. (d) $C_P - C_V = R$ for all gases.

35. (b) $\gamma = \frac{C_P}{C_V} = \frac{15}{10} = \frac{3}{2} \Rightarrow C_V = \frac{2}{3}C_P$

$C_P - C_V = \frac{R}{J} \Rightarrow C_P - \frac{2}{3}C_P = \frac{R}{J}$

$\Rightarrow C_P = \frac{3R}{J}$

36. (a) According to the formula

$\lambda = \frac{1}{\sqrt{2}n\pi d^2}$

37. (d) Mean free path $(\ell) = \frac{1}{\sqrt{2}\pi d^2 n} \propto \frac{1}{d^2}$

38. (b) Mean free path $\lambda_m = \frac{1}{\sqrt{2}\pi d^2 n}$

Where d = diameter of molecule $\Rightarrow \lambda_m \propto \frac{1}{r^2}$

CHAPTER 13

Oscillations

Periodic Motion, Superposition of Waves, Displacement, Velocity, Phase, Acceleration in SHM

1. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is: *(2020)*

a. $\frac{3\pi}{2}$ rad b. $\frac{\pi}{2}$ rad

c. Zero d. π rad

2. Identify the function which represents a periodic motion. *(2020-Covid)*

a. $\log_e(\omega t)$ b. $\sin\omega t + \cos\omega t$

c. $e^{-\omega t}$ d. $e^{\omega t}$

3. The displacement of a particle executing simple harmonic motion is given by $y = A_0 + A\sin\omega t + B\cos\omega t$

Then the amplitude of its oscillation is given by : *(2019)*

a. $A_0 + \sqrt{A^2 + B^2}$ b. $\sqrt{A^2 + B^2}$

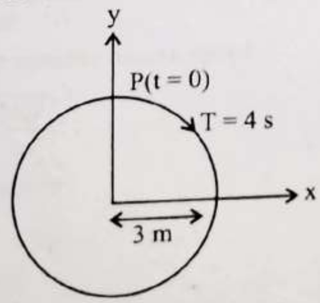
c. $\sqrt{A_0^2 + (A+B)^2}$ d. $A + B$

4. Average velocity of a particle executing SHM in one complete vibration is : *(2019)*

a. $\frac{A\omega}{2}$ b. $A\omega$

c. $\frac{A\omega^2}{2}$ d. Zero

5. The radius of circle, the period of revolution, initial position and sense of revolution are indicated in the



y - projection of the radius vector of rotating particle P is: *(2019)*

a. $y(t) = -3\cos 2\pi t$, where y in m

b. $y(t) = 4\sin\left(\frac{\pi t}{2}\right)$, where y in m

c. $y(t) = 3\cos\left(\frac{3\pi t}{2}\right)$, where y in m

d. $y(t) = 3\cos\left(\frac{\pi t}{2}\right)$, where y in m

6. A particle executes linear simple harmonic motion with an amplitude of 3 cm. When the particle is at 2 cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is: *(2017-Delhi)*

a. $\frac{\sqrt{5}}{2\pi}$ b. $\frac{4\pi}{\sqrt{5}}$

c. $\frac{2\pi}{\sqrt{3}}$ d. $\frac{\sqrt{5}}{\pi}$

7. When two displacements represented by $y_1 = a\sin(\omega t)$ and $y_2 = b\cos(\omega t)$ are superimposed, the motion is: *(2015)*

a. Simple harmonic with amplitude $\frac{a}{b}$

b. Simple harmonic with amplitude $\sqrt{a^2 + b^2}$

c. Simple harmonic with amplitude $\frac{(a+b)}{2}$

d. Not a simple harmonic

8. A particle is executing S.H.M. along a straight line. Its velocities at distances x_1 and x_2 from the mean position are v_1 and v_2 respectively. Its time period is: *(2015)*

a. $2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$ b. $2\pi\sqrt{\frac{v_1^2 - v_2^2}{x_1^2 + x_2^2}}$

c. $2\pi\sqrt{\frac{v_1^2 - v_2^2}{x_1^2 - x_2^2}}$ d. $2\pi\sqrt{\frac{x_1^2 + x_2^2}{v_1^2 + v_2^2}}$

9. A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then, its time period of vibration will be: *(2015 Re)*

a. $\frac{2\pi\beta}{\alpha}$ b. $\frac{\beta^2}{\alpha^2}$

c. $\frac{\alpha}{\beta}$ d. $\frac{\beta^2}{\alpha}$

10. The oscillation of a body on a smooth horizontal surface is represented by the equation,

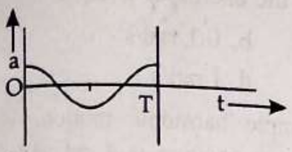
$X = A\cos(\omega t)$

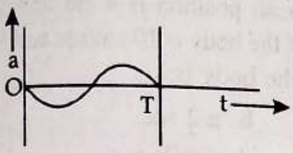
Where,

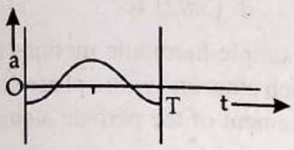
X = displacement at time t

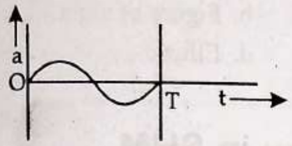
ω = frequency of oscillation

Which one of the following graphs shows correctly the variation a with t? *(2014)*

a. 

b. 

c. 

d. 

Here a = acceleration at time t

T = time period

11. Two particles are oscillating along two close parallel straight lines side by side, with the same frequency and amplitudes. They pass each other, moving in opposite directions when their displacement is half of the amplitude. The mean positions of the two particles lie on a straight line perpendicular to paths of the two particles. The phase difference is: *(2011 Mains)*

a. 0 b. $\frac{2\pi}{3}$

c. π d. $\frac{\pi}{6}$

12. Out of the following functions representing motion of a particle which represents S.H.M.: *(2011 Pre)*

(i) $y = \sin\omega t - \cos\omega t$

(ii) $y = \sin^3\omega t$

(iii) $y = 5\cos\left(\frac{3\pi}{4} - 3\omega t\right)$

(iv) $y = 1 + \omega t + \omega^2 t^2$

a. Only (i)

b. Only (iv) does not represent SHM

c. Only (i) and (iii)

d. Only (i) and (ii)

13. A particle moves in a circle of radius 5 cm with constant speed and time period 0.2π. The acceleration of the particle is: *(2011 Pre)*

a. 15 m/s^2 b. 25 m/s^2

c. 36 m/s^2 d. 5 m/s^2

14. The displacement of a particle along the x-axis is given by $x = a\sin^2\omega t$. The motion of the particle corresponds to: *(2010 Pre)*

a. Simple harmonic motion of frequency $\omega/2\pi$

b. Simple harmonic motion of frequency ω/π

c. Simple harmonic motion of frequency $3\omega/2\pi$

d. Non simple harmonic motion

15. Which one of the following equations of motion represents simple harmonic motion? *(2009)*

a. Acceleration = $-k(x)$

b. Acceleration = $k(x + a)$

c. Acceleration = kx

d. Acceleration = $-k_0x + k_1x^2$

where k, k_0, k_1 and a are all positive.

16. A simple pendulum performs simple harmonic motion about $x = 0$ with an amplitude a and time period T. The speed of pendulum at $x = a/2$ will be: *(2009)*

a. $\frac{\pi a}{T}$ b. $\frac{3\pi^2 a}{T}$

c. $\frac{\pi a\sqrt{3}}{T}$ d. $\frac{\pi a\sqrt{3}}{2T}$

17. A point performs simple harmonic oscillation of period T and the equation of motion is given by $x = a\sin(\omega t + \pi/6)$. After the elapse of what fraction of the time period the velocity of the point will be equal to half of its maximum velocity? *(2008)*

a. $\frac{T}{12}$ b. $\frac{T}{8}$

c. $\frac{T}{6}$ d. $\frac{T}{3}$

18. Two Simple Harmonic Motions of angular frequency 100 rads^{-1} and 1000 rads^{-1} have the same displacement amplitude. The ratio of their maximum accelerations is: *(2008)*

a. $1 : 10^4$ b. $1 : 10$

c. $1 : 10^2$ d. $1 : 10^3$

19. A particle executes simple harmonic oscillation with an amplitude *a*. The period of oscillation is T. The minimum time taken by the particle to travel half of the amplitude from the equilibrium position is: *(2007)*

a. T/8 b. T/12

c. T/2 d. T/4

20. The phase difference between the instantaneous velocity and acceleration of a particle executing simple harmonic motion is: *(2007)*

a. π b. 0.707π

c. Zero d. 0.5π

21. A rectangular block of mass m and area of cross-section. A floats in a liquid of density ρ. If it is given a small vertical displacement from equilibrium it undergoes with a time period T, then *(2006)*

a. $T \propto \frac{1}{\sqrt{m}}$ b. $T \propto \sqrt{\rho}$

c. $T \propto \frac{1}{\sqrt{A}}$ d. $T \propto \frac{1}{\rho}$

22. A particle executing simple harmonic motion of amplitude 5 cm has maximum speed of 31.4 cm/s. The frequency of its oscillation is: *(2005)*

a. 2 Hz b. 1.5 Hz
c. 0.5 Hz d. 1 Hz

23. The circular motion of a particle with constant speed is: *(2005)*

a. Simple harmonic but not periodic
b. Periodic and simple harmonic
c. Neither periodic nor simple harmonic
d. Periodic but not simple harmonic

24. Which one of the following statements is true for the speed 'v' and the acceleration 'a' of a particle executing simple harmonic motion? *(2004)*

a. Value of 'a' is zero, whatever may be the value of 'v'
b. When 'v' is zero, 'a' is zero
c. When 'v' is maximum, 'a' is zero
d. When 'v' is maximum, 'a' is maximum

25. The equations of two waves given as $x = a\cos(\omega t + \delta)$ and $y = a\cos(\omega t + \alpha)$, Where $\delta = \alpha + \pi/2$, then resultant wave represent: *(2000)*

a. a circle (c.w) b. a circle (a.c.w)
c. an Ellipse (c.w) d. an ellipse (a.c.w)

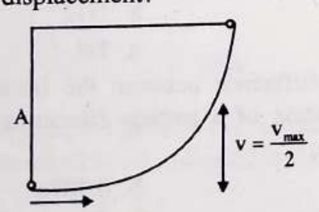
26. If time of mean position from amplitude (extreme) position is 6s. Then the frequency of S.H.M. will be: *(1998)*

a. 0.01 Hz b. 0.02 Hz
c. 0.03 Hz d. 0.04 Hz

27. Two S.H.M.s with same amplitude and time period, when acting together in perpendicular directions with a phase difference of $\pi/2$, give rise to: *(1997)*

a. Straight motion
b. Elliptical motion
c. Circular motion
d. None of these.

28. A particle starts with S.H.M. from the mean position as shown in the figure. Its amplitude is A and its time period is T. At one time, its speed is half that of the maximum speed. What is this displacement? *(1996)*



a. $\frac{2A}{\sqrt{3}}$ b. $\frac{3A}{\sqrt{2}}$
c. $\frac{\sqrt{2}A}{3}$ d. $\frac{\sqrt{3}A}{2}$

29. A particle executes S.H.M. along x-axis. The force acting on it is given by: *(1994, 88)*

a. $A\cos(kx)$ b. Ae^{-kx}
c. Akx d. $-Akx$

30. A simple harmonic oscillator has an amplitude A and time period T. The time required by it to travel from X = A to X = A/2 is: *(1992)*

a. T/6 b. T/4
c. T/3 d. T/2

31. If a simple harmonic oscillator has got a displacement of 0.02 m and acceleration equal to 2 m/s^2 at any time, the angular frequency of the oscillator is equal to: *(1992)*

a. 10 rad/s b. 0.1 rad/s
c. 100 rad/s d. 1 rad/s

32. A body is executing simple harmonic motion. When the displacements from the mean position is 4 cm and 5 cm the corresponding velocities of the body is 10 cm/sec and 8 cm/sec. Then the time period of the body is: *(1991)*

a. 2π sec b. $\pi/2$ sec
c. π sec d. $(3\pi/2)$ sec

33. The composition of two simple harmonic motions of equal periods at right angle to each other and with a phase difference of π results in the displacement of the particle along: *(1990)*

a. Circle b. Figure of eight
c. Straight line d. Ellipse

Energy in SHM (P.E., K.E. and T.E.)

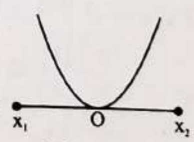
34. A body is executing simple harmonic motion with frequency 'n', the frequency of its potential energy is: *(2021)*

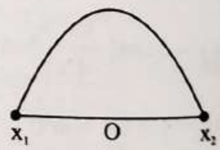
a. 2n b. 3n
c. 4n d. n

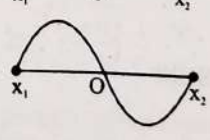
35. The particle executing simple harmonic motion has a kinetic energy $K_0 \cos^2 \omega t$. The maximum values of the potential energy and the total energy are respectively: *(2007)*

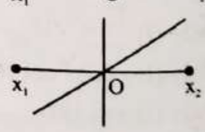
a. $K_0/2$ and K_0 b. K_0 and $2K_0$
c. K_0 and K_0 d. 0 and $2K_0$

36. A particle of mass m oscillates with simple harmonic motion between points x_1 and x_2, the equilibrium position being O. Its potential energy is plotted. It will be as given below in the graph: *(2003)*

a. 

b. 

c. 

d. 

37. The potential energy of a simple harmonic oscillator when the particle is half way to its end point is: *(2003)*

a. $\frac{2}{3}E$ b. $\frac{1}{8}E$
c. $\frac{1}{4}E$ d. $\frac{1}{2}E$

38. Displacement between max. P.E. position and max. K.E. position for a particle excuting simple harmonic motion is: *(2002)*

a. $\pm\frac{a}{2}$ b. + a
c. ± a d. – 1

39. The total energy of particle performing S.H.M. depend on: *(2001)*

a. K, a, m b. K, a
c. K, a, x d. K, x

40. A linear harmonic oscillator of force constant 2×10^6 N/m and amplitude 0.01 m has a total mechanical energy of 160 J. Its *(1996)*

a. P.E. is 160 J b. P.E. is zero
c. P.E. is 100 J d. P.E. is 120 J

41. In a simple harmonic motion, when the displacement is one-half of the amplitude, what fraction of the total energy is kinetic? *(1995)*

a. 1/2 b. 3/4
c. Zero d. 1/4

42. A body executes simple harmonic motion with an amplitude A. At what displacement from the mean position is the potential energy of the body is one fourth of its total energy? *(1992)*

a. A/4 b. A/2
c. 3A/4 d. Some other fraction of A

Simple Pendulum and Loaded Springs

43. Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their means position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the means position is : *(2022)*

a. 8 b. 11
c. 9 d. 10

44. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is: *(2021)*

a. 6.28 s b. 3.14 s
c. 0.628 s d. 0.0628 s

45. A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s² at a distance of 5 m from the mean position. The time period of oscillation is: *(2018)*

a. 2s b. π s
c. 2π s d. 1 s

46. A spring of force constant k is cut into lengths of ratio 1 : 2 : 3. They are connected in series and the new force constant is K′. Then they are connected in parallel and force constant is K″. Then K′ : K″ is: *(2017-Delhi)*

a. 1 : 9 b. 1 : 11
c. 1 : 14 d. 1 : 6

47. A body of mass m is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is slightly pulled down and released, it oscillates with a time period of 3 s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of m in kg is: *(2016 - II)*

a. $\frac{16}{9}$ b. $\frac{9}{16}$
c. $\frac{3}{4}$ d. $\frac{4}{3}$

48. The period of oscillation of a mass M suspended from a spring of negligible mass is T. If along with it another mass M is also suspended, the period of oscillation will now be: *(2010 Pre)*

a. $\sqrt{2}T$
b. T
c. $\frac{T}{\sqrt{2}}$
d. 2T

49. A mass of 2.0 kg is put on a flat pan attached to a vertical spring fixed on the ground as shown in the figure. The mass of the spring and the pan is negligible. When pressed slightly and released the mass executes a simple harmonic motion. The spring constant is 200 N/m. What should be the minimum amplitude of the motion so that the mass gets detached from the pan (take g = 10 m/s²)? *(2007)*

a. 10.0 cm
b. Any value less than 12.0 cm
c. 4.0 cm
d. 8.0 cm

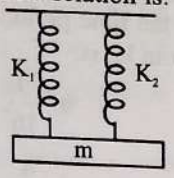
50. Two springs of spring constants k_1 and k_2 are joined in series. The effective spring constant of the combination is given by: *(2004)*

a. $\frac{(k_1+k_2)}{2}$
b. $k_1 + k_2$
c. $\frac{k_1 k_2}{(k_1+k_2)}$
d. $\sqrt{k_1 k_2}$

51. The time period of a mass suspended from a spring is T. If the spring is cut into four equal parts and the same mass is suspended from one of the parts, then the new time period will be: *(2003)*

a. $\frac{T}{4}$ b. T
c. $\frac{T}{2}$ d. 2T

52. A mass is suspended separately by two different springs in successive order then time period is t_1 and t_2 respectively. If it is connected by both spring as shown in figure then time period is t_0. The correct relation is: *(2002)*



a. $t_0^2 = t_1^2 + t_2^2$ b. $t_0^{-2} = t_1^{-2} + t_2^{-2}$
c. $t_0^{-1} = t_1^{-1} + t_2^{-1}$ d. $t_0 = t_1 + t_2$

53. The bob of simple pendulum having length is displaced from mean position to an angular position θ with respect to vertical. If it is released, then velocity of bob at lowest position: *(2000)*

a. $\sqrt{2g\,(1-\cos\theta)}$ b. $\sqrt{2g\ell(1+\cos\theta)}$
c. $\sqrt{2g\ell\cos\theta}$ d. $\sqrt{2g\ell}$

54. Two sphrical bob of masses M_A and M_B are hung vertically from two strings of length ℓ_A and ℓ_B respectively. They are executing SHM with frequency relation $f_A = 2f_B$, Then: *(2000)*

a. $\ell_A = \frac{\ell_B}{4}$ b. $\ell_A = 4\ell_B$
c. $\ell_A = 2\ell_B$ & $M_A = 2M_B$ d. $\ell_A = \frac{\ell_B}{2}$ & $M_A = \frac{M_B}{2}$

55. A spring elongated by length L when a mass M is suspended to it. Now a tiny mass m is attached and then released, its time period of oscillation is: *(1999)*

a. $2\pi\sqrt{\frac{(M+m)\ell}{Mg}}$ b. $2\pi\sqrt{\frac{m\ell}{Mg}}$
c. $2\pi\sqrt{L/g}$ d. $2\pi\sqrt{\frac{M\ell}{(m+M)g}}$

56. Frequency of simple pendulum in a free falling lift is: *(1999)*

a. Zero b. Infinite
c. Can't be say d. Finite

57. Two pendulums suspended from same point having length 2 m and 0.5 m. If they displaced slightly and released then they will be in same phase, when small pendulum will have completed: *(1998)*

a. 2 oscillation b. 4 oscillation
c. 3 oscillation d. 5 oscillation

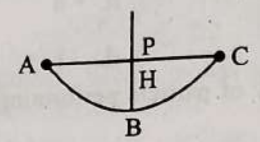
58. If the frequency of a spring is n after suspending mass M, now 4M mass is suspended from spring then the frequency will be: *(1998)*

a. 2n b. n/2
c. n d. None of the above

59. If the length of a simple pendulum is increased by 2%, then the time period: *(1997)*

a. Increases by 1% b. Decreases by 1%
c. Increases by 2% d. Decreases by 2%

60. A simple pendulum with a bob of mass m oscillates from A to C and back to A such that PB is H. If the acceleration due to gravity is g, then the velocity of the bob as it passes through B is: *(1995)*



a. *mgH* b. $\sqrt{2gH}$
c. Zero d. 2gH

61. A body of mass 5 kg hangs from a spring and oscillates with a time period of 2π seconds. If the ball is removed, the length of the spring will decrease by: *(1994)*

a. g/k metres b. k/g metres
c. 2π metres d. g metres

62. A seconds pendulum is mounted in a rocket. Its period of oscillation will decrease when rocket is: *(1994)*

a. Moving down with uniform acceleration
b. Moving around the earth in geostationary orbit
c. Moving up with uniform velocity
d. Moving up with uniform acceleration.

63. A loaded vertical spring executes S.H.M. with a time period of 4 sec. The difference between the kinetic energy and potential energy of this system varies with a period of: *(1994)*

a. 2 sec b. 1 sec
c. 8 sec d. 4 sec

64. A simple pendulum is suspended from the roof of a trolley which moves in a horizontal direction with an acceleration *a*, then the time period is given by $T = 2\pi\sqrt{(l/g)}$, where g is equal to: *(1991)*

a. g b. g − a
c. g + a d. $\sqrt{(g^2 + a^2)}$

65. The angular velocity and the amplitude of a simple pendulum is ω and *a* respectively. At a displacement *x* from the mean position if its kinetic energy is *T* and potential energy is *V*, then the ratio of *T* to *V* is: *(1991)*

a. $\frac{(a^2 - x^2\omega^2)}{x^2\omega^2}$ b. $\frac{x^2\omega^2}{(a^2 - x^2\omega^2)}$
c. $\frac{(a^2 - x^2)}{x^2}$ d. $\frac{x^2}{(a^2 - x^2)}$

66. A mass *m* is suspended from the two coupled springs connected in series. The force constant for springs are k_1 and k_2. The time period of the suspended mass will be: *(1990)*

a. $T = 2\pi\sqrt{\frac{m}{k_1 - k_2}}$ b. $T = 2\pi\sqrt{\frac{mk_1k_2}{k_1 + k_2}}$
c. $T = 2\pi\sqrt{\frac{m}{k_1 + k_2}}$ d. $T = 2\pi\sqrt{\frac{m(k_1 + k_2)}{k_1k_2}}$

Damped Oscillations, Forced Oscillations and Resonance

67. In case of a forced vibration, the resonance wave becomes very sharp when the: *(2003)*

a. Damping force is small
b. Restoring force is small
c. Applied periodic force is small
d. Quality factor is small

68. When an oscillator completes 100 oscillation its amplitude reduced to $\frac{T}{3}$ of initial value. What will be its amplitude, oscillation when it completes 200 oscillation? *(2002)*

a. $\frac{1}{8}$ b. $\frac{2}{3}$
c. $\frac{1}{6}$ d. $\frac{1}{9}$

69. The amplitude of a S.H.M. reduces to 1/3 in first 20 secs, then in first 40 sec. its amplitude becomes: *(1999)*

a. $\frac{1}{3}$
b. $\frac{1}{9}$
c. $\frac{1}{27}$
d. $\frac{1}{\sqrt{3}}$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	b	b	d	d	b	b	a	a	c	b	c	d	d	a	c	a
18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
c	b	d	c	d	d	c	b	d	c	d	d	a	a	c	c	a
35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
c	a	c	c	b	c	b	b	b	c	b	b	b	a	a	c	c
52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
b	a	a	a	a	a	b	a	b	d	d	a	d	c	d	a	d
69																
b																

Explanations

1. **(d)** Displacement is defined as the distance of the particle from the mean position at that instant.

Displacement (x) equation is

$x = A\sin(\omega t + \phi)$, ...(i)

where A is amplitude and ω is angular velocity.

$$\frac{dx}{dt} = A\omega\cos(\omega t + \phi)$$

$$\text{Acceleration } a = \frac{d\,x}{dt} = -\omega\ A\sin(\omega t + \phi)$$

$a = \omega^2 A\sin(\omega t + \phi + \pi)$ ...(ii)

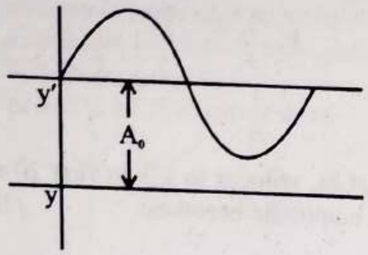
From equation (i) and (ii) phase difference between displacement and acceleration is π.

2. **(b)** For periodic function F(t) = F(t + T), where T is time period of function.

sin ωt and cos ωt both are periodic function and their period is also same so its resultant is also periodic.

$\therefore \sin(\omega t + 2\pi) + \cos(\omega t + 2\pi) = \sin\omega t + \text{cost}\ \omega t$

3. **(b) Simple Hormonic Motion:** If the restoring force or torque acting on the body in oscillatory motion is directly proportional to the displacement of body/particle and is always directed towards equilibrium position then the motion is called SHM.



$y = A_0 + A \sin \omega t + B \cos \omega t$

Here, two waves are superimposing therefore the phase difference is 90°.

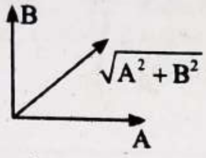
This S.H.M can be reduced to

$y' = y - A_0 = A \sin \omega t + B \cos \omega t$

Hence, resultant amplitude

$R = \sqrt{A^2 + B^2 + 2AB\cos 90°}$

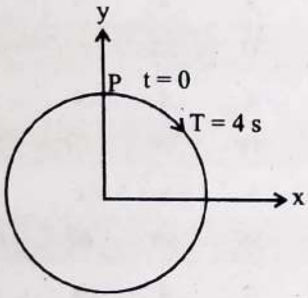
$= \sqrt{A^2 + B^2}$



4. **(d)** As in one complete vibration, the particle comes back to the initial position. So the displacement of the particle is zero. Hence, average velocity in one complete vibration

$$= \frac{\text{Displacement}}{\text{Time interval}} = \frac{y_f - y_i}{T} = 0$$

5. **(d)** At t = 0, y displacement is maximum, so equation will be cosine function



As, T = 4 s

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2} \text{rad / s}$$

$y = a \cos\omega t$, where a is the radius of the circle.

$$y = 3\cos\frac{\pi}{2}t$$

6. **(b)** Given: A = 3 cm, x = 2 cm

a = v

For particle undergoing SHM, the velocity of the particle at any position X is given by

$v = \omega\sqrt{A^2 - X^2}$

and acceleration $a = \omega^2 x$

$\Rightarrow \omega^2 x = \omega\sqrt{A^2 - x^2} \Rightarrow \omega x = \sqrt{A^2 - x^2}$

$\Rightarrow 2\omega = \sqrt{3^2 - 2^2} \Rightarrow 2\omega = \sqrt{5}$

$$\omega = \frac{\sqrt{5}}{2} \Rightarrow \frac{2\pi}{T} = \frac{\sqrt{5}}{2}$$

Time period for the given particle

$$T = \frac{4\pi}{\sqrt{5}}$$

7. **(b)** Given: $y_1 = a \sin \omega t$

$$y_2 = b \cos \omega t = b \sin\left(\omega t + \frac{\pi}{2}\right)$$

Since the frequencies for both S.H.M. are same, resultant motion will be S.H.M. Now

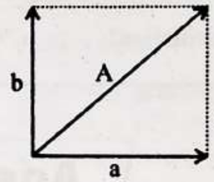
Amplitude $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$

Here $A_1 = a$, $A_2 = b$, $\phi = \frac{\pi}{2}$

So $A = \sqrt{a^2 + b^2}$

OR

$y_2 = b \cos \omega t$



$A = \sqrt{a^2 + b^2}$

8. **(a)** For particle undergoing S.H.M.

velocity $v = \omega\sqrt{A^2 - x^2}$

So, $v_1 = \omega\sqrt{A^2 - x_1^2}$ & $v_2 = \omega\sqrt{A^2 - x_2^2}$

$v_1^2 = \omega^2(A^2 - X_1^2)$ ...(i)

$v_2^2 = \omega^2(A^2 - X_2^2)$ ...(ii)

Subtracting (ii) from (i)

$v_1^2 - v_2^2 = \omega^2(X_2^2 - X_1^2)$

$$\Rightarrow \omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}} = \frac{2\pi}{T} \Rightarrow T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

9. **(a)** For S.H.M.

Acceleration will be maximum when $X = \pm A$ (towards mean position)

Maximum acceleration $a_{max} = \omega^2 A = \alpha$

Maximum velocity $V_{max} = \omega A = \beta$

$$\frac{\alpha}{\beta} = \frac{A\omega^2}{\omega A} = \omega = \frac{2\pi}{T}$$

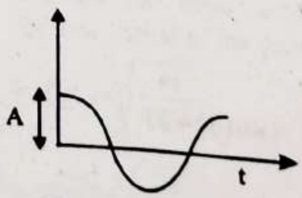
$\Rightarrow$ Time period of vibration $T = \frac{2\pi\beta}{\alpha}$

10. **(c)** Displacement, $x = A \cos(\omega t)$

at t = 0

$x = A \cos(\omega \times 0)$

$x = A$

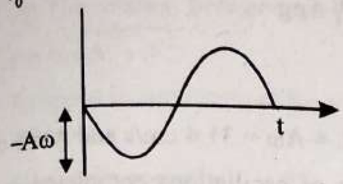


$$v = \frac{dx}{dt} = -A\omega\sin\omega t$$

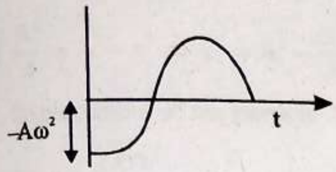
on comparing this with

$v = v_0 \sin \omega t$ [$\because$ sin function starts from origin]

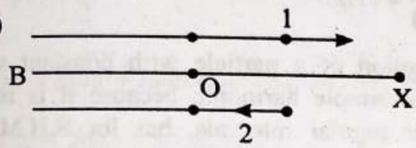
$v_0 = -A\omega$



$a = \frac{dv}{dt} = -A\omega^2 \cos\omega t$ [$\because$ cos function do not starts from origin]



11. (b)



$x = A\sin(\omega t + \phi)$

Let the displacement of the two particles 1 and 2 be given by

$x_1 = A \sin(\omega t + \phi_1)$

$x_2 = A \sin(\omega t + \phi_2)$

$x_1 = \frac{A}{2}$ [$\because$ displacement always measured from mean position]

$\frac{A}{2} = A\sin(\omega t + \phi)$

$\Rightarrow \omega t + \phi_1 = \frac{\pi}{6}$

For particle 2

$x_2 = \frac{A}{2}$

As particle 2 is moving towards mean position at displacement $\frac{A}{2}$

$\frac{A}{2} = A\sin(\omega t + \phi_2)$

$\Rightarrow \omega t + \phi_2 = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

Phase difference $= \phi_2 - \phi_1$

$= \frac{5\pi}{6} - \omega t - \frac{\pi}{6} + \omega t = \frac{2\pi}{3}$

12. (c) The necessary and sufficient condition for SHM is $F = -kx$, where k = positive constant or restoring force constant

x = displacement from mean position

(i) $y = \sin\omega t - \cos\omega t$

$= \sqrt{2}\left[\frac{1}{\sqrt{2}}\sin\omega t - \frac{1}{\sqrt{2}}\cos\omega t\right]$

$= \sqrt{2}\sin\left(\omega t - \frac{\pi}{4}\right)$

This is in the form of $Y = A\sin(\omega t \pm \phi)$ So, this is S.H.M.

(ii) $y = \sin^3\omega t$

$= \frac{1}{4}[3\sin\omega t - \sin 3\omega t]$ [$\therefore \sin 3x = 3\sin x - 4\sin^3 x$]

This is periodic motion with 2 different frequency

(iii) $y = 5\cos\left(\frac{3\pi}{4} - 3\omega t\right)$

$= 5\cos\left(3\omega t - \frac{3\pi}{4}\right)$ $(\because \cos(-\theta) = \cos\theta)$

It is S.H.M. with time period $T = \frac{2\pi}{3\omega}$

(iv) $y = 1 + \omega t + \omega^2 t^2$

It is non-periodic motion y will increase monotonously with time So, (i) and (iii) represents S.H.M.

13. (d) Given: Radius $r = 5$ cm $= 5 \times 10^{-2}$m

Time period $T = 0.2\pi$s

When a particle moves in a circular motion, the acceleration is due to centripetal force (directed towards the centre).

$\Rightarrow$ centripetal acceleration $a_c = r\omega^2$

where ω = angular velocity $= \frac{2\pi}{T} = \frac{2\pi}{0.2\pi} = 10$ rad / s

Therefore $a_c = 0.05 \times 10 = 5$m/s^2

14. (d) Necessary condition for simple harmonic motion is that its acceleration should be directly proportional to displacement.

$x = a\sin^2\omega t = \frac{a}{2}\left(\frac{1-\cos 2\omega t}{2}\right)$ [$\therefore \cos 2\theta = 1 - 2\sin^2\theta$]

$v = \frac{dx}{dt} = a\omega \sin 2\omega t$

acceleration $a = \frac{dv}{dt} = 2\omega^2 a\cos 2\omega t$

Hence the condition is not satisfied.

15. (a) For a motion to be S.H.M.

Force directly proportional to $-x$

i.e., $F = -kx$

16. (c) Using equation speed of oscillation.

$v = \omega\sqrt{a^2 - x^2}$

At $x = \frac{a}{2}$

$\Rightarrow v = \frac{2\pi}{T}\sqrt{a^2 - \frac{a^2}{4}}$

$= \frac{2\pi}{T} \times \sqrt{\frac{3a^2}{4}}$

$= \frac{2\pi}{T} \times \frac{\sqrt{3a^2}}{2} = \frac{\sqrt{3}\pi a}{T}$ m/s

17. (a) Given displacement of the particle

$x = a\sin\left(\omega t + \frac{\pi}{6}\right)$

$v = \frac{dx}{dt} = a\omega\cos\left(\omega t + \frac{\pi}{6}\right)$

According to question, velocity is equal to half of maximum velocity.

$$\Rightarrow \frac{a\omega}{2} = a\omega\cos\left(\omega t + \frac{\pi}{6}\right) \Rightarrow \frac{1}{2} = \cos\left(\omega t + \frac{\pi}{6}\right)$$

$$\omega t + \frac{\pi}{6} = \frac{\pi}{3} \Rightarrow \omega t = \frac{\pi}{6}$$

$$\Rightarrow \frac{2\pi}{T}t = \frac{\pi}{6} \Rightarrow t = \frac{T}{12}$$

18. (c) Maximum acceleration (a) $= \omega^2 A$

Where A is the displacement amplitude

$$\frac{a_1}{a_2} = \frac{\omega_1^2}{\omega_2^2} \Rightarrow \frac{a_1}{a_2} = \frac{(100)^2}{(1000)^2}$$

$$\Rightarrow \frac{a_1}{a_2} = \left(\frac{1}{10}\right)^2 \Rightarrow a_1 : a_2 = 1 : 10^2$$

19. (b)

$-a \quad 0 \quad +a$

If particle starts motion from equilibrium position

$x = a\sin\omega t$

$$\Rightarrow \frac{a}{2} = a\sin\omega t \Rightarrow \omega t = \frac{\pi}{6} \qquad \left[\text{Since } \sin\frac{\pi}{6} = \frac{1}{2}\right]$$

$$\Rightarrow \left(\frac{2\pi}{T}\right)t = \frac{\pi}{6} \Rightarrow t = \frac{T}{12}$$

Particle will take time $\frac{T}{12}$ to travel half of the amplitude from the equilibrium position.

20. (d) General equation for particle displacement

$x = x_0 \sin\omega t$

Velocity for particle

$$v = \frac{dx}{dt} = \omega v_0 \cos\omega t$$

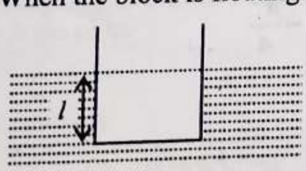
Acceleration of a particle

$$a = \frac{dv}{dt} = -a_0\omega^2 \sin\omega t$$

$$a = +\omega^2 a_0 \cos\left[\frac{\pi}{2} + \omega t\right]$$

Phase difference $(\Delta\phi) = \frac{\pi}{2} = 0.5\,\pi$

21. (c) Let l be the length of block immersed in liquid as shown in the figure. When the block is floating.



$\therefore\ mg = Al\rho g$

If the block is given vertical displacement y then the effective restoring force is

$F = -[A(l+y)\rho g - mg] = -[A(l+y)\rho g - Al\rho g] = -A\rho gy$

Restoring force $= -[A\rho g]y$. As this F is directed towards its equilibrium position of block, so it the block is left free, it will execute simple harmonic motion.

Here inertia factor = mass of block = m

Spring factor $= A\rho g$

$$\therefore \text{ Time period } = T = 2\pi\sqrt{\frac{m}{A\rho g}}$$

i.e., $T \propto \frac{1}{\sqrt{A}}$

22. (d) Max^m speed in S.H.M. $= A\omega = 31.4$ cm/s and $A = 5$ cm

Frequency (f) : Number of oscillations completed in unit time interval is called frequency of oscillations.

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

Hence $A\omega = A(2\pi f)$

$\Rightarrow 31.4 = 5 \times 2 \times 3.14 \times f$

$$\Rightarrow f = \frac{31.4}{10 \times 3.14} = 1\text{Hz}.$$

23. (d) Circular motion of a particle with constant speed is periodic but not simple harmonic because it is repeating its motion after regular intervals, but for S.H.M. Force $\propto -$ y(displacement).

24. (c) In SHM, velocity $v = \omega\sqrt{A^2 - X^2}$, acceleration $a = -\omega^2 X$

where A is amplitude and X is displacement.

when $X = 0$, $V_{max} = \omega A$ and $a = 0$

But when $X = A$, $V = 0$ and $a_{max} = -\omega^2 A$

25. (b) Given $x = a\cos(\omega t + \delta)$

and $y = a\cos(\omega t + \alpha)$ ...(i)

where, $\delta = \alpha + \pi/2$

$\therefore\ x = a\cos(\omega t + \alpha + \pi/2)$

$= -a\sin(\omega t + \alpha)$ ...(ii)

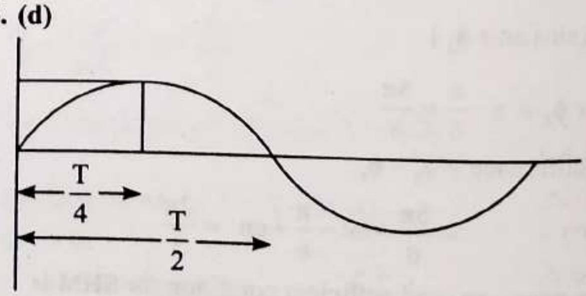
Given the two waves are acting in perpendicular direction with the same frequency and phase difference $\pi/2$.

From equations (i) and (ii)

$x^2 + y^2 = a^2$

which represents the equation of a circle.

26. (d)



$$\frac{T}{4} = 6\,\text{sec} \Rightarrow T = 24\text{ sec}$$

$$\text{Frequency} = \frac{1}{T} = \frac{1}{24}\text{Hz} = 0.04\,\text{Hz}$$

27. (c) $x = a\sin\omega t$

$y = a\sin(\omega t + \pi/2) = a\cos\omega t$ $[\therefore \sin(90 + \theta) = \cos\theta]$

$y^2 = a^2\cos^2\omega t$

$= a^2(1 - \sin^2\omega t)$ $[\therefore \cos^2\theta + \sin^2\theta = 1]$

$y^2 = a^2 - a^2 \sin^2\omega t$
$= a^2 - x^2$
$y^2 + x^2 = a^2$, represent equation of circle.

28. (d) The relation between speed and displacement in SHM is $v = \omega\sqrt{A^2 - X^2}$

As speed is maximum at the mean position and displacement is zero.

Maximum speed, $v_{max} = A\omega$

According to question, $\frac{v_{max}}{2} = \frac{A\omega}{2} = \omega\sqrt{A^2 - y^2}$

$\frac{A^2}{4} = A^2 - y^2 \Rightarrow y^2 = A^2 - \frac{A^2}{4}$

displacement of the particle from mean position is

$\Rightarrow y = \frac{\sqrt{3}A}{2}$

29. (d) For simple harmonic motion, $\frac{d^2x}{dt^2} \propto -x$.

Force = mass × acceleration

Thus $F \propto -x$

Therefore force acting on the particle = – Akx.

where A and K are positive constant.

30. (a) Amplitude is the maximum value of displacement of the particle from its equilibrium position.

Amplitude = $\frac{1}{2}$[distance between extreme points]

For S.H.M. $x = A\sin\left(\frac{2\pi}{T}t\right)$

When $x = A$, $A = A\sin\left(\frac{2\pi}{T}t\right)$

$\therefore \sin\left(\frac{2\pi}{T}\cdot t\right) = 1 \Rightarrow \sin\left(\frac{2\pi}{T}\cdot t\right) = \sin\left(\frac{\pi}{2}\right) \Rightarrow t = (T/4)$

When $x = \frac{A}{2}, \frac{A}{2} = A\sin\left(\frac{2}{T}\cdot t\right)$

Or $\sin\frac{\pi}{6} = \sin\left(\frac{2\pi}{T}t\right)$ or $t = (T/12)$

Therefore, time taken to travel from x = A to x = A/2 = T/4 – T/12 = T/6

31. (a) Given: $a = 2\text{ms}^{-2}$

$y = 0.02$ m

Acceleration = $-\omega^2$ displacement

$\omega^2 = \frac{\text{acceleration}}{\text{displacement}} = \frac{2.0}{0.02}$

$\omega^2 = 100$ or $\omega = 10$ rad/s

32. (c) In case of simple harmonic motion

velocity $v = \omega\sqrt{a^2 - x^2}$ at displacement x.

$10 = \omega\sqrt{a^2 - 16}$ ...(i)

$8 = \omega\sqrt{a^2 - 25}$ ...(ii)

$\frac{100}{\omega^2} = a^2 - 16$ ...(iii)

$\frac{64}{\omega^2} = a^2 - 25$ ...(iv)

Subtracting equation (iii) from (iv) we get,

$\frac{36}{\omega^2} = 9 \Rightarrow \omega = 2$ rad/s

Or $T = \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$ sec

33. (c) $x = a\sin(\omega t)$ ...(i)

And $y = b\sin(\omega t + \pi) = -b\sin\omega t$. ...(ii)

By dividing eq (i) and eq (ii)

$\frac{x}{y} = \frac{-a}{b}$

$\Rightarrow \frac{x}{a} = -\frac{y}{b}$

As equation of straight line $y = mx + c$

where m = slope

c = constant

Thus $y = \frac{-bx}{a}$ is a straight line equation.

34. (a) A particle executing SHM has displacement equation,

$x = A\sin(\omega t + \phi)$ ...(i)

where, angular frequency $\omega = 2\pi n$

Restoring force: When a system is displaced from its equilibrium position, then a force acts on it in such a direction so as to restore the system back to its equilibrium position. It is directed opposite to the displacement.

Potential Energy: Work done by the restoring force while displacing the particle from the mean position (x = 0) to x = x.

A particle executing SHM has potential energy,

$U = \frac{1}{2}Kx^2 \Rightarrow U = \frac{1}{2}KA^2\sin^2(\omega t + \phi)$

$U = \frac{1}{4}KA^2 - \frac{1}{4}KA^2\cos(2\omega t + \phi)$ ...(ii)

$\left[\because \sin^2\theta = \frac{1 - \cos 2\theta}{2}\right]$

It is clear from eq (i) and (ii) that

For $x = A\sin(\omega t + \phi)$

$T_1 = \frac{2\pi}{\omega} \Rightarrow$ frequency $n = \frac{\omega}{2\pi}$

For $x^2 = A^2\sin^2(\omega t + \phi)$

$T_2 = \frac{2\pi}{2\omega} \Rightarrow$ frequency $n' = \frac{\omega}{\pi}$

$\therefore n' = 2n$

Also, for each oscillation, the particle reaches the maximum point energy twice. Thus frequency becomes 2n.

35. **(c)** $K.E = K_0 \cos^2\omega t$

As maximum value of $\cos^2\omega t = 1$

$\therefore K_{max} = K_0$

As total energy is constant, so when potential energy is zero Kinetic energy will be maximum.

Hence at P.E = 0, $K.E_{max} = k_0$

So total energy $T.E = K.E._{max} = k_0$

Similarly when K.E = 0, the total energy is given as

$\Rightarrow$ T.E = P.E

Hence $P.E = K_0$ [Because the total energy of SHM is constant]

Therefore, the maximum potential energy and maximum total energy will be k_0 and k_0.

36. **(a)** Simple Hormonic Motion

F = – Kx [Hooke's law in a spring]

As, we know $F = -\frac{dU}{dx} \Rightarrow \int dU = -\int F dx$

$\int dU = \int_0^x Kx\,dx$ or $= \frac{1}{2}Kx^2 + c$

Now at x = 0, $c = U_0 \Rightarrow \frac{1}{2}Kx^2 + U_0$

or $U_0 = 0$, i.e., P.E. at equilibrium position

$\Rightarrow U = \frac{1}{2}Kx^2$

$\therefore$ P.E. $\propto x^2$

At x = 0, P.E. = 0

At $x = \pm A$, $U = U_{max} = \frac{1}{2}Kx^2$

Thus, option a is correct answer which represent equation of parabola.

37. **(c)** P.E. for SHM $= \frac{1}{2}Kx^2$

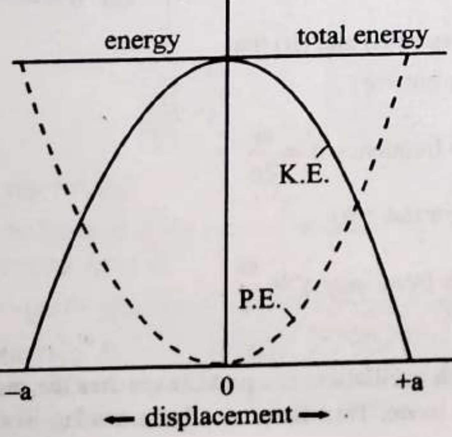
For $x = \frac{a}{2}$

$P.E. = \frac{1}{2}K\left(\frac{a}{2}\right)^2 = \frac{E}{4}$

38. **(c)** Fora SHM the P.E. of the particle is zero at equilibrium position where K.E. is maximum.

Again at the extreme positions for displacement = + a or – a, the K.E. of the particle is zero and it has got maximum PE.

Hence displacement between these two points is + a or – a, where a is the amplitude of the SHM.



39. **(b)** Kinetic energy (K.E) for SHM $= \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - X^2)$

$\Rightarrow \frac{1}{2}m\omega^2(A^2 - X^2) = \frac{1}{2}K(A^2 - X^2)$

potential energy for SHM $= \frac{1}{2}Kx^2$

$\therefore$ Total mechanical energy

= Kinetic energy + potential energy

$= \frac{1}{2}K(A^2 - X^2) + \frac{1}{2}KX^2 = \frac{1}{2}KA^2$

40. **(c)** Force constant (k) = 2×10^6 N/m; Amplitude (x) = 0.01 m and total mechanical energy = 160 J.

Potential energy $= \frac{1}{2}kx^2 = \frac{1}{2} \times (2 \times 10^6) \times (0.01)^2$

= 100 J.

41. **(b)** Displacement $(x) = \frac{a}{2}$. Total energy $= \frac{1}{2}m\omega^2 a^2$ and kinetic energy when displacement is $(x) = \frac{1}{2}m\omega^2(a^2 - x^2)$

$= \frac{1}{2}m\omega^2\left(a^2 - \left(\frac{a}{2}\right)^2\right) = \frac{3}{4}\left(\frac{1}{2}m\omega^2 a^2\right)$.

Therefore fraction of the total energy at

$x = \frac{\frac{3}{4}\left(\frac{1}{2}m\omega^2 a^2\right)}{\frac{1}{2}m\omega^2 a^2} = \frac{3}{4}$.

42. **(b)** According to the given condition in question

$P.E = \frac{1}{4}(T.E)$

$\therefore \frac{1}{2}M\omega^2 x^2 = \frac{1}{4}\left(\frac{1}{2}M\omega^2 A^2\right)$

where x = displacement

$x^2 = \frac{A^2}{4} \Rightarrow x = \frac{A}{2}$

43. **(b)** Let the time period of pendulums having longer length and shorter length arc T_ℓ and T_s.

Given: length of long pendulum = 121 cm

length of short pendulum = 100 cm

$\Rightarrow n(T)_\ell = (n + 1)T_s$

$(n)2\pi\sqrt{\frac{1.21}{g}} = (n+1)2\pi\sqrt{\frac{1}{g}}$

(n)(1.1) = (n + 1)

n = 10

No. of oscillation of smaller one

= n + 1 = 11

44. **(c)** Hooke's law = It states that within certain limits, the force required to stretch an elastic object is directly proportional to the extension of the spring. F = –kx where x is the length of extension/ compression and K is a constant of proportionality (spring constant).

Given: x = 5 cm, F = 10 N

$$\Rightarrow K = \frac{10}{5\times10^{-2}} = \frac{10\times10^2}{5} \Rightarrow K = 200\text{ N/m.}$$

The time period of the oscillations in a spring,

$$T = 2\pi\sqrt{\frac{m}{K}}$$

Given M = 2 kg

$$\Rightarrow T = 2\pi\sqrt{\frac{2}{200}} = 2\pi\sqrt{\frac{1}{100}} \Rightarrow T = \frac{2\pi}{10} = 0.628\text{ sec}$$

45. (b) If a heavy point-mass is suspended by a weightless, inextensible and perfectly flexible string from a rigid support, then this arrangement is called a simple pendulum.

Simple pendulum performs SHM if displacement is very small.

We know that for SHM

Acceleration $a = \omega^2 x$

Given $a = 20\text{ m/s}^2$

$x = 5$ m

$\therefore 20 = \omega^2(5)$

$\omega = 2$ rad/s

and Time – Period for oscillation $T = \dfrac{2\pi}{\omega} = \pi\,\text{sec}$

46. (b) Spring constant $K \propto \dfrac{1}{\ell}$

Where ℓ = natural length of spring

$K = \dfrac{c}{\ell}$ c = constant

It is cut into lengths of ratio 1 : 2 : 3 then ratio of spring constant,

$$\frac{c}{1}:\frac{c}{2}:\frac{c}{3} \Rightarrow \frac{1}{1}:\frac{1}{2}:\frac{1}{3}$$

$K_1 : K_2 : K_3 \Rightarrow 6 : 3 : 2$

Now,

parallel combination

$K'' = 6K + 3K + 2K \Rightarrow K'' = 11K$

series combination

$$\frac{1}{K'} = \frac{1}{6K} + \frac{1}{2K} + \frac{1}{3K}$$

$$\frac{1}{K'} = \frac{1}{6K} + \frac{(3+2)}{6K} \Rightarrow \frac{1}{K'} = \frac{1}{K}$$

$K' = K$

$$\frac{K'}{K''} = \frac{K}{11K} \Rightarrow \frac{K'}{K''} = \frac{1}{11}$$

47. (b) For hanging spring mass system $F = -kx$

where $k = m\omega^2$

$$\Rightarrow \omega = \sqrt{\frac{k}{m}}$$

Time period for SHM $T = \dfrac{2\pi}{\omega} = 2\pi\sqrt{\dfrac{m}{k}}$

When mass is slightly pulled down time period = 3s

$$3 = 2\pi\sqrt{\frac{m}{k}} \quad ...(i)$$

when mass is increased by 1 kg time period = 5s

$$5 = 2\pi\sqrt{\frac{m+1}{k}} \quad ...(ii)$$

From $\dfrac{\text{eq.(i)}^2}{\text{eq.(ii)}^2} \Rightarrow \dfrac{9}{25} = \dfrac{m}{m+1} \Rightarrow m = \dfrac{9}{16}$

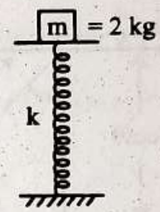
48. (a) Time period of oscillation of spring – mass system

$$T = 2\pi\sqrt{\frac{M}{k}},$$

If another mass M is also suspended with the existing mass, then time – period of oscillation would be

$$T' = 2\pi\sqrt{\frac{2M}{k}} = \sqrt{2}\,T$$

49. (a)



Mass of block = 2 kg

Spring constant = 200 N/M.

Let the minimum amplitude of the motion be a, so that the mass gets detached from the pan.

For given condition,

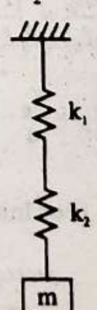
$mg = m\omega^2 a = ka$

$$\Rightarrow a = \frac{mg}{k}$$

$$= \frac{2\times10}{200} = 0.1\text{m} = 10\text{cm}$$

If the amplitude will be less than 10 cm, the restoring force of spring will not be equal to the weight of the block and the mass gets detached from the pan.

50. (c) In series $\dfrac{1}{k_{eff}} = \dfrac{1}{k_1} + \dfrac{1}{k_2} + \ldots$



Hence for two springs of spring constant k_1 and k_2 connected in series $k_{eff} = \dfrac{k_1k_2}{k_1+k_2}$

51. (c) Time period of oscillation $T = 2\pi\sqrt{\frac{M}{k}}$, where m is the mass of the particle suspended with a spring.

As kl = constant

Thus if length is reduced to one-fourth, the stiffness (k) will become four times.

$$\therefore T' = 2\pi\sqrt{\frac{M}{4k}} = \frac{1}{2}2\pi\sqrt{\frac{M}{k}} = \frac{1}{2}T$$

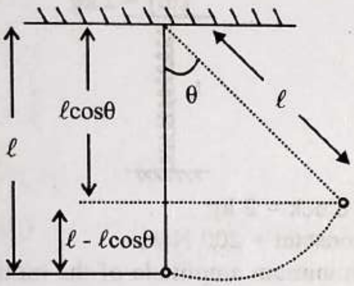
52. (b) $\because T = 2\pi\sqrt{\frac{m}{K}} \Rightarrow K \propto \frac{1}{T^2}$

For parallel combination $K = K_1 + K_2$

$$\frac{1}{t_0^2} = \frac{1}{t_1^2} + \frac{1}{t_2^2} \quad \Rightarrow t_0^{-2} = t_1^{-2} + t_2^{-2}$$

53. (a) When the bob is at equilibrium position, total energy is due to kinetic energy $= \frac{1}{2}mv^2$

Potential energy at extreme position = Kinetic energy at mean position



when bob will be lifted through an angle θ, the height through which it has been lifted is $h = l - l\cos\theta = l(1 - \cos\theta)$

As energy is conserved

$$\Rightarrow mg\ell(1-\cos\theta) = \frac{1}{2}mv^2$$

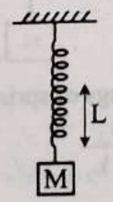
$$v = \sqrt{2g\ell(1-\cos\theta)}$$

54. (a) Frequency of oscillation where g is acceleration due to gravity $f = \frac{1}{2\pi}\sqrt{\frac{g}{\ell}} \Rightarrow f \propto \frac{1}{\sqrt{\ell}}$

$$\frac{f_A}{f_B} = \sqrt{\frac{\ell_B}{\ell_A}} \qquad \Rightarrow \frac{2f_B}{f_B} = \sqrt{\frac{\ell_B}{\ell_A}}$$

$$\Rightarrow 4 = \frac{\ell_B}{\ell_A} \qquad \Rightarrow \ell_A = \frac{\ell_B}{4}$$

55. (a) $\because T = 2\pi\sqrt{\frac{M}{K}}$ [For Initial equilibrium $\because Mg = KL$]



After attaching mass m, $T' = 2\pi\sqrt{\frac{(M+m)L}{Mg}}$

56. (a) Frequency of simple pendulum $n = \frac{1}{2\pi}\sqrt{\frac{g_{eff.}}{\ell}}$

where, g_{eff} = effective acceleration due to gravity.
In a freely falling lift $g_{eff} = g - g = 0$ then $n = 0$

57. (a) Time period of oscillation $T = 2\pi\sqrt{\frac{l}{g}}$

For first pendulum $T_1 = 2\pi\sqrt{\frac{l_1}{g}} = 2\pi\sqrt{\frac{5}{10}}$

$= 2\pi\sqrt{\frac{1}{2}}$ ...(i)

For second pendulum $T_2 = 2\pi\sqrt{\frac{20}{10}} = 2\pi\sqrt{2}$ ...(ii)

From equation (i) and (ii)

$T_2 = 2T_1$

Thus small pendulum has completed 2 oscillation.

58. (b) Frequency $n = \frac{1}{2\pi}\sqrt{k/m}\,;\, n \propto \frac{1}{\sqrt{m}}$

$$\frac{n}{n_2} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{4m}{m}} \Rightarrow n_2 = \frac{n}{2}$$

59. (a) According to given $l_2 = 1.02 l_1$,

Time period $(T) = 2\pi \times \sqrt{\frac{l}{g}} \propto \sqrt{l}$

Therefore $\frac{T_2}{T_1} = \sqrt{\frac{l_2}{l_1}} = \sqrt{\frac{1.02 l_1}{l_1}} = 1.01$. Thus time period increased by 1%.

60. (b) As we know that at mean position kinetic energy is maximum where as at extreme position potential energy is maximum.

Since total mechanical energy is constant in SHM

So, potential energy at A (or C) = Kinetic energy at B. Thus $\frac{1}{2}mv_B^2 = mgH$ or $v_B = \sqrt{2gH}$.

61. (d) Mass (m) = 5 kg and time period (T) = 2π sec.

Therefore time period $T = 2\pi \times \sqrt{\frac{m}{k}} \Rightarrow \sqrt{\frac{5}{k}} = 1$ or $k = 5$ N/m

According to Hooke's Law, $F = -kx$

Therefore decrease in length.

As restoring force should be equal to the weight of the body

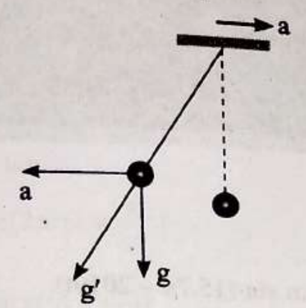
$$\Rightarrow (x) = \frac{F}{k} = \frac{5g}{5} = g \text{ metres}$$

62. (d) Time period of oscillation $T = 2\pi\sqrt{\frac{l}{g}}$.

Therefore T will decrease when acceleration (g) increases. And g will increase when the rocket moves up with a uniform acceleration.

63. (a) Time period = 4 sec. In one oscillation, the same kinetic and potential energies are repeated two times in S.H.M. So the difference will be 2 seconds.

64. (d) The effective value of acceleration due to gravity is $\sqrt{(a^2+g^2)}$.



65. (c) P.E, $U = \frac{1}{2} m\omega^2 x^2$ and

KE, $T = \frac{1}{2} m\omega^2 (a^2 - x^2)$

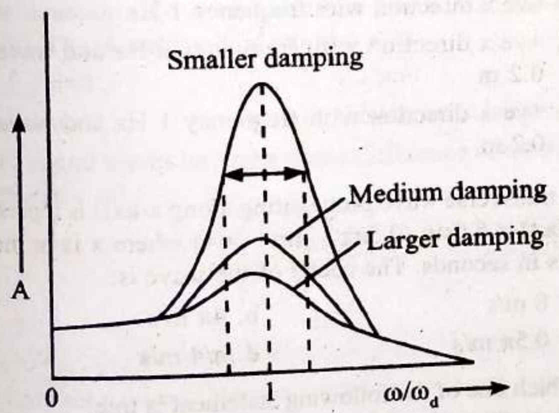
$\therefore \frac{T}{U} = \frac{a^2 - x^2}{x^2}$

66. (d) For two springs in series the effective spring constant is

$k_{eff} = \frac{k_1 k_2}{k_1 + k_2}$

Time period, $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{m(k_1+k_2)}{k_1 k_2}}$

67. (a) The phenomena of increase in amplitude when the frequency of driving force is equal to the natural frequency of the oscillator is called resonance.



In the presence of resistive forces proportional to velocity ad the applied periodic force.

$F = F_0 \sin \omega_d t$ equation becomes

$$m\frac{d^2x}{dt^2} + kx + b\frac{dx}{dt} = F_0 \sin \omega_d t$$

Thus, $x = A \sin(\omega_d t + \phi)$

where $A = \dfrac{F_0}{m\sqrt{(\omega^2 - \omega_d^2)^2 + \left(\dfrac{b\omega_d}{m}\right)^2}}$

i.e., a function of natural frequency ω and driving frequency ω_d.

Thus in forced vibration, the resonance wave become very sharp when damping force is small (i.e., negligible)

68. (d) For damped oscillation amplitude $A = A_0 e^{-bt}$

where a_0 = Initial amplitude, b = Clamping constant

Let the time-period for each oscillation i & T

According to given condition $\frac{A_0}{3} = A_0 e^{-b(100T)} \Rightarrow e^{-100\,bT} = \frac{1}{3}$

at t = 200 T, $A = A_0\, e^{-b(200\,T)} = A_0 e^{(-100bT)^2}$

$\Rightarrow A = A_0\left(\frac{1}{3}\right)^2 = \frac{A_0}{9}$

69. (b) Amplitude of damped oscillation at time t

$x = x_0 e^{-\lambda t}$ Where λ is a constant

after 20 sec

$\frac{x_0}{3} = x_0 e^{-\lambda(20)} \Rightarrow e^{-\lambda(20)} = \frac{1}{3}$ ....(1)

After 40 sec

$x' = x_0 e^{-\lambda(40)} \Rightarrow x_0 e^{-\lambda(2\times 20)}$

From Eq. (1)

$x' = x_0\left(\frac{1}{3}\right)^2 = \frac{x_0}{9}$

CHAPTER 14

Waves

Mechanical Waves, Progressive and Stationary Waves; Intensity and Speed of Sound

1. A wave travelling in the +ve x-direction having displacement along y-direction as 1 m, wavelength 2π m and frequency of $1/\pi$ Hz is represented by: *(2013)*
   a. $y = \sin(2\pi x + 2\pi t)$
   b. $y = \sin(x - 2t)$
   c. $y = \sin(2\pi x - 2\pi t)$
   d. $y = \sin(10\pi x - 20\pi t)$

2. The equation of a simple harmonic wave is given by $y = 3\sin\frac{\pi}{2}(50t - x)$ where x and y are in metres and t is in seconds. The ratio of maximum particle velocity to the wave velocity is: *(2012 Mains)*
   a. 2π
   b. $3\pi/2$
   c. 3π
   d. $2/3\pi$

3. Two waves are represented by the equation $y_1 = a\sin(\omega t + kx + 0.57)$ m and $y_2 = a\cos(\omega t + kx)$ m, where x is in meter and t in sec. The phase difference between them is *(2011 Pre)*
   a. 1.0 radian
   b. 1.25 radian
   c. 1.57 radian
   d. 0.57 radian

4. Sound waves travel at 350 m/s through a warm air and at 3500 m/s through brass. The wavelength of a 700 Hz acoustic wave as it enters brass from warm air: *(2011 Pre)*
   a. Decreases by a factor 10
   b. Increases by a factor 20
   c. Increases by a factor 10
   d. Decreases by a factor 20

5. A transverse wave is represented by $y = A\sin(\omega t - kx)$. For what value of the wavelength is the wave velocity equal to the maximum particle velocity? *(2010 pre)*
   a. $\pi A/2$
   b. πA
   c. $2\pi A$
   d. A

6. A wave in a string has an amplitude of 2 cm. The wave travels in the +ve direction of x axis with a speed of 128 m/sec and it is noted that 5 complete waves fit in 4 m length of the string. The equation describing the wave is *(2009)*
   a. $y = (0.02)$ m $\sin(15.7x - 2010t)$
   b. $y = (0.02)$ m $\sin(15.7x + 2010t)$
   c. $y = (0.02)$ m $\sin(7.85x - 1005t)$
   d. $y = (0.02)$ m $\sin(7.85x + 1005t)$

7. Two points are located at a distance of 10 m and 15 m from the source of oscillation. The period of oscillation is 0.05 sec and the velocity of the wave is 300 m/sec. What is the phase difference between the oscillations of two points? *(2008)*
   a. $\frac{\pi}{6}$
   b. $\frac{\pi}{3}$
   c. $\frac{2\pi}{3}$
   d. π

8. The wave described by $y = 0.25\sin(10\pi x - 2\pi t)$ where x and y are in meters and t in seconds, is a wave travelling along the: *(2008)*
   a. –ve x direction with amplitude 0.25 m and wavelength = 0.2 m
   b. –ve x direction with frequency 1 Hz
   c. +ve x direction with frequency π Hz and wavelength = 0.2 m
   d. +ve x direction with frequency 1 Hz and wavelength = 0.2 m

9. A transverse wave propagating along x-axis is represented by $y(x,t) = 8.0\sin(0.5\pi x - 4\pi t - \pi/4)$ where x is in metres and t is in seconds. The speed of the wave is: *(2006)*
   a. 8 m/s
   b. 4π m/s
   c. 0.5π m/s
   d. $\pi/4$ m/s

10. Which one of the following statement is true? *(2006)*
    a. Both light and sound waves can travel in vacuum
    b. Both light and sound waves in air are transverse
    c. The sound waves in air are longitudinal while the light waves are transverse
    d. Both light and sound waves in air are longitudinal

11. A point source emits sound equally in all directions in a non-absorbing medium. Two points P and Q are at distance of 2 m and 3 m respectively from the source. The ratio of the intensities of the waves at P and Q is: *(2005)*
    a. 9 : 4
    b. 2 : 9
    c. 9 : 2
    d. 4 : 9

12. The phase difference between two waves, represented by: $y_1 = 10^{-6} \sin\{100t + (x/50) + 0.5\}$m

$y_2 = 10^{-6} \cos\{100t + \left(\frac{x}{50}\right)\}$m where x is expressed in metres and t is expressed in seconds, is approximately: *(2004)*

a. 2.07 radians
b. 0.5 radians
c. 1.5 radians
d. 1.07 radians

13. A wave travelling in positive x-direction with A = 0.2 m velocity = 360 m/s and λ = 60 m, then correct expression for the wave is: *(2002)*

a. $y = 0.2\sin[2\pi(6t + \frac{x}{60})]$
b. $y = 0.2\sin[\pi(6t + \frac{x}{60})]$
c. $y = 0.2\sin[2\pi(6t - \frac{x}{60})]$
d. $y = 0.2\sin[\pi(6t - \frac{x}{60})]$

14. The equation of a wave is represented by

$y = 10^{-4} \sin\left(100 - \frac{x}{10}\right)$ m, then the velocity of wave will be: *(2001)*

a. 100 m/s
b. 4 m/s
c. 1000 m/s
d. 0.00 m/s

15. For a wave $y = y_0 \sin(\omega t - kx)$, for what value of λ is the maximum particle velocity equal to two times the wave velocity: *(1998)*

a. πy_0
b. $2\pi y_0$
c. $\pi y_0/2$
d. $4\pi y_0$

16. The equation of a sound wave is y = 0.0015 sin (62.4*x* + 316t). The wavelength of this wave is: *(1996)*

a. 0.3 unit
b. 0.2 unit
c. 0.1 unit
d. Cannot be calculated

17. Two sound waves having a phase difference of 60° have path difference of: *(1996)*

a. $\frac{\lambda}{6}$
b. $\frac{\lambda}{3}$
c. 2λ
d. $\frac{\lambda}{2}$

18. A hospital uses an ultrasonic scanner to locate tumours in a tissue. The operating frequency of the scanner, is 4.2 MHz. The speed of sound in a tissue is 1.7 km/s. The wavelength of sound in the tissue is close to: *(1995)*

a. 4×10^{-3} m
b. 8×10^{-3} m
c. 4×10^{-4} m
d. 8×10^{-4} m

19. Which one of the following represents a wave? *(1994)*

a. $y = A \sin(\omega t - kx)$
b. $y = A \cos(at - bx + c)$
c. $y = A \sin kx$
d. $y = A \sin \omega t$

20. A stationary wave is represented by $y = A \sin(100t) \cos(0.01x)$, where y and A are in millimetres, t is in seconds and x is in metres. The velocity of the wave is: *(1994)*

a. 10^4 m/s
b. Not derivable
c. 1 m/s
d. 10^2 m/s

21. A wave of frequency 100 Hz travels along a string towards its fixed end. When this wave travels back, after reflection, a node is formed at a distance of 10 cm from the fixed end. The speed of the wave (incident and reflected) is: *(1994)*

a. 20 m/s
b. 40 m/s
c. 5 m/s
d. 10 m/s

22. The temperature at which the speed of sound becomes double as was at 27°C is: *(1993)*

a. 273°C
b. 0°C
c. 927°C
d. 1027°C.

23. The frequency of sinusoidal wave y = 0.40 cos [2000 t + 0.80] would be: *(1992)*

a. 1000π Hz
b. 2000 Hz
c. 20 Hz
d. $\frac{1000}{\pi}$ Hz

24. With the propagation of a longitudinal wave through a material medium, the quantities transmitted in the propagation direction are: *(1992)*

a. Energy, momentum and mass
b. Energy
c. Energy and mass
d. Energy and linear momentum

25. Velocity of sound waves in air is 330 m/s. For a particular sound wave in air, a path difference of 40 cm is equivalent to phase difference of 1.6 π. The frequency of this wave is: *(1990)*

a. 165 Hz
b. 150 Hz
c. 660 Hz
d. 330 Hz

26. If the amplitude of sound is doubled and the frequency reduced to one fourth, the intensity of sound at the same point will be: *(1989)*

a. Increasing by a factor of 2
b. Decreasing by a factor of 2
c. Decreasing by a factor of 4
d. Unchanged

27. The velocity of sound in any gas depends upon: *(1988)*

a. Wavelength of sound only
b. Density and elasticity
c. Intensity of sound waves only
d. Amplitude and frequency of sound

28. Equation of progressive wave is given by

$y = 4\sin\left[\pi\left(\frac{t}{5} - \frac{x}{9}\right) + \frac{\pi}{6}\right]$ where *y, x* are in cm and t is in seconds. Then which of the following is correct? *(1988)*

a. v = 5 cm
b. λ = 18 cm
c. a = 0.04 cm
d. f = 50 Hz

Superposition of Waves, Interference and Reflection of Wave

29. A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2. The ratio λ_2/λ_1 is: *(2016 - I)*

a. $\sqrt{\frac{m_1}{m_2}}$ b. $\sqrt{\frac{m_1+m_2}{m_2}}$

c. $\sqrt{\frac{m_2}{m_2}}$ d. $\sqrt{\frac{m_1+m_2}{m_1}}$

30. Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the same direction. The sum of the maximum and minimum intensities is *(2008)*

a. $\left(\sqrt{I_1}-\sqrt{I_2}\right)^2$ b. $2(I_1+I_2)$

c. I_1+I_2 d. $\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

31. Two waves having equation: $x_1 = a\sin(\omega t+\phi_1)$

$x_2 = a\sin(\omega t+\phi_2)$

If in the resultant wave the frequency and amplitude remains equals to amplitude of superimposing waves. Then phase difference between them: *(2001)*

a. $\frac{\pi}{6}$ b. $\frac{2\pi}{3}$

c. $\frac{\pi}{4}$ d. $\frac{\pi}{3}$

Vibration of String and Organ Pipe

32. If the initial tension on a stretched string is doubled, then the ratio of the initial and final speed of a transverse wave along the string is : *(2022)*

a. 1 : 2 b. 1 : 1

c. $\sqrt{2}:1$ d. $1:\sqrt{2}$

33. The length of the string of a musical instrument is 90 cm and has a fundamental frequency of 120 Hz. Where should it be pressed to produce fundamental frequency of 180 Hz? *(2020-Covid)*

a. 60 cm b. 45 cm

c. 80 cm d. 75 cm

34. The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is: *(2018)*

a. 12.5 cm b. 8 cm

c. 13.2 cm d. 16 cm

35. A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is: *(2018)*

a. 350 m/s b. 339 m/s

c. 330 m/s d. 350 m/s

36. The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system? *(2017-Delhi)*

a. 20 Hz b. 30 Hz

c. 40 Hz d. 10 Hz

37. The second overtone of an open organ pipe has the same frequency as the first overtone of a closed pipe L meter long. The length of the open pipe will be: *(2016 - II)*

a. $\frac{L}{2}$ b. 4 L

c. L d. 2 L

38. An air column, closed at one end and open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is: *(2016 - I)*

a. 66.7 cm b. 100 cm

c. 150 cm d. 200 cm

39. The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is: *(2015)*

a. 100 cm b. 120 cm

c. 140 cm d. 80 cm

40. A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequencies for this string is: *(2015 Re)*

a. 105 Hz b. 155 Hz

c. 205 Hz d. 10.5 Hz

41. The number of possible natural oscillations of air column in a pipe closed at one end of length 85 cm whose frequencies lie below 1250 Hz are (velocity of sound = 340 ms^{-1}): *(2014)*

a. 4 b. 5

c. 7 d. 6

42. If n_1, n_2 and n_3 are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequency n of the string is given by: *(2014)*

a. $\frac{1}{n}=\frac{1}{n_1}+\frac{1}{n_2}+\frac{1}{n_3}$ b. $\frac{1}{\sqrt{n}}=\frac{1}{\sqrt{n_1}}+\frac{1}{\sqrt{n_2}}+\frac{1}{\sqrt{n_3}}$

c. $\sqrt{n}=\sqrt{n_1}+\sqrt{n_2}+\sqrt{n_3}$ d. $n=n_1+n_2+n_3$

43. If we study the vibration of a pipe open at both ends, then the following statement is not true: *(2013)*

a. Pressure change will be maximum at both ends
b. Open end will be anti-node
c. Odd harmonics of the fundamental frequency will be generated
d. All harmonics of the fundamental frequency will be generated

44. When a string is divided into three segments of length ℓ_1, ℓ_2 and ℓ_3 the fundamental frequencies of these three segments are v_1, v_2 and v_3 respectively. The original fundamental frequency (v) of the string is: *(2012 Pre)*

a. $\sqrt{v} = \sqrt{v_1} + \sqrt{v_2} + \sqrt{v_3}$ b. $v = v_1 + v_2 + v_3$
c. $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3}$ d. $\frac{1}{\sqrt{v}} = \frac{1}{\sqrt{v_1}} + \frac{1}{\sqrt{v_2}} + \frac{1}{\sqrt{v_3}}$

45. If the tension and diameter of a sonometer wire of fundamental frequency n is doubled and density is halved then its fundamental frequency will become: *(2001)*

a. $\frac{n}{4}$ b. $\sqrt{2}n$
c. n d. $\frac{n}{\sqrt{2}}$

46. A string is cut into three parts, having fundamental frequencies n_1, n_2 and n_3 respectively. Then original fundamental frequency 'n related by the expression as: *(2000)*

a. $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$ b. $n = n_1 \times n_2 \times n_3$
c. $n = n_1 + n_2 + n_3$ d. $n = \frac{n_1 + n_2 + n_3}{3}$

47. A cylindrical tube (L = 125 cm) is resonant with a tuning fork of frequency 330 Hz. If it is filling by water then to get resonance again, minimum length of water column is (v = 330 m/s): *(1999)*

a. 50 cm b. 60 cm
c. 25 cm d. 20 cm

48. A standing wave having 3 nodes and 2 antinodes is formed between 1.21 Å distance then the wavelength is: *(1998)*

a. 1.21 Å b. 2.42 Å
c. 0.605 Å d. 4.84 Å

49. Standing waves are produced in 10 m long stretched string. If the string vibrates in 5 segments and wave velocity is 20 m/s, the frequency is: *(1997)*

a. 5 Hz b. 10 Hz
c. 2 Hz d. 4 Hz

50. A cylindrical tube, open at both ends has fundamental frequency in air. The tube is dipped vertically in water, so that half of it is in water. The fundamental frequency of air column is now: *(1997)*

a. $f/2$ b. $3f/4$
c. $2f$ d. f

51. The length of a sonometer wire AB is 110 cm. Where should the two bridges be placed from A to divide the wire in 3 segments whose fundamental frequencies are in the ratio of 1 : 2 : 3? *(1995)*

a. 60 cm and 90 cm b. 30 cm and 60 cm
c. 30 cm and 90 cm d. 40 cm and 80 cm

52. A stretched string resonates with tuning fork frequency 512 Hz when length of the string is 0.5 m. The length of the string required to vibrate resonantly with a tuning fork of frequency 256 Hz would be: *(1993)*

a. 0.25 m b. 0.5 m
c. 1 m d. 2 m

53. A closed organ pipe (closed at one end) is excited to support the third overtone. It is found that air in the pipe has: *(1991)*

a. Three nodes and three antinodes
b. Three nodes and four antinodes
c. Four nodes and three antinodes
d. Four nodes and four antinodes

54. A 5.5 metre length of string has a mass of 0.035 kg. If the tension in the string in 77 N, the speed of a wave on the string is: *(1989)*

a. 110 ms^{-1} b. 165 ms^{-1}
c. 77 ms^{-1} d. 102 ms^{-1}

Beats

55. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be : *(2020)*

a. 524 Hz b. 536 Hz
c. 537 Hz d. 523 Hz

56. Three sound waves of equal amplitudes have frequencies (n – 1), n, (n + 1). They superimpose to give beats. The number of beats produced per second will be: *(2016 - II)*

a. 3 b. 2
c. 1 d. 4

57. A source of unknown frequency gives 4 beats/s, when sounded with a source of known frequency 250 Hz. The second harmonic of the source of unknown frequency gives five beats per second, when sounded with a source of frequency 513 Hz. The unknown frequency is: *(2013)*

a. 260 Hz b. 254 Hz
c. 246 Hz d. 240 Hz

58. Two sources of sound placed close to each other are emitting progressive waves given by $y_1 = 4 \sin 600\pi t$ and $y_2 = 5 \sin 608\pi t$. An observer located near these two sources of sound will hear: *(2012 Pre)*

a. 4 beats per second with intensity ratio 25 : 16 between waxing and waning.
b. 8 beats per second with intensity ratio 25 : 16 between waxing and waning
c. 8 beats per second with intensity ratio 81 : 1 between waxing and waning
d. 4 beats per second with intensity ratio 81 : 1 between waxing and waning

59. Two identical piano wires, kept under the same tension T have a fundamental frequency of 600 Hz. The fractional increase in the tension of one of the wires which will lead to occurrence of 6 beats/s when both the wires oscillate together would be: *(2011 Mains)*

a. 0.02 b. 0.03
c. 0.04 d. 0.01

60. A tuning fork of frequency 512 Hz makes 4 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per sec when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was: *(2010 Pre)*

a. 508 Hz b. 510 Hz
c. 514 Hz d. 516 Hz

61. Each of the two strings of length 51.6 cm and 49.1 cm are tensioned separately by 20 N force. Mass per unit length of both the strings is same and equal to 1 g/m. When both the strings vibrate simultaneously the number of beats is: *(2009)*

a. 7 b. 8
c. 3 d. 5

62. Two sound waves with wavelengths 5.0 m and 5.5 m respectively, each propagate in a gas with velocity 330 m/s. We expect the following number of beats per second: *(2006)*

a. 12 b. 0
c. 3 d. 6

63. Two vibrating tuning forks produce progressive waves given by $y_1 = 4 \sin 500\,\pi t$ and $y_2 = 2 \sin 506\,\pi t$. Number of beats produced per minute is: *(2005)*

a. 360 b. 180
c. 120 d. 30

64. A source of sound gives 5 beats per second, when sounded with another source of frequency 100 $second^{-1}$. The second harmonic of the source, together with a source of frequency 205 sec^{-1} gives 5 beats per second. What is the frequency of the source? *(1995)*

a. 105 $second^{-1}$ b. 205 $second^{-1}$
c. 95 $second^{-1}$ d. 100 $second^{-1}$

65. A source of frequency υ gives 5 beats/second when sounded with a source of frequency 200 Hz. The second harmonic of frequency 2υ of source gives 10 beats/second when sounded with a source of frequency 420 Hz. The value of υ is: *(1994)*

a. 205 Hz b. 195 Hz
c. 200 Hz d. 210 Hz

66. Wave has simple harmonic motion whose period is 4 seconds while another wave which also possesses simple harmonic motion has its period 3 sec. If both are combined, then the resultant wave will have the period equal to: *(1993)*

a. 4 sec b. 5 sec
c. 12 sec d. 3 sec

67. For production of beats the two sources must have: *(1992)*

a. Different frequencies and same amplitude
b. Different frequencies
c. Different frequencies, same amplitude and same phase
d. Different frequencies and same phase

Musical Sound and Doppler's Effect

68. Two cars moving in opposite directions approach each other with speed of 22 m/s and 16.5 m/s respectively. The driver of the first car blows a horn having a frequency 400 Hz. The frequency heard by the driver of the second car is [velocity of sound 340 m/s]: *(2017-Delhi)*

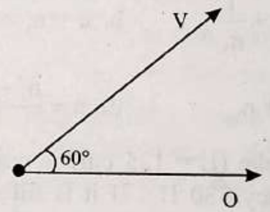
a. 361 Hz b. 411 Hz
c. 448 Hz d. 350 Hz

69. A siren emitting a sound of frequency 800 Hz moves away from an observer towards a cliff at a speed of 15 ms^{-1}. Then, the frequency of sound that the observer hears in the echo reflected from the cliff is: *(2016 - I)*

(Take velocity of sound in air = 330 ms^{-1})

a. 765 Hz b. 800 Hz
c. 838 Hz d. 885 Hz

70. A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 ms^{-1} at an angle of 60° with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air 330 ms^{-1}) is: *(2015 Re)*



a. 106 Hz b. 97 Hz
c. 103 Hz d. 100 Hz

71. A speeding motorcyclist sees traffic jam ahead him. He slows down to 36 km $hour^{-1}$. He finds that traffic has eased and a car moving ahead of him at 18 km $hour^{-1}$ is honking at a frequency of 1392 Hz. If the speed of sound is 343 ms^{-1}, the frequency of the honk as heard by him will be *(2014)*

a. 1332 Hz b. 1372 Hz
c. 1412 Hz d. 1454 Hz

72. A train moving at a speed of 220 ms^{-1} towards a stationary object, emits a sound of frequency 1000 Hz. Some of the sound reaching the object gets reflected back to the train as echo. The frequency of the echo as detected by the driver of the train is (Speed of sound in air is 330 ms^{-1}): *(2012 Mains)*

a. 3500 Hz b. 4000 Hz
c. 5000 Hz d. 3000 Hz

73. The driver of a car travelling with speed 30 m/sec towards a hill sounds a horn of frequency 600 Hz. If the velocity of sound in air is 330 m/s, the frequency of reflected sound as heard by driver is: *(2009)*

a. 555.5 Hz b. 720 Hz
c. 500 Hz d. 550 Hz

74. The time of reverberation of a room A is one second. What will be the time (in seconds) of reverberation of a room, having all the dimensions double of those of room A? *(2006)*

a. 2 b. 4
c. 1/2 d. 8

75. A car is moving towards a high cliff. The car driver sounds a horn of frequency 'f'. The reflected sound heard by the driver has a frequency 2f. If 'v' be the velocity of sound then the velocity of the car, in the same velocity units, will be: *(2004)*

a. $\frac{v}{3}$ b. $\frac{v}{4}$
c. $\frac{v}{2}$ d. $\frac{v}{\sqrt{2}}$

76. An observer moves towards a stationary source of sound with a speed 1/5th of the speed of sound. The wavelength and frequency of the source emitted are λ and f respectively. The apparent frequency and wavelength recorded by the observer are respectively: *(2003)*

a. 1.2f, 1.2λ
b. 1.2f, λ
c. f, 1.2λ
d. 0.8f, 0.8λ

77. A whistle revolves in a circle with angular speed $\omega = 20$ rad/sec using a string of length 50 cm. If the frequency of sound from the whistle is 385 Hz, then what is the minimum frequency heard by an observer which is far away from the centre: ($v_{sound} = 340$m/s) *(2002)*

a. 385 Hz b. 374 Hz
c. 394 Hz d. 333 Hz

78. Two stationary sources each emitting waves of wave length λ. An observer moves from one source to other with velocity u. Then number of beats heared by him: *(2000)*

a. $\frac{2u}{\lambda}$ b. $\frac{u}{\lambda}$
c. $\sqrt{u\lambda}$ d. $\frac{u}{2\lambda}$

79. If a source moves perpendicularly from listener then the change in frequency will be: *(1998)*

a. 2n b. n
c. n/2 d. Zero

80. Two trains move towards each other with the same speed. The speed of sound is 340 m/s. If the height of the tone of the whistle of one of them heard on the other changes to 9/8 times, then the speed of each train should be: *(1991)*

a. 20 m/s b. 2 m/s
c. 200 m/s d. 2000 m/s

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	b	a	c	c	c	c	d	a	c	a	d	c	c	a	c	a
18	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**
c	a,b	a	a	c	d	d	c	c	b	b	b	b	b	d	a	c
35	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**	**51**
b	a	d	c	b	a	d	a	a	c	c	a	a	a	a	d	a
52	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**	**61**	**62**	**63**	**64**	**65**	**66**	**67**	**68**
c	d	a	a	b	b	d	a	a	a	d	b	a	a	c	b	c
69	**70**	**71**	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**					
c	c	c	c	b	a	a	b	b	a	d	a					

Explanations

1. (b) The standard equation of a wave travelling along +ve x-direction is given by $y = A\sin(kx - \omega t)$

where A = amplitude of the wave, k = angular wave number ω = angular frequency of the wave

Given: $A = 1m, \lambda = 2\pi m, v = \frac{1}{\pi} Hz$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{2\pi} = 1 \text{ and } \omega = 2\pi f = (2\pi)\left(\frac{1}{\pi}\right) = 2$$

So equation of wave $y = \sin(x - 2t)$

2. (b) We have $y = 3\sin\frac{\pi}{2}(50t - x)$

$= 3\sin\left(25\pi t - \frac{\pi}{2}x\right)$

$= A\sin(\omega t - kx)$

On doing comparison, we have

$k = \frac{\pi}{2}$ and $A = 3$

particle velocity $(v_p) = \frac{dy}{dt} = 3 \times 25\pi\cos\left(25\pi t - \frac{\pi}{2}x\right)$

$(v_p)_{max} = A\omega = 75\pi \text{ m sec}^{-1}$

Wave velocity $v_{wave} = \frac{\omega}{k} = 50 \text{m sec}^{-1}$

Thus, required ratio $= \frac{75\pi}{50} = \frac{3}{2}\pi$

3. (a) Phase of a wave is given by cosine or sine equation of the wave. It is denoted by ϕ.

$y_1 = a\sin(\omega t + kx + 0.57)$

$\therefore$ Phase, $\phi_1 = \omega t + kx + 0.57$

$y_2 = a\cos(\omega t + kx) = a\sin\left(\omega t + kx + \frac{\pi}{2}\right)$

$\therefore$ Phase, $\phi_2 = \omega t + kx + \frac{\pi}{2}$

Phase difference, $\Delta\phi = \phi_2 - \phi_1$

$= \frac{\pi}{2} - 0.57$

$= (1.57 - 0.57)$ radian = 1 radian

4. (c) When a sound wave travels from one medium to another medium its frequency remains the same

$\Rightarrow v_{air} = v_{brass}$

$\because$ Frequency $v = \frac{v}{\lambda}$

$$\Rightarrow \frac{v_{air}}{\lambda_{air}} = \frac{v_{brass}}{\lambda_{brass}} \Rightarrow \frac{350}{700} = \frac{3500}{\lambda_{brass}}$$

$\Rightarrow \lambda_{brass} = 7000$ Hz

So, wave length of wire increases by a factor of 10.

5. (c) The given wave equation is $y = A\sin(\omega t - kx)$

wave velocity $v = \frac{\omega}{k}$ ...(i)

Particle velocity, $v_p = \frac{dy}{dt} = A\omega\cos(\omega t - kx)$

Maximum particle velocity, $(v_p)_{max} = A\omega$ ...(ii)

According to the given question

$v = (v_p)_{max}$

$\frac{\omega}{k} = A\omega$ (Using (i) and (ii))

$\frac{1}{k} = A$ or $\frac{\lambda}{2\pi} = A$ $\left(\because k = \frac{2\pi}{\lambda}\right)$

$\Rightarrow \lambda = 2\pi A$

6. (c) Amplitude = 2 cm = 0.02 m, v = 128 m/s

$\lambda = \frac{4}{5} = 0.8$ m; $v = \frac{128}{0.8} = 160$ Hz

$\omega = 2\pi v = 2\pi \times 160 = 1005$; $k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.8} = 7.85$

$\therefore y = 0.02m \sin(7.85x - 1005t)$

7. (c) Phase difference between the oscillations of two points

$(\Delta\phi) = \frac{2\pi}{\lambda}(\Delta x)$

λ = Velocity of wave × period of oscillation

$\lambda = 300 \times 0.05 = 15$ m $\Rightarrow \Delta\phi = \frac{2\pi}{15} \times 5$

$\Delta\phi = \frac{2\pi}{3}$

8. (d) $y = 0.25\sin(10\pi x - 2\pi t)$

General equation $\Rightarrow y = A\sin(kx - \omega t)$

If sign between kx and ωt is positive then wave travel in negative direction and vice-versa.

Inference drawn on comparing

(i) A = 0.25

(ii) Wave travel in positive x-direction

(iii) $k = \frac{2\pi}{\lambda}$

$k = 10\pi$

$\lambda = \frac{2}{10} = 0.2$ m

(iv) $\omega = 2\pi$

$2\pi f = 2\pi$

f = 1 Hz

9. (a) $y(x, t) = 8.0\sin(0.5\pi x - 4\pi t - \frac{\pi}{4})$

Compare with a standard wave equation,

$y = A\sin\left(\frac{2\pi x}{\lambda} - \frac{2\pi t}{T} + \phi\right)$

we get

$\frac{2\pi}{\lambda} = 0.5\pi \quad \Rightarrow \lambda = \frac{2\pi}{0.5\pi} = 4\text{ m}$

$\frac{2\pi}{T} = 4\pi \quad \Rightarrow T = \frac{2\pi}{4\pi} = \frac{1}{2}\text{ s}$

$\therefore$ Frequency: $\nu = 1/T = 2$ Hz

Wave velocity, $v = \lambda\nu = 4 \times 2 = 8$ m/s

10. (c) Transverse waves are waves in which the direction of vibration of particles in a medium is perpendicular to the direction of propagation of the wave. Light waves are electromagnetic waves. Light waves are transverse in nature and do not require a medium to travel, hence they can travel in vacuum. Longitudinal waves are waves in which direction of vibration of particles in a medium is along the direction of propagation of the wave. Sound waves are longitudinal waves and require a medium to travel. They do not travel in vacuum.

11. (a) As we know, intensity varies with the distance from the source and power of the source.

$\Rightarrow I = \frac{P}{4\pi r^2}$

where p = power and r = distance from source

$\therefore$ Point source $\propto \frac{1}{(\text{distance})^2}$

Here $d_P = 2$ m and $d_Q = 3$ m

$\therefore \frac{I_P}{I_Q} = \left(\frac{d_Q}{d_P}\right)^2 \Rightarrow \frac{I_P}{I_Q} = \left(\frac{3}{2}\right)^2$

Thus, the required ratio is 9 : 4.

12. (d) $y_1 = 10^{-6}\sin\{100t + \frac{x}{50} + 0.5\}$

$y_2 = 10^{-6}\cos\{100t + \frac{x}{50}\}$

$= 10^{-6}\sin\{\frac{\pi}{2} + 100t + \frac{x}{50}\}$

Phase difference between waves

$$= \frac{\pi}{2} - 0.5 = 1.57 - 0.5 = 1.07 \text{ radians}$$

13. (c) The equation of progressive wave traveling in positive x-direction is given by

$y = A\sin\frac{2\pi}{\lambda}(vt - x)$

Here $A = 0.2$ m, $v = 360$ m/sec, $\lambda = 60$ m

$\therefore y = 0.2\sin\frac{2\pi}{60}(360t - x) = 0.2\sin\left[2\pi\left(6t - \frac{x}{60}\right)\right]$.

14. (c) Comparing the given equation with general equation,

$y = A\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)$, we get

$T = \frac{2\pi}{100}$ and $\lambda = 20\pi$

$\therefore v = \nu\lambda = \frac{100}{2\pi} \times 20\pi = 1000$ m/s

15. (a) $v_{wave} = \frac{\omega}{k}$

$v_{particle} = \frac{dy}{dt} = \underbrace{y_0\omega}\cos(\omega t - kx)$

$(v_p)_{max} = y_0\omega$

$(v_p)_{max} = 2(v_{wave})$

$y_0\omega = 2\frac{\omega}{k} \Rightarrow k = \frac{2}{y_0} = \frac{2\pi}{\lambda} \Rightarrow \lambda = \pi y_0$

16. (c) Sound wave equation is

$Y = 0.0015\sin(62.4x + 316t)$.

Comparing it with the general equation of motion

$y = A\sin 2\pi\left[\frac{x}{\lambda} + \frac{t}{T}\right]$, we get $\frac{2\pi}{\lambda} = 62.4$

$\therefore \lambda = \frac{2\pi}{62.4} = 0.1$ unit.

17. (a) Phase difference $\phi = 60° = \frac{\pi}{3}$ rad.

Phase difference $(\phi) = \frac{\pi}{3} = \frac{2\pi}{\lambda} \times$ Path difference.

Therefore path difference $= \frac{\pi}{3} \times \frac{\lambda}{2\pi} = \frac{\lambda}{6}$.

18. (c) Frequency (ν) = 4.2 MHz = 4.2×10^6 Hz and speed of sound (v) = 1.7 km/s = 1.7×10^3 m/s.

Wave length of sound in tissue $(\lambda) = \frac{v}{\nu} = \frac{1.7 \times 10^3}{4.2 \times 10^6}$

$= 4 \times 10^{-4}$ m

19. (a, b) (a) Represents a harmonic progressive wave in the standard form where as (b) also represents a harmonic progressive wave, both travelling in the positive x-direction (d) represents only S.H.M.

20. (a) $y = A\sin(100t)\cos(0.01x)$.

Comparing it with standard equation

$y = A\sin\left(\frac{2\pi}{T}t\right)\cos\left(\frac{2\pi}{\lambda}x\right)$,

we get $T = \frac{\pi}{50}$ and $\lambda = 200\pi$.

Therefore velocity, $(v) = \frac{\lambda}{T} = \frac{200\pi}{\pi/50} = 200 \times 50$

$= 10000 = 10^4$ m/s.

21. (a) Frequency (ν) = 100 Hz and distance from fixed end = 10 cm = 0.1 m. When a stationary wave is produced, the fixed end behaves as a node. Thus wavelength (λ) = 2 × 0.1 = 0.2 m.

Therefore velocity $v = \nu\lambda = 100 \times 0.2 = 20$ m/s.

22. (c) Speed of sound is directly proportional to square root of temperature.

$$\Rightarrow v = \sqrt{\frac{\gamma RT}{M}}$$

$$\Rightarrow v \propto \sqrt{T}$$

$T_1 = 27 + 273 = 300$ K

At T_2, v becomes 2v

$$\therefore \frac{v}{2v} = \frac{\sqrt{T_1}}{\sqrt{T_2}} \quad \therefore T_2 = 4T_1$$

Thus $T_2 = 1200$ K $= 927°$C

23. (d) Comparing with the equation

$y = A\cos(\omega t + kx)$

$2\pi f = 2000$

$$f = \frac{1000}{\pi} \text{Hz}$$

24. (d) During the transmission of longitudinal waves, energy and linear momentum transported to the medium's particles in the wave propagation direction due to the interatomic forces acting on the atom.

25. (c) Path difference $\Delta x = \frac{\lambda}{2\pi}\Delta\phi$,

$$\lambda = 2\pi\frac{\Delta x}{\Delta\phi} = \frac{2\pi(0.4)}{1.6\pi} = 0.5 \text{ m}$$

$$v = \frac{v}{\lambda} = \frac{330}{0.5} = 660 \text{ Hz}$$

26. (c) Intensity $I = 2\pi^2a^2n^2v\rho$

Thus intensity $\propto$ (amplitude)^2 and also intensity $\propto$ (frequency)^2. Therefore

$I \propto A^2\omega^2$

$$I' \propto (4A^2)\frac{\omega^2}{16} \Rightarrow I' = \frac{I}{4}$$

27. (b) According to Newton Laplace's correction relation

$$v = \sqrt{\frac{\gamma P}{\rho}} \text{ or } \sqrt{\frac{B}{\rho}}$$, where adiabatic Bulk modulus $B = \gamma P$

Thus, velocity of sound in any gas depends upon the density and elasticity of the gas.

28. (b) For a standard progressive wave

$$y = A\sin\left[2x\left(\frac{t}{T} - \frac{x}{\lambda}\right) + \phi\right]$$

The given equation can be written as

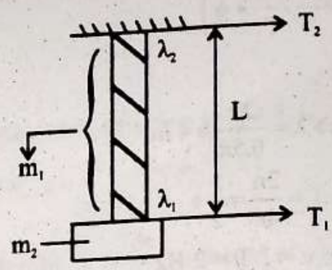
$$y = 4\sin\left[2x\left(\frac{t}{10} - \frac{x}{18}\right) + \frac{\pi}{6}\right]$$

On comparing, $\phi = \frac{\pi}{6}$

$\therefore A = 4$ cm, $T = 10$ s,

$\lambda = 18$ cm and $\phi = \pi/6$.

29. (b)



Tension in the string $T_1 = m_2g$

Tension in the string $T_2 = (m_1 + m_2)g$

We know that velocity $v \propto \sqrt{T}$ $[\therefore F = v \times \lambda]$

Also $\lambda \propto \sqrt{T}$

$$\frac{\lambda_1}{\lambda_2} = \frac{\sqrt{T_1}}{\sqrt{T_2}} = \sqrt{\frac{m_2}{m_1 + m_2}}$$

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{m_1 + m_2}{m_2}}$$

30. (b) When two periodic waves of intensities I_1 and I_2 pass across a region at the same time in same direction, then resultant intensity of superposed wave is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

where ϕ is the phase difference between waves.

For maximum intensity

$$I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$$

For minimum intensity

$$I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$$

$$\therefore I_{max} + I_{min} = 2I_1 + 2I_2 = 2(I_1 + I_2)$$

31. (b) Given, $x_1 = a\sin(\omega t + \phi_1)$

$x_2 = a\sin(\omega t + \phi_2)$

Resultant superposed wave,

$x' = x_1 + x_2$

$x' = a[\sin(\omega t + \phi_1) + \sin(\omega t + \phi_2)]$

$$x' = 2a\sin\left(\omega t + \frac{\phi_1 + \phi_2}{2}\right)\cos\left(\frac{\phi_1 - \phi_2}{2}\right)$$

$$\left[\therefore \sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)\right]$$

Thus, resultant amplitude $= 2a\cos\left(\frac{\phi_1 - \phi_2}{2}\right)$

$$\Rightarrow \cos\left(\frac{\phi_1 - \phi_2}{2}\right) = \frac{1}{2} \Rightarrow \frac{\phi_1 - \phi_2}{2} = \frac{\pi}{3}$$

$$\Rightarrow \phi_1 - \phi_2 = \frac{2\pi}{3}$$

32. (d) Velocity of transverse wave in a string is

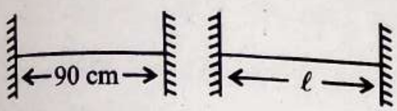
$$v = \sqrt{\frac{T}{m}}$$

T is tension in the string and m is mass per unit length of the string.

$$\Rightarrow v \propto \sqrt{T} \Rightarrow \frac{v_i}{v_f} = \sqrt{\frac{T_i}{T_f}}$$

$$\Rightarrow \frac{v_i}{v_f} \sqrt{\frac{T}{2T}} \Rightarrow \frac{v_i}{v_f} = \frac{1}{\sqrt{2}}$$

33. (a) The velocity of the travelling wave remains constant. Initially for fundamental mode, the length of the string is half the wavelength.



$\therefore \lambda = 180$ cm

we know that wave velocity $v = \nu\lambda$

Thus $\nu = \frac{v}{180}$

Here $\nu = 120 \Rightarrow 120 = \frac{v}{180}$

For $\nu = 180\,Hz = \frac{v}{2\ell}$

$\Rightarrow 180 = \frac{120 \times 180}{2\ell}$

Thus $l = 60$ cm

34. (c) Organ pipes are those cylindrical pipes which are used to produce musical (longitudinal) sounds.

Open organ pipe are open at both ends, where as closed pipe is the one in which only one end is open and the other is closed.

For open organ pipe, fundamental frequency $f_1 = \frac{v}{2l_1}$

The frequency for third harmonic for closed organ pipe is $f_3 = \frac{3v}{4l_2}$

According to question both are equal

$\frac{v}{2\ell_1} = 3\frac{v}{4\ell_2} \Rightarrow \frac{1}{\ell_1} = \frac{3}{2 \times 20} \quad (\because \ell_2 = 20)$

$\ell_1 = \frac{40}{3} = 13.33$ cm

35. (b) According to question, two successive resonance is produced.

Here length of I[st] resonance, $l_1 = 20$ cm

Length of II[nd] resonance, $l_2 = 73$ cm

Frequency $\nu = 320$ Hz and temperature = 270°C.

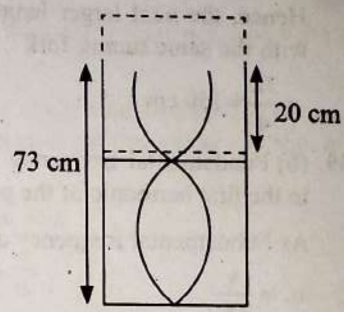
As we know that closed end tube resonates if the length of the tube is one fourth the wavelength of the sound.

$l_1 = \frac{\lambda}{4}$

$l_2 = \frac{\lambda}{4} + \frac{\lambda}{2} = \frac{3\lambda}{4}$

$l_2 - l_1 = \frac{3\lambda}{4} - \frac{-\lambda}{4} = \frac{2\lambda}{4}$

$\Rightarrow 73 - 20 = \frac{\lambda}{2}$



As, wave velocity $v = \nu\lambda = 320\,(53 \times 2 \times 10^{-2}) = 339.2$ m/s

36. (a) Resonant frequency for a closed organ pipe is $f = \frac{nv}{4l}$

and fundamental frequency $f_0 = \frac{v}{4\ell}$

Here $f_n = 220$ Hz

and successive frequency,

$f_{n+2} = \frac{(n+2)v}{4l} = 260\,Hz$ [as only odd harmonics produce]

so difference between any two successive harmonics

$260 - 220 = \frac{2v}{4\ell}$

$40 = \frac{2v}{4\ell}$

$40 = 2f_0 \quad \Rightarrow \quad f_0 = 20$ Hz

37. (d) The frequencies that are produced by an instrument successively are called over tones and integral multiples of fundamental frequency are called Harmonies.

Now, frequency of closed organ pipe is $f = \frac{(2n-1)v}{4l}$

where n = number of Harmonics, v = velocity of sound

and n − 1 = number of overtones

According to question $n - 1 = 1 \Rightarrow n = 2$

$\therefore f = \frac{3v}{4l_C}$

Now, frequency of open organ pipe $f = \frac{nv}{2l}$

According to question $n - 1 = 2 \Rightarrow n = 3$

$\therefore f = \frac{3v}{2l_0}$

According to given condition $\frac{3v}{2l_0} = \frac{3v}{4l_C}$

$\Rightarrow l_0 = 2l_C = 2L$ $\quad$ [l_C = L = length of closed organ pipe]

38. (c) According to question, the air column is a closed organ pipe.

Now, first minimum resonating length for closed organ pipe.

$= \frac{\lambda}{4} = 50$ cm, where λ is the wave length of the sound waves that is generated by tuning fork.

For closed organ pipe the next resonating length occurs at a distance which is equal to three times the length of the smallest length at which the resonance occurs.

Hence, the next larger length of the air column resonating with the same tuning fork

$=\frac{3\lambda}{4}=150$ cm

39. (b) Fundamental frequency is the frequency corresponding to the first harmonic of the pipe.

As Fundamental frequency of closed organ pipe

$n_1=\frac{v}{4\ell_c}$

Fundamental frequency of open organ pipe is $n_1=\frac{v}{2l_0}$

Now, third harmonic or second overtone of open organ pipe

$n_3=\frac{3v}{2l_0}$

According to given condition

$\frac{v}{4\ell_c}=\frac{3v}{2\ell_0}\Rightarrow \ell_0=6\ell_c=6(20\text{ cm})=120$ cm

40. (a) Two consecutive resonant frequencies for a string fixed at both ends will be

$\frac{nv}{2\ell}$ and $\frac{(n+1)v}{2\ell}$.

$315=\frac{nv}{2l}$ ...(i)

$420=\frac{(n+1)v}{2l}$ ...(ii)

[where l is the length of the string and v is the speed of the wave]

Now, by dividing equation (i) and (ii)

$\frac{315}{420}=\frac{n}{n+1}$

$\Rightarrow n=3$

Thus the lowest frequency is $\frac{315}{n}=\frac{315}{3}=105$ Hz.

41. (d) Oscillation can be defined as the motion that is moving back and forth repeatedly between two positions or states. It is a periodic motion that repeated itself in a regular cycle.

The fundamental frequency for a closed organ pipe

$\nu=\frac{v}{4l}$

It is given that $v=340\text{ ms}^{-1}$ and $l=85$ cm $=0.85$ m

$\therefore \nu=\frac{340\text{ ms}^{-1}}{4\times0.85\text{ m}}=100$ Hz

Now, the natural frequencies of a closed organ pipe can be derived from the given equation as :

$\nu_n=(2n-1)\nu$

$=\nu, 3\nu, 5\nu, 7\nu, 9\nu, 1/\nu, 13\nu$

= 100Hz, 300Hz, 500Hz, 700Hz, 900Hz, 1100Hz, 1300Hz

As we can see the natural frequencies that lie below 1250 Hz is 6.

42. (a) When sonometer wire vibrates in 1 loop, then its frequency of vibration is called fundamental frequency.

As, $l=\frac{1}{2n}\sqrt{\frac{T}{\mu}}$

$n=\frac{1}{2l}\sqrt{\frac{T}{\mu}}$ and l is the length of the string.

According to given condition total length of the string

$l=l_1+l_2+l_3+....$ ...(i)

Thus $l_1=\frac{1}{2n}\sqrt{\frac{T}{\mu}}$

$l_2=\frac{1}{2n_2}\sqrt{\frac{T}{\mu}}$

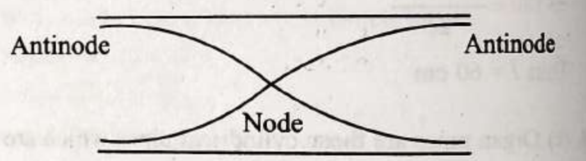
$l_3=\frac{1}{2n_3}\sqrt{\frac{T}{\mu}}$

From (i)

$\frac{1}{2n}\sqrt{\frac{T}{\mu}}=\frac{1}{2n_1}\sqrt{\frac{T}{\mu}}+\frac{1}{2n_2}\sqrt{\frac{T}{\mu}}+\frac{1}{2n_3}\sqrt{\frac{T}{\mu}}$

$\Rightarrow \frac{1}{n}=\frac{1}{n_1}+\frac{1}{n_2}+\frac{1}{n_3}$

43. (a)



Wave-formation in open-ended pipe

So, from diagram, we can see open ends are antinodes.

As frequency $\nu=\frac{nv}{\lambda}$, where n is the number of harmonies.

n = 1, 2, 3 ....

Hence odd and even both harmonics will present.

Since the pipe is open ended, so the ends will have constant pressure which will be equal to the atmospheric pressure.

44. (c) Fundamental frequency of string $\nu=\frac{1}{2l}\sqrt{\frac{T}{\mu}}$

$\therefore$ Fundamental frequency $\nu\propto\frac{1}{\ell}$

$\therefore \nu_1\propto\frac{1}{\ell_1}, \nu_2\propto\frac{1}{\ell_2}$ and $\nu_3\propto\frac{1}{\ell_3}$

Also $\ell_1+\ell_2+\ell_3=\ell$ so $\frac{1}{\nu}=\frac{1}{\nu_1}+\frac{1}{\nu_2}+\frac{1}{\nu_3}$

45. (c) Sonometer consist of a stretched wire whose length and tension can be changed.

Fundamental frequency of sonometer wire

$n=\frac{1}{2l}\sqrt{\frac{T}{\mu}}=\frac{1}{2l}\sqrt{\frac{T}{A\rho}}=\frac{1}{2l}\sqrt{\frac{T}{\pi r^2\rho}}$

According to given condition when

$\rho_1 = \frac{\rho}{2}, T_1 = 2T$ and $D_1 = 2D$ or $r_1 = 2r$

$$n_1 = \frac{1}{2\ell}\sqrt{\frac{2T}{\pi(2r)^2 \frac{\rho}{2}}} = \frac{1}{2\ell}\sqrt{\frac{T}{\pi r^2 \rho}} = n$$

$\Rightarrow$ No change

46. (a) $\ell = \ell_1 + \ell_2 + \ell_3$

$$\frac{k}{n} = \frac{k}{n_1} + \frac{k}{n_2} + \frac{k}{n_3} \Rightarrow \frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$$

47. (a) Given: $v = 330\ ms^{-1}$

$v = 330$ Hz

Frequency of closed organ pipe $v = \frac{(2n-1)v}{4\ell}$

$$\ell = \frac{(2n-1)v}{4v} = \frac{(2n-1)\times 330}{4\times 330} = \frac{(2n-1)}{4}$$

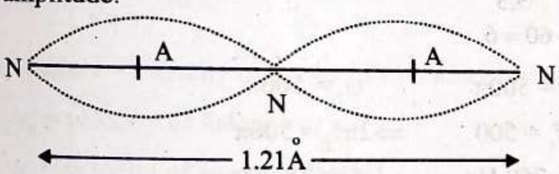
$$\ell = \frac{1}{4}m, \frac{3}{4}m = 25cm, 75cm.$$

$\therefore$ Minimum height of water column

$= 125 - 75 = 50$ cm.

48. (a) Standing waves are formed when two harmonic waves of equal amplitude and frequency travelling through the medium in opposite direction superimposed.

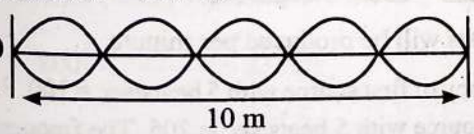
Nodes are the points in a standing wave that has minimum amplitude whereas antinodes are the points that has maximum amplitude.



From the figure we can see distance between position having 3 nodes and 2 antinodes = wavelength = 1.21Å

Therefore $\lambda = 1.21$ Å

49. (a)



As the string produce standing waves and vibrates in five parts so length of the string can be show as

$$\frac{5\lambda}{2} = 10m$$

$\Rightarrow \lambda = 4$ m

Now, the frequency $v = \frac{v}{\lambda} = \frac{20}{4} = 5Hz$ [Given v = 20 m/s]

50. (d) For the cylindrical tube open at both ends, f = v/2L.

When half of the tube is in water, it behaves as a closed pipe of length l/2.

$$\therefore\ f' = \frac{v}{4(l/2)} = \frac{v}{2l}.\ \therefore\ f' = f.$$

Hence frequency remains unchanged.

51. (a) Length of sonometer wire (l) = 110 cm and ratio of frequencies = 1 : 2 : 3.

A sonometer consists of a string that is attached by some weight which provides tension to the string on top of a wooden box.

As we know for sonometer wire

Frequency (ν) $\propto \frac{1}{l}$ or $l \propto \frac{l}{\nu}$.

Therefore AC : CD : DB $= \frac{1}{1} : \frac{1}{2} : \frac{1}{3} = 6 : 3 : 2.$

Therefore $AC = 6\times\frac{110}{11} = 60$ cm

and $CD = 3\times\frac{110}{11} = 30$ cm.

Thus AD = 60 + 30 = 90 cm.

52. (c) For a stretched string, frequency of oscillation is

$$\nu = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$$

$$\Rightarrow \nu \propto \frac{1}{l}$$

When ν is halved, the length is doubled

$\therefore$ Length is 1 m.

53. (d) Third overtone has a frequency of 4n.

4th harmonic = three full loops + one half loop, which Would make four nodes and four antinodes.

54. (a) $\mu = \frac{0.035}{5.5} kg/m, T = 77$ N

where μ is mass per unit length.

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{77\times 5.5}{0.035}} = 110\ m/s$$

55. (a) The number of beats per second is called beat frequency.

Frequency of string A, $f_A = 530$ Hz

$|f_A - f_B|$ = Beat frequency

It is given that $|f_A - f_B| = 6$ Hz

$\Rightarrow f_B = 530 \pm 6Hz$

As we know that frequency is directly proportional to square root of tension T in string.

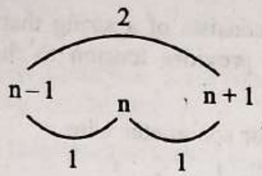
$\nu \propto \sqrt{T}$

According to question when tension in B slightly decrease beat frequency increases to 7Hz.

This condition is only possible when $f_B = 524$ Hz.

Hence $f_B = 524$ Hz.

56. (b) The variation in the intensity of sound between successive maxima or minima is called one beat.



Therefore $(n + 1) - (n - 1) = 2$

57. (b) It is given that when unknown source is sounded with known source of frequency 250 Hz, it gives 4 beats/s.

So, frequency of unknown source = 250 ± 4 Hz = 254Hz or 246 Hz.

As second harmonic of unknown frequency source gives five beats per second when sounded with a frequency source of 513 Hz.

So, frequency of unknown source = 513 ± 5Hz = 508 Hz or 518 Hz

$\therefore$ frequency $= \frac{508}{2}$ Hz $= 254$ Hz

Therefore unknown frequency = 254 Hz.

58. (d) Given: $y_1 = 4 \sin 600\,\pi t$, $y_2 = 5 \sin 608\,\pi t$

$\therefore \omega_1 = 600\,\pi, A_1 = 4$

$\therefore \omega_2 = 608\,\pi, A_2 = 5$

$$\text{Number of beat} = \frac{608\pi}{2\pi} - \frac{600\pi}{2\pi} \Rightarrow 304 - 300 = 4$$

$$\text{and } \frac{I_{max}}{I_{min}} = \left(\frac{A_1 + A_2}{A_1 - A_2}\right) = \left(\frac{4+5}{4-5}\right)^2 = 81$$

59. (a) As we know frequency of vibration

$$\nu = \frac{1}{2L}\sqrt{\frac{T}{\mu}} \quad \text{...(i)}$$

From given condition $\nu + 6 = \frac{1}{2L}\sqrt{\frac{T'}{\mu}}$ ...(ii)

Dividing equation (i) and (ii)

$$\frac{\nu}{\nu+6} = \sqrt{\frac{T}{T'}}$$

$$\frac{600}{606} = \sqrt{\frac{T}{T'}}$$

$$\left(\frac{100}{101}\right)^2 = \frac{T}{T'}$$

$$\frac{T'}{T} = \left(\frac{101}{100}\right)^2 = \left(1 + \frac{1}{100}\right)^2$$

$$\frac{T'}{T} \approx 1 + \frac{2}{100}$$

$$\frac{T'}{T} - 1 = \frac{2}{100}$$

$$\frac{T'-T}{T} = \frac{1}{50} = 0.02$$

60. (a) Given:

Frequency of tuning fork = 512 Hz = n_f

Number of beats = 4

Reduced beats = 2

Since beats = $n_f - n_p$

$\Rightarrow 4 = 512 - n_p$

$\Rightarrow n_p = 508$ Hz

61. (a) As frequency $\propto \sqrt{\text{Tension}}$.

$$\because \nu = \frac{1}{2\ell}\sqrt{\frac{T}{m}} \quad \text{where m is mass per unit length}$$

Number of beats = $\nu_1 - \nu_2$

$$= \frac{1}{2l_1}\sqrt{\frac{T}{m}} - \frac{1}{2l_2}\sqrt{\frac{T}{m}}$$

$$= \frac{1}{2\times\left(\frac{49.1}{100}\right)}\sqrt{\frac{20}{10^{-3}}} - \frac{1}{2\times\left(\frac{51.6}{100}\right)}\sqrt{\frac{20}{10^{-3}}} = 6.97 \approx 7$$

62. (d) Basically, the number of beats per second is equal to the difference in the frequencies of two sound waves.

$$n = \nu_1 - \nu_2 \qquad \left[\because \nu = \frac{v}{\lambda}\right]$$

$$= \frac{v}{\lambda_1} - \frac{v}{\lambda_2} \qquad \begin{bmatrix} v = 300 \text{ m/s} \\ \lambda_1 = 5 \text{ m} \\ \lambda_2 = 5.5 \text{ m} \end{bmatrix}$$

$$= \frac{330}{5} - \frac{330}{5.5}$$

$$= 66 - 60 = 6$$

63. (b) $\omega_1 = 500\pi \qquad \omega_2 = 506\pi$

$\Rightarrow 2\pi f_1 = 500 \qquad \Rightarrow 2\pi f_2 = 506\pi$

$\Rightarrow f_1 = 250$ Hz $\qquad \Rightarrow f_2 = 253$ Hz

No. of beats produced per second

$= f_2 - f_1 = 253 - 250 = 3$ beats/sec

So, 180 beats will be produced per minute

64. (a) Frequency of first source with 5 beats/sec = 100. Frequency of second source with 5 beats/sec = 205. The frequency of the first source = 100 ± 5 = 105 or 95 Hz. Therefore frequency of second harmonic source = 2 × (100 ± 5) = 210 Hz or 190 Hz. As the second harmonic gives 5 beats/second with the sound of frequency 205 Hz, therefore frequency of second harmonic source should be 210 Hz. The frequency of source

$$= \frac{210}{2} = 105 \text{ Hz.}$$

65. (a) First case: Frequency = ν;

No. of beats/sec. = 5 and

frequency (sounded with) = 200 Hz.

Second case: Frequency = 2ν; No. of beats/sec = 10 and frequency (sounded with) = 420 Hz.

In the first case, frequency $(\nu) = 200 \pm 5 = 205$ or 195 Hz.

And in the second case, frequency $(2\nu) = 420 \pm 10$

Or $\nu = 210 \pm 5 = 205$ or 215.

So common value of frequency in both the cases is 205 Hz.

66. **(c)** The time period of one beat or the time interval between two successive maxima or minima is $n_1 - n_2$.

Beats are produced. Frequency of beats will be $\frac{1}{3} - \frac{1}{4} = \frac{1}{12}$.

Hence time period = 12 s.

67. **(b)** For production of beats different frequencies are essential. If frequencies are different, phases will be different.

68. **(c)**

$f_0 = 400$ Hz $\qquad f_1 = ?$

$v_0 = 22$ m/s $\qquad v_1 = 16.5$ m/s

As both source and observer are moving towards each other frequency heard by driver of second car.

$$f_1 = f_0\left[\frac{v + v_1}{v - v_0}\right]$$

$$f_1 = 400\left[\frac{340 + 16.5}{340 - 22}\right]$$

$$f_1 = 448.4 \text{ Hz}$$

69. **(c)** Apparent frequency (n_1) in terms of actual frequency (n)

$$n_1 = \left(\frac{v \pm v_0}{v \pm v_s}\right)n$$

where v = velocity of sound in stationary medium

v_0 = velocity of listener

v_s = velocity of source of sound

As obeser is at rest so $v_0 = 0$ and source moves towards cliff.

Hence frequency received by cliff

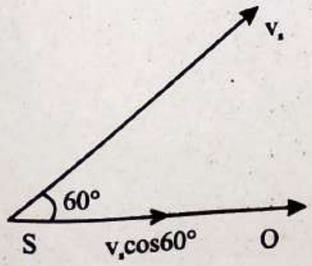
$$n' = 800\left(\frac{330}{330 - 15}\right) = 838 \text{ Hz}$$

Therefore, the frequency which cliff will reflect = 838 Hz.

70. **(c)** The source has a component of velocity along 'SO'. So, frequency emitted by source

$$f_0 = f_s\left[\frac{v}{v - v_s \cos 60°}\right]$$

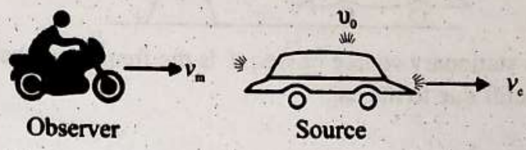
$v = 330 \text{ ms}^{-1}$



$$f_0 = f_s\left(\frac{v}{v - v_s}\right) = 100\left(\frac{330}{330 - \frac{19.4}{2}}\right) \approx 103 \text{ Hz}$$

The apparent frequency observed by the observer will increase.

71. **(c)**



Here, speed of motorcyclist,

$$v_m = 36 \text{ km hour}^{-1} = 36 \times \frac{5}{18} \text{ ms}^{-1} = 10 \text{ ms}^{-1}$$

Speed of car,

$$v_c = 18 \text{ km hour}^{-1} = 18 \times \frac{5}{18} \text{ ms}^{-1} = 5 \text{ ms}^{-1}$$

Frequency of source, $v_0 = 1392$ Hz

Speed of sound, $v = 343 \text{ ms}^{-1}$

The frequency of the honk heard by the motorcyclist is

$$v' = v_0\left(\frac{v + v_m}{v + v_c}\right) = 1392\left(\frac{343 + 10}{343 + 5}\right)$$ [As both are moving in same direction]

$$= \frac{1392 \times 353}{348} = 1412 \text{ Hz}$$

72. **(c)** Here, speed of train $v_t = 220 \text{ ms}^{-1}$

speed of sound in air $= 330 \text{ ms}^{-1}$

$$v' = v\left(\frac{v + v_t}{v - v_t}\right) = 1000\left(\frac{330 + 220}{330 - 220}\right)$$

$$v = 1000 \times \frac{550}{110} = 5000 \text{ Hz}$$

73. **(b)** Let n' be the frequency of sound at the hill and n'' be the frequency of reflected sound as heard by the driver.

Using Doppler effect,

$$n' = \left(\frac{v}{v - v_1}\right)n$$

$$n'' = \left(\frac{v + v_1}{v}\right)n' = \left(\frac{v + v_1}{v}\right) \times \left(\frac{v}{v - v_1}\right)n$$

$$= \left(\frac{v + v_1}{v - v_1}\right)n$$

$$= \left(\frac{330 + 30}{330 - 30}\right) \times 600 = 720 \text{ Hz}$$

74. **(a)** Reverberation time $T = \frac{6.61V}{aS}$, where V is the volume, a is the absorption coefficient, S is the total surface area.

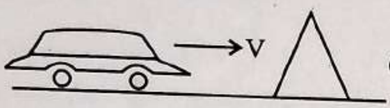
$$T \propto \frac{V}{S}$$

$$\Rightarrow \frac{T_1}{T_2} = \left(\frac{V_1}{V_2}\right)\left(\frac{S_2}{S_1}\right) = \left(\frac{V}{8V}\right)\left(\frac{4S}{S}\right) = \frac{1}{2}$$

As $T_1 = 1$ sec.

$\Rightarrow T_2 = 2T_1 = 2$ sec.

75. (a) According to given condition



cliff is stationary source of freq. f' is the frequency observed at the cliff due to the car.

$$f' = \frac{v}{v - v_s} f$$

$$f'' = \left(\frac{v + v_{car}}{v}\right) f' = \left(\frac{v + v_{car}}{v - v_{car}}\right) f = 2f$$

$$\Rightarrow v + v_{car} = 2v - 2v_{car} \Rightarrow 3v_{car} = v \Rightarrow v_{car} = v/3$$

76. (b) Here, speed of observer $v = \frac{1}{5} v_s$

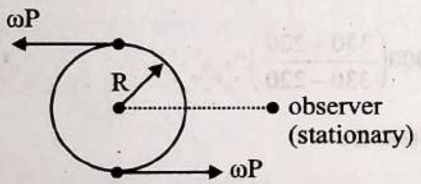
Source is stationary $\Rightarrow \lambda =$ constant

Since observer moves towards a source.

$$f' = \left(\frac{v + v_s}{v}\right) f$$

$$f = \left(1 + \frac{v_s}{v}\right) f = \left(1 + \frac{1}{5}\right) f = 1.2f$$

77. (b) Frequency heard by observe will be minimum when the source recedes away from him as shown in the figure.



Speed of the source $= \omega R = 0.5 \times 20 = 10\ ms^{-1}$

$$n_{max} = n_0\left(\frac{V}{V - \omega R}\right); n_{min} = n_0\left(\frac{V}{V + \omega R}\right)$$

$$= 385 \times \frac{340}{340 + 10} = \frac{34}{35} \times 385 = 374 \text{ Hz}$$

78. (a) Let frequency of both the sources = n

Now, frequency of each stationary sources $n = \frac{v}{\lambda}$

S_1 $0 \xrightarrow{u}$ S_2

(i) When the observer is moving away from the source

$$n_1 = \left(\frac{v - u}{v}\right) n = \left(1 - \frac{u}{v}\right) n$$

(ii) When the observer is moving towards the source

$$n_2 = \left(\frac{v + u}{v}\right) n = \left(1 + \frac{u}{v}\right) n$$

Beat freq. $|n_1 = n_2| = n + \frac{nu}{v} - n + \frac{nu}{v}$

$$= \frac{2nu}{v} = 2\frac{u}{\lambda} \qquad \left[\because \frac{1}{\lambda} = \frac{n}{v}\right]$$

79. (d) If source moves perpendicular to observer's motion then change in freq. = 0 (No doppler's effect)

80. (a) Given $v' = \frac{9}{8} v$

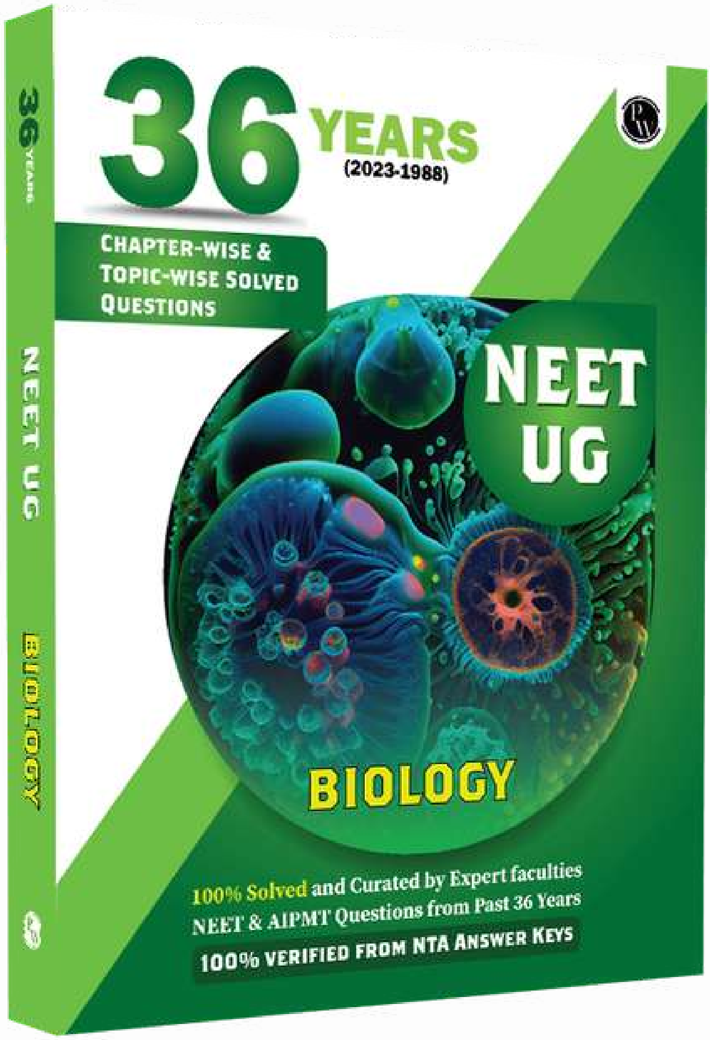
Source and observer are moving in opposite direction, ∴ apparent frequency

$$v' = v \times \frac{(v + u)}{(v - u)}$$

$$\frac{9}{8} v = v \times \frac{340 + u}{340 - u}$$

$9(340) - 9u = 8(340) + 8u$

$340 = 17\,u \quad \Rightarrow \quad u = 20$ m/s



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